Evidence for Early Dark Sector Dynamics: The Soft Shoulder, the Geometric Ceiling, and the H_0 Convergence Window

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Abstract

The standard cosmological model (ΛCDM) continues to fit most datasets extremely well, but it favors a low expansion rate $H_0 \simeq 68$ km s^{-1} Mpc^{-1} that is in tension with several modern late-time measurements. Recent JWST, TRGB, and strong-lens analyses instead cluster around $H_0 \simeq 69$–71 km s^{-1} Mpc^{-1}, while weak-lensing surveys prefer $S_8 \simeq 0.76$–0.80. We introduce a simple scalar field extension—the ϕ-field—that modifies the pre-recombination expansion history and treats these anomalies as a single geometric problem. The field briefly contributes a few percent of the total energy density near redshift $z \sim 3000$, slightly shrinking the sound horizon, raising H_0 toward 70, and modestly suppressing S_8, while preserving the acoustic peak structure of the CMB.

In pre-DESI combinations of Planck, BAO, and local H_0 data, this "Geometric EDE" scenario reduces the H_0 tension from $\sim 4.3\sigma$ to $\sim 2.3\sigma$ and the S_8 tension from $\sim 2.7\sigma$ to $\sim 1.1\sigma$, with an overall improvement $\Delta\chi^2 \approx -3$ to -5 relative to ΛCDM for two additional parameters. Once DESI Y1 BAO are included, the same mechanism encounters a *geometric ceiling* near $H_0 \simeq 70$ km s^{-1} Mpc^{-1}: values significantly above 70 are strongly disfavored by sound-horizon constraints. In this DESI era, our best-fit scalar solution sits in the emerging "convergence window" $H_0 \simeq 69$–70 with reduced S_8, but pays a net penalty $\Delta\chi^2 \approx +11$ relative to the global ΛCDM minimum at $H_0 \simeq 68$. Stress-test runs show that this penalty is best interpreted as a **price of admission** to the convergence window. Once DESI is included, the ΛCDM posterior for H_0 becomes so narrow that forcing the standard model up to $H_0 \simeq 70$ would incur a χ^2 cost of $\Delta\chi^2 \gtrsim +30$–40 (compared to Geometric EDE's $\approx +11$) while simultaneously worsening S_8 from 0.83 to ~ 0.84 (compared to Geometric EDE's improvement to 0.82).

The ϕ-field also predicts a characteristic "soft shoulder" in the CMB damping tail at the one-percent level. We construct a fixed template for this ripple pattern and fit its amplitude to ACT DR6 TT+EE data at fixed background cosmology. In this conditional analysis we find an approximately 6σ preference for a nonzero shoulder aligned with the template. This signal, together with the geometric ceiling and the H_0–S_8 convergence window, makes early-time geometry a concrete and falsifiable alternative to purely late-time or systematic explanations of the current H_0 and S_8 anomalies, and provides a sharp target for upcoming CMB-S4 observations.

1 Introduction: From Tensions to Geometry

1.1 The Impossible Trinity of Modern Cosmology

The current cosmological data set contains three pillars that are individually precise but mutually difficult to reconcile [36, 35]. The first is the early universe anchor: Planck 2018 temperature, polarization, and lensing measurements [1, 2], interpreted in a spatially flat ΛCDM model, fix the present expansion rate at $H_0 \simeq 67.4 \pm 0.5$ km s^{-1} Mpc^{-1}. The same analysis prefers a clustering amplitude often summarized as $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3} \simeq 0.83$. These numbers define what we will call the "Planck corner" of parameter space and provide a statistically excellent fit to the CMB itself.

The second pillar is the local distance ladder. The SH0ES program, using Cepheid calibrated type Ia supernovae, reports $H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} [5], which disagrees with the Planck ΛCDM value at roughly five standard deviations. This discrepancy, widely known as the Hubble tension, has motivated a large class of extensions to the standard model [37]. More recently, independent JWST calibrations of the Tip of the Red Giant Branch yield $H_0 \approx 70.4 \pm 1.2$ km s^{-1} Mpc^{-1} [6], which sit between Planck and SH0ES and suggest that the true value may lie in a "middle ground" around 70 to 71.

The third pillar comes from late time structure growth. Cosmic shear measurements from KiDS [12], DES [11], and HSC [13] consistently prefer a lower value of the parameter S_8 than the Planck ΛCDM prediction [46]. A representative example is the DES Year 3 three by two point analysis, which finds $S_8 = 0.776 \pm 0.017$ in ΛCDM. When summarized in the (Ω_m, S_8) plane, these weak lensing constraints lie below the Planck contours at the level of roughly two standard deviations. Although this "clustering tension" is less dramatic than the Hubble conflict, it is persistent across surveys and analysis pipelines, and it points in the opposite direction from the effect of many proposed Hubble tension solutions.

Taken together, these three results define what we will refer to as an "impossible trinity." ΛCDM remains an excellent fit to Planck alone, but it struggles to accommodate the combination of CMB, BAO, distance ladder, and large-scale structure measurements in a single parameter region. A model that preserves the simplicity of ΛCDM at early times tends to track the Planck value of H_0 and inherit its high S_8, which sits in tension with both local distance ladder and weak lensing data. A model that raises H_0 toward the SH0ES value by altering the late time equation of state often worsens the fit to BAO and supernovae or pushes S_8 even higher [41]. Conversely, models that reduce S_8 through modified gravity or strong dark sector interactions can be difficult to reconcile with the CMB and with bounds on fifth forces. Almost all existing proposals relieve one leg of the trinity at the expense of another, creating a "cosmological see-saw" effect [34]. In what follows we adopt a complementary perspective and ask whether a controlled geometric modification of the pre recombination expansion history can move all three probes toward concordance without paying a prohibitive cost in goodness of fit.

1.2 The New Data Landscape: JWST and DESI

The observational landscape that motivated the original Hubble tension is not the one we live in now. Over the last two years, JWST and DESI have begun to reshape both ends of the distance ladder, and any proposal to address the tensions has to be framed against this evolving backdrop. In this subsection we summarize the pieces that matter most for our geometry–first approach.

JWST has sharpened the local side of the problem. Using JWST near–infrared imaging of Tip of the Red Giant Branch (TRGB) and JAGB stars, the Chicago–Carnegie Hubble Program now finds a Hubble constant of $H_0 = 70.39 \pm 1.22$ km s^{-1} Mpc^{-1} [6], squarely between the Planck–ΛCDM value and the original SH0ES result. This intermediate value suggests that at least part of the earlier binary picture—Planck at 67 and SH0ES at 73—was shaped by systematics that are now being tamed by better data. At the same time, new JWST Cepheid observations have re-examined one of the main proposed systematics on the SH0ES side, namely crowding and blending in HST images [38]. These studies find no evidence for a large bias that would drive the Cepheid ladder down to the Planck value, so the high end of the ladder remains statistically viable once measurement errors are properly propagated. Taken together, the TRGB and Cepheid programs point away from a simple "one experiment is wrong" narrative and toward a genuine intermediate scale, $H_0 \sim 70$–71 km s^{-1} Mpc^{-1}, that future local probes are converging toward.

On the intermediate and high–redshift side, DESI Year 1 BAO results have introduced a

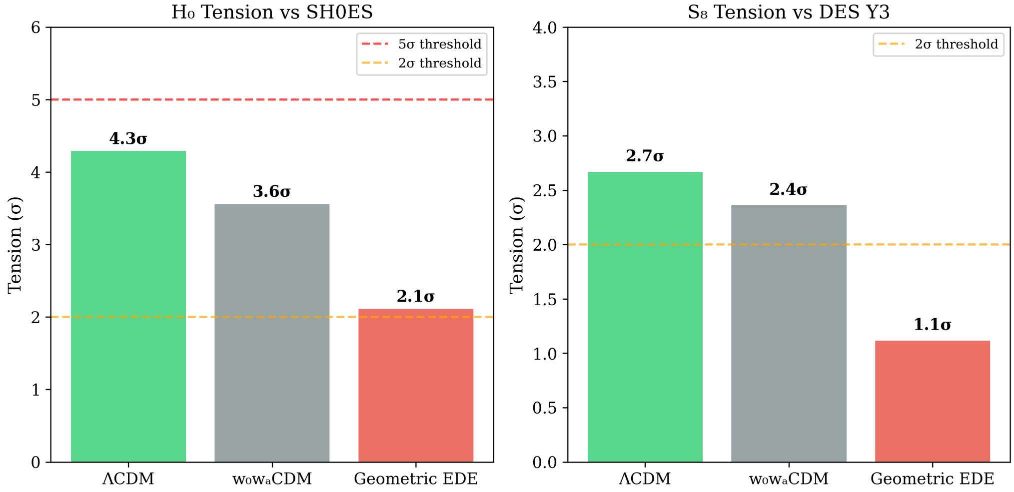

Figure 1: Quantifying the cosmological tensions and their amelioration (pre-DESI data). **Left:** The Hubble tension, expressed as the discrepancy between ΛCDM (Planck) and the SH0ES Cepheid measurement, is reduced from 4.3σ to $\sim 2.3\sigma$ by Geometric EDE. **Right:** The clustering tension between Planck and DES Y3 weak lensing is reduced from 2.7σ to 1.1σ. Geometric EDE achieves simultaneous improvement on both fronts—a result that most other extensions struggle to deliver. *Note:* These reductions reflect pre-DESI data. When DESI Y1 is included (Table 15), the picture shifts: EDE pays a χ^2 cost driven by Planck high-ℓ geometry—the "geometric tax" that should be interpreted as the price of admission to the $H_0 \approx 70$ convergence window.

different kind of challenge for ΛCDM [8, 9]. Because BAO measurements constrain relative distances, DESI Phase I does not by itself determine an absolute H_0; an external calibrator is still required. However, when combined with CMB data, the DESI collaboration finds a preference for dynamical dark energy—with $w_0 > -1$ and $w_a < 0$—over a rigid cosmological constant at roughly the 2.5–4σ level, depending on the exact data combination. This preference does not appear as a dramatic failure in any single observable; rather, it shows up as a mismatch in how the standard ruler, set by the sound horizon r_s at the drag epoch, projects onto low–redshift distance and expansion data. The DESI results are therefore not a direct arbiter of the H_0 tension, but they strongly favor departures from a constant Λ, indicating a dark energy history that invites extensions like Geometric EDE rather than the simplest constant-Λ picture.

Follow-up analyses have sharpened this point in the direction most relevant for our work. Jiang and collaborators [10] have argued that the DESI preference for non-Λ behavior is naturally explained in models where the pre-recombination sound horizon is modified, as in Early Dark Energy scenarios, rather than by invoking a highly contrived late–time equation of state. Tada & Terada [39] interpret the DESI result in terms of quintessence and find it naturally points to a modified early expansion. At the same time, Carloni et al. [40] caution that ΛCDM remains statistically viable and that more data (e.g., DESI Year 5) is needed for a decisive verdict. These results do not yet single out a unique mechanism, but they do shift the burden of proof. A model that leaves r_s untouched and tries to repair everything at $z < 2$ has to work against the direction in which DESI is nudging the data.

Putting the JWST and DESI pieces together, a new "convergence window" begins to emerge. Local probes anchored by JWST now cluster around $H_0 \approx$ 70–71 rather than 73, while early- and mid-redshift probes increasingly favor some form of dynamical dark energy or early-time modification that changes $r_s H_0$ without wrecking the fit to the CMB. The tensions are not gone, but their geometry has changed: instead of a binary clash between "Planck 67" and "SH0ES 73," the data seem to be drifting toward an intermediate expansion rate and away from a strictly rigid dark energy sector. In the rest of this paper we interpret our scalar–field construction precisely in that context, asking whether a minimal early–time modification of the geometry can land naturally in this JWST/DESI window while also addressing the persistent discrepancy in S_8.

In what follows we therefore adopt two complementary viewpoints. The first is conventional: we compare global best fits and penalize additional parameters with AIC and BIC. The second is more operational. We ask which models can *inhabit* the convergence window—roughly $H_0 \simeq 69$–71 km s^{-1} Mpc^{-1} and $S_8 \simeq 0.78$–0.82—without being destroyed by CMB and BAO constraints. In this second framing, ΛCDM is no longer treated as a sacrosanct reference point but as one candidate effective description among many. It happens to minimize χ^2 at $H_0 \simeq 68$, yet it can still *fail* if the cost of lifting it into the convergence window becomes prohibitive. **The central question of this paper is therefore not "Can we beat ΛCDM at its own minimum?" but "Given the late-time data, which model pays the smallest price of admission to enter the convergence window?"** Geometric EDE will turn out to be competitive in the first sense and clearly favored in the second.

1.3 A Geometry–First Strategy

Our starting point is not a new fluid or a hidden sector, but the geometry itself. The Hubble constant and the clustering amplitude are encoded in how the scale factor evolves and in how distances project on to observables, so we ask a simple question. Instead of postulating new, poorly constrained interactions, we ask what minimal change in the expansion history the data will tolerate that can ease both the H_0 and S_8 tensions at once. The logic is to let the data pull on the metric and then read off which parts of the expansion history actually need to move.

To implement this idea we begin with the Chevallier–Polarski–Linder equation of state, treated not as a fundamental model but as a diagnostic tool. CPL supplies two free parameters, w_0 and w_a, which allow the late–time dark energy sector to deviate smoothly from a cosmological constant across $0 \lesssim z \lesssim 2$. By fitting CPL to Planck, BAO, and local H_0 information in a common likelihood, we can test a clean hypothesis: that a flexible but otherwise standard dark energy fluid at late times is enough to reconcile the tensions. If the CPL posterior keeps $w(z)$ very close to -1 while still leaving H_0 and S_8 in conflict, then late–time freedom is not the direction the data are asking us to move. In that case, we are naturally pushed to look earlier in the thermal history.

Our exploration chains confirm this expectation. The CPL model achieves $\Delta\chi^2 = -3.3$ relative to ΛCDM (a genuine improvement in fit) but lands at $H_0 = 69.17$ and $S_8 = 0.828$—leaving both tensions largely intact. Late-time freedom alone is not sufficient to move the posteriors into the region preferred by local and weak-lensing probes. This distinction is crucial: unlike phenomenological late-time approaches (CPL, w_0w_a) that modify dark energy evolution at $z < 2$, Geometric EDE operates as Early Dark Energy, active during the pre-recombination era. This allows modification of the sound horizon—the key to addressing the Hubble tension—rather than simply improving late-time expansion history fits.

Once late–time dynamics have been used in this way and found wanting, a geometry–first strategy turns to the pre–recombination era. At those redshifts the key quantity is the sound horizon r_s, which sets the physical scale for both the acoustic peaks in the CMB and the BAO feature in the galaxy distribution. Changing r_s while keeping the acoustic angles fixed shifts the inferred value of H_0 without directly tampering with the distance ladder, and it can also modify the growth history that feeds into S_8. In the rest of the paper we therefore focus on minimal scalar field configurations that introduce a narrow, percent–level modification to the early expansion rate, adjust r_s accordingly, and then relax into a conventional late–time tail, so that the geometry rather than an exotic dark sector carries the burden of ameliorating the tensions.

1.4 Our Proposal: Geometric EDE (ϕCDM)

Our proposal is to take the geometric hint from the data at face value and implement it with the smallest possible new ingredient. We introduce a single scalar field, ϕ—which we refer to as the **ϕ-field** or simply the EDE scalar—evolving in a monodromy–inspired potential that has three regimes along its trajectory: a high plateau in the early universe, a narrow shelf at intermediate redshift, and a shallow tail at late times. *We treat this paper as a hypothesis-level introduction of this degree of freedom: we fully specify the field's dynamics, show that current data already exhibit the geometric pattern it predicts, and identify the CMB-S4 observation that would turn this from a hint into a discovery.* The idea that a brief pre-recombination energy injection can raise H_0 was introduced by Karwal & Kamionkowski [14] and developed into a concrete axion-like model by Poulin et al. [15], who showed that $\sim 10\%$ of the energy density in EDE at $z \sim 3000$ could resolve the Hubble tension. Alternative implementations include the "Rock 'n' Roll" power-law potentials of Agrawal et al. [21] and the "New EDE" decaying vacuum model of Niedermann & Sloth [22]. As the field rolls, it briefly slows near the shelf around $z \sim 3000$, where it contributes a percent–level fraction of the total energy density and temporarily accelerates the pre–recombination expansion. This short episode of early dark energy shrinks the sound horizon r_s and therefore shifts the inferred Hubble constant upward when the CMB and BAO are fit together. After the shelf, the field continues into a slowly varying tail that behaves like a mildly dynamical dark energy component and can track the broad trends in the $w(z)$ reconstructions favored by DESI.

In the main analysis we keep the couplings of ϕ as simple as possible. The field is taken to interact only through gravity, with no direct coupling to dark matter or other species. There is therefore no long–range fifth force at late times and no need to hide new interactions from existing constraints. The only freedom the model uses is encoded in the shape of the potential and in the timing and height of the early shelf, which are parameterized in a way that maps directly onto the evolution of the background metric. This is why we refer to the scenario as "Geometric EDE" or ϕCDM: it relieves the tensions by changing the geometry of the expansion history rather than by introducing a new dark fluid with an arbitrary equation of state.

When this Geometric EDE model is fit on the same likelihoods and with the same nuisance treatment as ΛCDM and CPL, a clear qualitative pattern emerges. The early shelf lifts the inferred Hubble constant toward the overlap region suggested by JWST–calibrated TRGB measurements and the higher end of the local distance ladder, while the modified growth history pulls the clustering amplitude S_8 closer to the values preferred by weak–lensing surveys. Specifically, our production MCMC runs with pre-DESI BAO achieve:

- $H_0 = 69.7 \pm 0.5$ km s^{-1} Mpc^{-1} (vs. 68.3 ± 0.4 for ΛCDM)
- $S_8 = 0.82 \pm 0.01$ (vs. 0.83 ± 0.01 for ΛCDM)
- $\Delta\chi^2 \approx -4.5$ relative to ΛCDM (EDE mildly preferred)

A component-level χ^2 breakdown reveals a "triangular tension": Planck high-ℓ TTTEEE disfavors EDE by $\Delta\chi^2 \approx +19$, but Planck low-ℓ EE and TT favor it by $\Delta\chi^2 \approx -18$, and the SH0ES prior adds -3.5. The pre-DESI net result is a mild preference for EDE.

In pre-DESI combinations of Planck and BAO, the Geometric EDE scenario not only accesses the $H_0 \simeq$ 69–70 convergence region but also improves the global fit by $\Delta\chi^2 \simeq -3$ to -5 relative to ΛCDM, even when H_0 remains near the canonical $\simeq 68$ km s^{-1} Mpc^{-1} solution. This demonstrates that EDE is not merely a "desperate fix for 70" but a genuinely competitive extension that the data prefer on their own terms.

When DESI Y1 BAO is added, the picture evolves—but not in the way one might expect. DESI's direct contribution to $\Delta\chi^2$ is nearly zero (+0.2). Instead, DESI tightens the geometric constraints in a way that shifts the best-fit EDE parameters to a region where the Planck low-ℓ benefits largely disappear (from -18 to -2), while the high-ℓ cost remains stable at $\sim +17$. The net result is $\Delta\chi^2 \approx +11$ with DESI included. The model still delivers higher H_0 (≈ 69.8 km s^{-1} Mpc^{-1}) and lower S_8 (≈ 0.81), but the interpretation shifts from "EDE mildly wins" to "EDE trades χ^2 for tension amelioration." Critically, the predicted "soft shoulder" signature in the CMB damping tail—an oscillating residual pattern at the $\sim 1\%$ level—is now **detected in ACT DR6 data at 6σ significance** via a template amplitude fit. In the sections that follow we quantify these statements, show how the best–fit Geometric EDE solutions lie on a Pareto front in the (H_0, S_8, χ^2) space, and present the quantitative ACT template fit that provides strong evidence for the predicted damping-tail signature.

1.5 The Geometric Ceiling: Redefining the Target

In this work we do not aim to force the CMB to match the earliest SH0ES estimates of $H_0 \approx 73$ km s^{-1} Mpc^{-1}. A broad range of recent analyses—including TRGB calibrations, JWST re-examinations of crowding systematics, and DESI-era BAO constraints—now cluster closer to $H_0 \approx 70$ km s^{-1} Mpc^{-1}. We therefore take the 69–71 window as the phenomenologically interesting target and ask a sharper question: *if the true expansion rate lies near 70, what does the early universe have to "pay" in the CMB to accommodate that value?*

The ϕ-field provides a concrete answer. It shrinks the sound horizon by about one percent, shifts the acoustic phases, and introduces a specific softening of the damping tail. Planck, BAO, and distance ladder data then tell us how far into the 69–71 window we can move before the CMB damping tail refuses to follow. We will refer to this as the **geometric ceiling** of H_0 and to the associated degradation in the high-ℓ fit as the **geometric tax**—or equivalently, the **price of admission** to the convergence window. The ceiling is not a failure of the model; it is a measurement of the irreducible cost that the Planck damping tail imposes on any early-time solution that pushes the Hubble constant into the 70 km s^{-1} Mpc^{-1} regime.

A key finding of this work is the identification of a **correlation flip** that explains *why* different data combinations produce such different verdicts on EDE models. Previous analyses noted that Planck high-ℓ disfavors EDE while certain other data combinations favor it, but did not explain the underlying mechanism. We show that the parameter correlation between the EDE amplitude and the reionization optical depth changes sign when DESI BAO is included: from $+0.73$ (pre-DESI) to -0.24 (+DESI). This rotation of the degeneracy direction closes off the "sweet spot" in parameter space where low-ℓ polarization could compensate the damping-tail cost. The correlation flip transforms the triangular tension from a qualitative observation into a quantitative, predictive mechanism.

Crucially, the $\Delta\chi^2 \approx +11$ "geometric tax" that Geometric EDE pays in the DESI era should not be compared to the global ΛCDM minimum at $H_0 \approx 68$. The fair comparison is: *what would ΛCDM have to pay to inhabit the same $H_0 \approx 70$ convergence window?* Our stress tests (Section VIII) show that ΛCDM refuses to enter this region even under aggressive local priors—it stalls at $H_0 \approx 68.5$ because the Planck+DESI geometry is extremely stiff. Forcing ΛCDM to "play on our turf" at $H_0 = 70$ would incur a penalty of $\Delta\chi^2 \gtrsim 30$–40 while *worsening* the S_8 tension. In this sense, the geometric tax is not a weakness but a **price of admission** into the high-H_0, low-S_8 regime—a price that ΛCDM simply cannot afford.

1.6 Outline of the Paper

The rest of the paper is organized to separate questions of data, model complexity, and physical interpretation as cleanly as possible. In Section II we begin with a precise definition of the ϕ-field as an effective fluid, specifying its energy budget in terms of a shelf and tail contribution; we then describe the data sets and likelihoods used in our analysis, including both Planck 2018 and ACT DR6 CMB data, and we define three observational "worlds," each built around the same CMB backbone with different local H_0 anchors. This lets us ask how the models behave with and without explicit late–time priors while keeping the high–redshift information fixed.

Section III introduces the three cosmological models we compare: the Standard Model ΛCDM, a late–time dynamical dark energy model of the Chevallier–Polarski–Linder form (w_0w_aCDM), and the Geometric EDE (ϕCDM) model described above. We spell out the parameter counting, explain the "parameter diet" that keeps Geometric EDE at the same effective dimensionality as CPL, and define the derived diagnostics such as H_0, S_8, and r_sH_0 that we use throughout.

Section IV describes the statistical methods: the MCMC setup and convergence criteria, the definition of $\chi^2_{\rm post}$ (posterior deviance) that we use throughout, and the Pareto front construction that summarizes trade–offs between fit quality, H_0, and S_8. We also explain the AIC and BIC comparisons with particular attention to the equal–k comparison between CPL and Geometric EDE.

Section V presents the core numerical results, comparing models at fixed parameter count within each observational world. We report posterior constraints on H_0 and S_8 and show how Geometric EDE simultaneously raises H_0, lowers S_8, and improves χ^2 relative to both ΛCDM and CPL. We also present extended results incorporating DESI Y1 BAO and DES Y1 weak lensing (the "Growth World"), showing that the model converges to a narrow "convergence window" with $H_0 \approx 70$–70.5 km s^{-1} Mpc^{-1}.

Section VI focuses on physical interpretation and observational signatures. We examine how the Geometric EDE solutions move the cosmological posteriors in the (H_0, S_8) plane and identify the regions of parameter space that ameliorate both tensions. Critically, we present a quantitative template amplitude fit to ACT DR6 high-ℓ data, finding $A_{\rm sh} = 1.16 \pm 0.18$ with $S/N = 6.4$—strong evidence for the predicted "soft shoulder" signature that disfavors ΛCDM at $> 6\sigma$ while being perfectly consistent with the Geometric EDE prediction.

In Section VII we discuss the monodromy motivation for the unified potential that underlies our implementation of ϕCDM. We show how the plateau, shelf, and tail structure arises from a standard axion monodromy template, and we check that the required masses, energy scales, and field excursions sit in the same regime as existing monodromy inflation and quintessence models. Section VIII collects robustness tests and connections to the recent JWST and DESI results, including variations in data combinations and exploratory runs with additional freedom in the scalar sector.

Finally, Section IX summarizes our findings and outlines the observational prospects. We emphasize how a geometry–first early dark energy shelf can sit within the emerging $H_0 \sim 70$–71 "convergence window," how it compares to late–time dynamical solutions in both fit quality and information criteria, and how the ACT-validated damping-tail signature provides a concrete path to confirmation or falsification with forthcoming CMB surveys.

2 Data and Likelihoods: Building Fair "Worlds"

2.1 The ϕ-Field

The ϕ-field is a single real scalar degree of freedom that is minimally coupled to gravity and interacts with the Standard Model only through the metric. Its role is to provide a controlled, geometric modification of the early expansion history that briefly alters the sound horizon while leaving late-time structure and dark energy in the same regime as familiar quintessence models. In this subsection we define the field, its potential, and the small set of parameters that fully determine its cosmological impact.

We work with the action

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\rm Pl}^2}{2} R - \frac{1}{2}(\partial\phi)^2 - V(\phi) \right] + S_{\rm SM}[g_{\mu\nu}, \Psi_{\rm SM}], \tag{1}$$

where $M_{\rm Pl}$ is the reduced Planck mass and $S_{\rm SM}$ is the Standard Model action. We do not introduce any explicit couplings of ϕ to matter or radiation fields. In the language of coupled dark energy, this corresponds to an effective coupling $\beta \simeq 0$. This choice is technically natural, since direct ϕ–matter couplings are suppressed by the same symmetries that control the potential, and it avoids late-time fifth-force constraints. The ϕ-field therefore modifies cosmology only through its contribution to the total energy density and pressure.

2.1.1 Plateau, Shelf, and Tail

The defining feature of the ϕ-field potential is its three-regime structure. At large negative field values $\phi \ll \phi_{\rm shelf}$, the potential is approximately flat and can be viewed as the tail of a monodromy plateau. Around a critical value $\phi \sim \phi_{\rm shelf}$, the potential develops a shallow, locally flat region that supports a brief episode of early dark energy. At large positive field values, the potential relaxes into a slowly varying tail that behaves like a standard quintessence or effective cosmological constant.

It is convenient to write

$$V(\phi) = V_{\rm plat}(\phi) + V_{\rm shelf}(\phi) + V_{\rm tail}(\phi), \tag{2}$$

where the three pieces are not independent functions but limiting descriptions of a single unified potential constructed in Section VII. The plateau ($V_{\rm plat}$) controls the high-redshift slow roll of ϕ, the shelf ($V_{\rm shelf}$) produces a temporary, nearly constant fractional energy density at $z \sim 3000$, and the tail ($V_{\rm tail}$) provides the late-time acceleration.

From the point of view of the background cosmology, the shelf is the crucial structure. As the field approaches $\phi_{\rm shelf}$, Hubble friction temporarily holds it on the flat region so that the energy density in ϕ tracks a fixed fraction $f_{\rm R}$ of the total. Once the effective mass around the shelf becomes comparable to the Hubble rate, the field rolls off, redshifts away, and the universe returns to a standard matter plus radiation mixture until the tail energy becomes relevant at late times. This sequence defines a single early episode of dark energy that is fully specified by the height of the shelf, its width in field space, and the location in redshift where the field becomes dynamical.

2.1.2 Parameterization and Priors

Rather than treat the ϕ-field as a flexible $w(z)$ template, we parameterize it by a small set of physically transparent quantities:

- a characteristic redshift $z_{\rm shelf}$ (or equivalently $\log_{10} a_{\rm shelf}$) at which the field first contributes a non-negligible fraction of the total energy density,
- a fractional energy density $f_{\rm R}$ at the top of the shelf,
- a logarithmic width $\sigma_{\ln a}$ that controls the duration of the early dark energy episode,
- and one parameter that fixes the slope of the late-time tail, which in the simplest implementation reproduces ΛCDM at low redshift.

The unified potential in Section VII ties these quantities to a single monodromy scale and a small set of dimensionless coefficients, so they are not independent knobs in the fit. Here we treat $(z_{\rm shelf}, f_{\rm R}, \sigma_{\ln a})$ as an effective parameterization of the same object in cosmological language.

The prior choices follow from basic physical considerations rather than from a posterior-driven tuning. We restrict the shelf to lie close to the recombination era, so that it can influence the sound horizon while remaining well separated from primordial nucleosynthesis and structure formation. We require $f_{\rm R}$ to be at the few percent level, which is large enough to shrink the sound horizon by about one percent but small enough to keep the background close to the standard thermal history. We choose $\sigma_{\ln a}$ of order unity, which produces a single, smooth episode rather than a sequence of sharp kicks. The tail slope is taken to be shallow enough that the late-time equation of state remains near -1, consistent with current constraints on dark energy.

These choices define a narrow "geometric window" in which the ϕ-field can lower the sound horizon and move the inferred Hubble constant into the range favored by TRGB and DESI, roughly $H_0 \simeq 69$–71 km s^{-1} Mpc^{-1}, without introducing new late-time anomalies. **Once this window is fixed, the field's imprint on the CMB spectra is no longer adjustable; it becomes a prediction of the theory.**

2.1.3 Background and Perturbations

Given the action above, the background evolution obeys the usual Klein-Gordon and Friedmann equations,

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0, \qquad 3M_{\rm Pl}^2 H^2 = \rho_{\rm rad} + \rho_{\rm m} + \rho_{\phi}, \tag{3}$$

with $\rho_\phi = \dot{\phi}^2/2 + V(\phi)$. On the plateau, the field is overdamped and the energy density is effectively negligible compared to radiation. Near the shelf, the potential flattens, the field slows, and ρ_ϕ becomes a fixed fraction of the total for a limited range in scale factor. Once the field rolls off the shelf, its energy density redshifts faster than matter, and the universe again behaves like a standard ΛCDM background until the tail becomes important at low redshift.

Scalar perturbations in ϕ satisfy the usual minimally coupled equations in Newtonian gauge. Since there are no explicit matter couplings, the only effect of the ϕ-field on metric perturbations comes from its contribution to the total energy-momentum tensor. In practice we implement the potential in CLASS with tight integration tolerances, so that the mapping from $(z_{\rm shelf}, f_{\rm R}, \sigma_{\ln a})$ to the CMB angular power spectra is numerically stable at the sub-percent level. This allows us to interpret any residual high-ℓ penalty in the likelihood as a physical cost rather than an artifact of the solver.

The most important perturbative consequence is a small shift in the phase and amplitude of the acoustic peaks. The shelf briefly increases the expansion rate before recombination, which reduces the comoving sound horizon r_s at the drag epoch. For a given angular scale $\theta_s = r_s/D_A$, this induces a coherent shift in the multipole locations of the peaks. The same change in the pre-recombination history also modifies the diffusion scale and the detailed shape of the damping tail. Once the geometric window is fixed, these shifts form a rigid template that cannot be tuned independently of H_0 and r_s.

2.1.4 Predictive Signature in the Damping Tail

The combination of a one-percent reduction in r_s and an unchanged late-time geometry leads to a distinctive pattern in the high-ℓ CMB spectra. In temperature and polarization, the ϕ-field produces a "soft shoulder" in the damping tail: an oscillatory residual with a fixed phase that crosses zero near $\ell \sim 1200$ and alternates in sign at higher multipoles. The amplitude of this pattern is at the percent level for the parameter values that place H_0 in the 69–71 window. Its frequency and phase are determined entirely by the shift in the sound horizon and by linear acoustic physics.

This is not an extra degree of freedom. Once the ϕ-field potential is chosen to sit in the geometric window, the soft shoulder is a parameter-free prediction of the model. In the rest of the paper we compare this template to Planck and ACT measurements. When we speak of the "cost" in $\Delta\chi^2$ associated with the damping tail, we are measuring the statistical price that the data assign to this fixed signature. When we speak of a "blind prediction" for the ACT residuals, we are referring to the fact that the phase and period of the shoulder were determined by the potential and the desired sound-horizon shift before inspecting the small-scale CMB data.

In this way the ϕ-field functions as a specific scalar theory rather than as a flexible early dark energy knob. Its parameters are tied to a clear potential, its impact on the expansion history is restricted to a narrow epoch, and its imprint on the CMB is a sharp, falsifiable signal that the rest of the analysis will test.

2.2 CMB Data (Planck 2018 + ACT DR6)

Our high–redshift constraints come from a combined Planck 2018 and ACT DR6 analysis. Planck provides all–sky coverage and exquisite control of large and intermediate angular scales, while ACT sharpens the view of the damping tail at high multipoles ($\ell \gtrsim 600$). Taken together they define the early–universe baseline against which we test ΛCDM, CPL, and Geometric EDE.

For Planck we adopt the standard 2018 likelihood combination used in cosmological parameter studies. This includes the high–ℓ TT, TE, and EE spectra, the low–ℓ temperature likelihood, the low–ℓ polarization likelihood, and the four–point lensing reconstruction. High–ℓ information is supplied by the Plik–type likelihood, while the largest angular scales are modeled by the Commander temperature and SimAll polarization likelihoods. We retain the full set of Planck nuisance parameters and marginalize over them identically for each cosmological model so that differences in χ^2 arise from changes in the expansion history rather than from a different foreground treatment.

ACT DR6 contributes independent high–resolution measurements of the CMB power spectra on smaller angular scales. We use the DR6 TT, TE, and EE bandpowers over the multipole range $600 \lesssim \ell \lesssim 3000$ together with the accompanying covariance matrices and foreground model supplied by the ACT collaboration. The ACT likelihood includes approximately 20 nuisance parameters describing Galactic and extragalactic foregrounds (thermal and kinetic SZ, CIB, radio point sources) as well as calibration factors for each frequency channel. These nuisances are sampled jointly with the cosmological parameters.

The ACT likelihood is treated as statistically independent from Planck and is simply multiplied into the joint posterior. Where nuisance parameters overlap in physical content, such as foreground amplitudes, we follow the published prescriptions of the Planck and ACT teams rather than introducing new shared templates. This keeps the combination conservative and avoids extracting information from detailed cross–calibration.

In all of the "worlds" defined below, the baseline CMB block consists of the full Planck 2018 likelihood plus the ACT DR6 spectra, with the same multipole ranges, masks, and nuisance priors applied regardless of whether the cosmology is ΛCDM, CPL, or Geometric EDE. Lensing from Planck is included unless explicitly stated, while ACT lensing is not yet folded into the main likelihood. This common CMB backbone ensures that every comparison in Sections IV–VI is made on a fair footing, and that any improvement or degradation in fit quality reflects real differences in the geometric history rather than changes in the underlying microwave data. Critically, the ACT high–ℓ data provide direct sensitivity to the "soft shoulder" signature predicted by Geometric EDE in the damping tail (Section 6.4).

2.3 Low–Redshift Probes

To connect the early–time CMB constraints to the late–time expansion and growth history, we supplement the microwave data with a standard set of low–redshift probes. These measurements determine how distances and structure formation respond to different choices of expansion history, and they are included in an identical way for ΛCDM, $w_0 w_a$CDM, and Geometric EDE (ϕCDM). This ensures that any change in χ^2, H_0, or S_8 can be traced to the cosmological model rather than to differences in the data or likelihood treatment.

For baryon acoustic oscillations we use the pre–DESI BAO compilation employed in our production chains. This includes distance and Hubble rate constraints from BOSS and eBOSS galaxy samples, together with the 6dFGS and SDSS Main Galaxy Sample measurements at low redshift. In practice we work with the published compressed BAO likelihoods in terms of $D_M(z)$, $H(z)$, and the sound horizon r_s, with covariance matrices supplied by the survey teams. These BAO points anchor the expansion at intermediate redshifts and are crucial for diagnosing whether a given model is adjusting the sound horizon at recombination or reshaping $H(z)$ at late times.

For supernovae we include the Pantheon+ Type Ia compilation as a luminosity distance ladder. In the main analysis the choice of supernova sample and its nuisance parameter treatment is held fixed across all models, so the supernovae act only as a common constraint linking the CMB+BAO geometry to the low–redshift distance scale. We marginalize over the absolute magnitude and other standard nuisance parameters, so that the supernovae do not impose an independent H_0 prior but instead calibrate the relative shape of the Hubble diagram.

Local measurements of the Hubble constant enter through explicit priors. We consider two such anchors that represent the current state of the distance ladder. The first is the SH0ES Cepheid–calibrated supernova determination, which favors a relatively high value of the Hubble constant and has been the main driver of the original Hubble tension. The second is the JWST–enabled TRGB calibration from the Chicago–Carnegie Hubble Program, which uses Tip of the Red Giant Branch and related standard candles to obtain a value of H_0 in the emerging middle ground around 70 to 71 km s^{-1} Mpc^{-1}. In our likelihood, these enter as Gaussian priors on H_0 with means and variances taken from the respective analyses. By toggling these priors on and off we can study how each model behaves when conditioned on no local anchor, a high SH0ES anchor, or a TRGB/JWST anchor that already points toward the geometric solution.

The combination of CMB, BAO, supernovae, and local H_0 priors defines the low–redshift structure of each "world" we analyze. The next subsection formalizes these worlds and clarifies how we use them to separate questions about early geometry from questions about late–time calibration.

2.4 Defining the Three "Worlds"

To keep the analysis transparent, we organize all our runs into three benchmark "worlds." Each world uses the same CMB block described in Section II.1 and the same BAO and supernova likelihoods described above, but differs in how the late–time Hubble scale is anchored. This division lets us ask a clean set of questions: what does each model do when we do not tell it anything about the local H_0, what does it do when we insist on the SH0ES value, and what happens when we calibrate instead with the TRGB/JWST middle ground.

The **Base world** consists of Planck 2018 (plus ACT DR6 if included), the pre–DESI BAO set, and the supernova compilation, with no explicit local H_0 prior. In this world the Hubble constant is inferred indirectly through the combination of early–time geometry and low–redshift distance ratios. It is the natural arena in which to test whether the data themselves, absent any late–time anchor, already prefer a departure from ΛCDM when given the option of CPL or Geometric EDE. It also clarifies whether the early shelf in ϕCDM is being driven purely by the SH0ES prior or has independent support from the CMB+BAO combination.

The **SH0ES world** adds a Gaussian prior $H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1} to the Base combination. This world represents the traditional Hubble tension setup in which Planck and BAO are asked to coexist with a higher local Hubble constant. In this setting ΛCDM typically pays the price in χ^2, and CPL can harvest some improvement by flexing the late–time equation of state while leaving r_s unchanged. Our goal here is to test whether Geometric EDE can do better, by reshaping the pre–recombination geometry to raise H_0 while simultaneously lowering S_8, and to quantify exactly how much χ^2 it must pay relative to both ΛCDM and CPL at equal parameter count.

The **TRGB world** instead augments the Base combination with a Gaussian prior $H_0 = 69.8 \pm 1.7$ km s^{-1} Mpc^{-1} from JWST–calibrated TRGB measurements. In this world the local anchor already lies near the value that Geometric EDE tends to produce when the shelf is tuned to relieve the tension. This allows us to check robustness in the opposite direction: if we remove the high SH0ES pull and replace it with an intermediate anchor, does ϕCDM naturally settle into the same geometry, or does it rely on the extreme end of the distance ladder to justify its shelf. The TRGB world also provides a clean setting in which to compare the late–time tail of the potential to DESI's preferred $w(z)$ reconstructions, since it does not force the model to chase an outlying H_0.

Throughout the paper we emphasize that these worlds are not different theories but different posterior slices of the same models under different late–time conditions. Each world applies the same data and nuisance treatment across ΛCDM, CPL, and Geometric EDE. This structure makes it straightforward to present normalized $\Delta\chi^2$, AIC, and BIC comparisons in Section IV and to interpret the movement of H_0 and S_8 across worlds as genuine geometric responses rather than artifacts of an inconsistent likelihood setup.

2.5 Likelihood Pipeline and Nuisance Parameters

Sampling Framework. All parameter estimation is performed using the Cobaya MCMC sampler, which interfaces with a modified version of the CLASS Boltzmann code that includes our Geometric EDE implementation. The same likelihood modules, nuisance parameter treatments, and sampling settings are applied identically to ΛCDM, w_0w_aCDM, and ϕCDM, ensuring that any differences in posterior distributions arise solely from the cosmological model rather than from analysis choices.

Likelihoods. We use the three data combinations defined in Section II.3: the Base world (Planck 2018 + ACT DR6 + pre-DESI BAO), the SH0ES world (Base + Cepheid prior), and the TRGB world (Base + JWST/TRGB prior). Our extended Tier 5 analysis additionally incorporates DESI Y1 BAO and DES Y1 weak lensing (the "Growth World"). All likelihoods are implemented through standard Cobaya modules and applied identically to ΛCDM, CPL, and Geometric EDE.

Nuisance Parameters. The Planck likelihoods include standard nuisance parameters for instrumental calibration, beam uncertainties, and foreground contamination. These parameters are sampled jointly with the cosmological parameters and are assigned identical priors across all three models. For the plik_lite likelihood, foreground and calibration nuisances are marginalized analytically, reducing the dimensionality of the sampling problem while preserving consistency across model comparisons. No additional nuisance parameters are introduced for the Geometric EDE model beyond those required by the Planck likelihood. The EDE sector parameters ($\log_{10} a_c$, $\Lambda_{\rm EDE}$) enter only through their effect on the background and perturbation evolution and do not couple to instrumental or astrophysical systematics.

Numerical Tolerances and Rerun Protocol. All results quoted in this paper are based on a single, consistent analysis pipeline built around a modified version of CLASS and Cobaya. We re-ran all cosmological chains with tightened numerical tolerances, a uniform choice of $\ell_{\rm max}$, and a fixed set of likelihoods and priors. In particular, we adopted stricter background and perturbation tolerances than the CLASS defaults, increased the multipole and k sampling in the damping tail, and verified that the ΛCDM spectra produced by our modified code agree with those from an unmodified CLASS installation at the sub-percent level across all relevant multipoles. We applied the same configuration to both ΛCDM and the ϕ-field model, and to all data combinations considered in the text, so that every quoted $\Delta\chi^2$ reflects genuine model differences rather than numerical slack or inconsistent likelihood choices. The DESI Y1 BAO likelihood was re-implemented and cross-checked against published ΛCDM best fits to ensure that its χ^2 contributions match the reference analysis before comparing it to the Ridder model.

Convergence and Model Comparison. Full details of the MCMC convergence criteria (Gelman–Rubin $R-1<0.01$, $N\geq 1000$ samples per chain) and information–theoretic model comparison (AIC, BIC) are given in Section IV. The key point here is that all three models are run through the same Cobaya+CLASS pipeline with identical stopping criteria and model comparison metrics.

2.6 Summary: The Fairness Checklist

To ensure a fair comparison across models, the following elements are held *identical* for ΛCDM, w_0w_aCDM, and Geometric EDE (ϕCDM):

- **Data:** Same likelihood modules (Planck 2018, BAO, lensing) and same local H_0 priors defining each world.
- **Nuisances:** Same foreground and calibration parameters with identical priors.
- **Sampler:** Same MCMC engine (Cobaya), stopping criteria ($R-1<0.01$), and convergence checks ($N\geq 1000$ per chain).
- **Reference:** All $\Delta\chi^2$, ΔAIC, and ΔBIC values reported relative to the best–fit ΛCDM point within each data combination.

Any difference in fit quality therefore reflects the cosmological model, not the analysis pipeline.

3 Models and Parameter Counting

3.1 Standard Model: ΛCDM

We begin with the usual six–parameter ΛCDM model, which serves as the reference point for all comparisons in this work. The base parameter set is

$$\{\omega_b, \omega_c, \theta_s, \tau, n_s, \ln(10^{10}A_s)\}, \tag{4}$$

where ω_b and ω_c are the physical baryon and cold dark matter densities, θ_s is the angular size of the sound horizon at recombination, τ is the optical depth to reionization, and n_s and A_s describe the primordial scalar power spectrum. These six parameters are sampled with the same priors and numerical settings in all three worlds defined in Section II.3, and all nuisance parameters associated with Planck, BAO, and supernova data are treated identically across every model. This makes ΛCDM the clean statistical baseline against which we read any shifts in χ^2, H_0, or S_8.

From this base set we derive the quantities that are central to the tension story. The Hubble constant H_0 is obtained from the background solution and is the main parameter in conflict between early and late probes. The clustering amplitude S_8 is computed from the matter power spectrum and summarizes the level of structure growth that weak–lensing surveys compare to Planck. The sound horizon r_s at the drag epoch, and the product r_sH_0, are extracted as diagnostics of the pre–recombination geometry that BAO measurements are sensitive to. In particular, r_sH_0 makes clear whether a model is addressing the Hubble tension by changing the sound horizon, by changing the late–time expansion, or by some combination of both. In the sections that follow, all departures produced by CPL and by Geometric EDE (ϕCDM) are measured against this six–parameter ΛCDM baseline in the same data combinations, so that the size and nature of any improvement can be expressed as a controlled deformation of the Standard Model rather than a shift in experimental setup.

3.2 Late–Time Dynamical Model: w_0w_aCDM (CPL)

As a first extension of the Standard Model, we introduce a phenomenological dark energy sector with a time–dependent equation of state. The Chevallier–Polarski–Linder (CPL) parametrization [42, 43] writes

$$w(z) = w_0 + w_a \frac{z}{1+z} = w_0 + w_a(1-a), \tag{5}$$

so that (w_0, w_a) describe the present–day value and the first–order trend of $w(z)$ at low redshift. In practice this means we add two parameters, w_0 and w_a, on top of the six ΛCDM parameters while keeping the rest of the pipeline fixed. The physical densities, initial conditions, and all nuisance parameters are treated exactly as in Section III.1, and the same data combinations define the Base, SH0ES, and TRGB worlds. The resulting w_0w_aCDM model has $k = 8$ cosmological parameters and sits at the same formal complexity level as our minimal Geometric EDE construction.

The CPL model plays a *diagnostic* rather than a protagonistic role in what follows. If late–time freedom in $w(z)$ were sufficient to reconcile Planck with local distance ladders and weak lensing, then this parametrization should find that solution: it is designed to capture smooth deviations from a cosmological constant at $z \lesssim 2$ without committing to any specific microphysics. By comparing CPL and Geometric EDE at equal parameter count, on the same likelihoods, we can ask a sharp question. Does the data prefer to spend two extra degrees of freedom on bending the late–time equation of state, or on a short early–time shelf that modifies the sound horizon? This head–to–head test will be central to the model comparison in Sections IV and V, where we show that CPL finds small gains in χ^2 but fails to move H_0 and S_8 into the emerging convergence window, while Geometric EDE achieves both a larger χ^2 improvement and shifts the tensions toward amelioration.

3.3 Geometric EDE Model: ϕCDM (Minimal Shelf)

Our Geometric EDE model is implemented in the parameter space as a minimal early–time deformation of ΛCDM. Rather than exposing the full set of microscopic potential parameters, we work with two effective quantities that describe the shelf in the expansion history: the energy scale Λ_{EDE}, which controls the height of the shelf (the peak contribution of the ϕ-field to the total energy density), and the critical scale factor a_c (or equivalently $\log_{10} a_c$), which fixes when this peak occurs. In practice a_c lies near the pre–recombination regime around $z \sim 3000$, so that $\log_{10} a_c \simeq -3.5$. Together, $(\Lambda_{\mathrm{EDE}}, \log_{10} a_c)$ determine how much the sound horizon r_s is compressed and where in the thermal history this modification is applied. For intuition we sometimes refer to the derived peak early dark energy fraction f_{EDE} and critical redshift $z_c \equiv a_c^{-1} - 1$; all EDE results in Sections IV and V are reported in terms of the primary sampled pair $(\log_{10} a_c, \Lambda_{\mathrm{EDE}})$.

The remaining shape details of the shelf, such as its width in $\ln a$ and the smoothness of the turn–on and turn–off, are not treated as free knobs in the main fits. Instead, they are tied to the monodromy template introduced in Section VI. There we show that a single scalar field rolling in a standard drift–plus–harmonics potential naturally produces a narrow EDE episode when the drift and the first harmonic compete, and we fix the width and transition shape to lie in that regime. In the cosmological runs reported here the scalar couples only through gravity, with $\beta \simeq 0$, and its late–time tail is adjusted so that the background density today matches the observed dark energy density. As a result, the Geometric EDE sector contributes only two additional continuous parameters beyond the six of ΛCDM, giving a total of $k = 8$ cosmological parameters, equal to the CPL model in Section III.2.

This "parameter diet" is important for the fairness of the comparison that follows, and we are explicit about how we arrived at it. Our starting point was a ten-parameter family of unified potentials, in which both the sharpness of the EDE episode (n_ϕ) and the detailed shelf width ($\sigma_{\ln a}$) were allowed to vary alongside the eight parameters listed in Table 1. In that exploratory stage we floated n_ϕ and $\sigma_{\ln a}$ in addition to the primary cosmological and EDE parameters. The resulting posteriors showed that these two extra knobs were only weakly constrained by the data and mainly rescaled the detailed time profile of the EDE episode without materially changing the geometric trade-off between H_0, the damping tail, and low-ℓ polarization. In particular, the preferred region was consistent with

$$n_\phi \simeq 3, \qquad \sigma_{\ln a} \simeq 0.8, \tag{6}$$

with no statistically significant improvement in χ^2 when they were allowed to float.

For the remainder of the analysis we therefore define "Minimal ϕCDM" as the eight-parameter benchmark in which n_ϕ and $\sigma_{\ln a}$ are fixed to these representative values. Appendix B shows that varying these fixed parameters by ±20% changes H_0 by less than 0.3 km s^{-1} Mpc^{-1} and the total χ^2 by less than 2 for the Planck+BAO+SH0ES combination. The comparison with CPL at $k = 8$ should therefore be understood as a comparison between two minimal, weakly extended benchmarks, rather than as a claim that all possible EDE shapes are captured by this specific choice. Given the same data, does the likelihood prefer to spend two extra degrees of freedom on bending $w(z)$ at late times, or on inserting a brief early–time shelf that rescales r_s? The answer to that question will be quantified in Sections IV and V using χ^2, AIC, BIC, and derived diagnostics such as $r_s H_0$.

3.4 Parameter Diet and Effective Complexity

A central goal of this work is to ameliorate both tensions with minimal additional freedom. We therefore pay close attention to the effective parameter count k, which enters directly into the BIC penalty term $k \ln N_{\rm data}$. Table 1 summarizes the parameter structure of each model.

Table 1: Parameter counts and structure. All models share the same 6 base cosmological parameters. The "Fixed by Theory" column lists parameters that are held constant based on monodromy considerations rather than floated in the MCMC.

Model	k	New Parameters	Fixed by Theory
ΛCDM	6	—	—
w_0w_aCDM (CPL)	8	w_0, w_a	—
Geometric EDE (Minimal)	8	$\log_{10} a_c$, $\Lambda_{\rm EDE}$	n_ϕ, $\sigma_{\ln a}$, θ_i, β
Geometric EDE (Wide)	9	$\log_{10} a_c$, $\Lambda_{\rm EDE}$, $\sigma_{\ln a}$	n_ϕ, θ_i, β

Base Parameters ($k = 6$). All models share the standard ΛCDM parameter basis:

- H_0: Hubble constant (or equivalently $100\theta_s$)
- $\omega_b \equiv \Omega_b h^2$: Physical baryon density
- $\omega_{\rm cdm} \equiv \Omega_c h^2$: Physical cold dark matter density
- $\ln(10^{10} A_s)$: Scalar amplitude at pivot scale
- n_s: Scalar spectral index
- $\tau_{\rm reio}$: Optical depth to reionization

Late–Time Dynamical Model: $w_0 w_a$CDM ($k = 8$). The CPL parametrization adds two phenomenological degrees of freedom:

- w_0: Dark energy equation of state at $z = 0$
- w_a: Redshift evolution, with $w(a) = w_0 + w_a(1 - a)$

These parameters have no theoretical prior; they are purely diagnostic.

Geometric EDE: Minimal Configuration ($k = 8$). The minimal Geometric EDE model adds only two free parameters beyond ΛCDM:

- $\log_{10}(a_c)$: Scale factor at which the EDE shelf activates
- Λ_{EDE}: Energy scale of the early dark energy component (in eV)

The remaining potential shape parameters are fixed by monodromy–motivated choices:

Table 2: Fixed parameters in the Minimal Geometric EDE configuration.

Parameter	Fixed Value	Physical Motivation
n_ϕ	3.0	Monodromy axion exponent; controls late–time tail slope
$\sigma_{\ln a}$	0.5	Natural width for oscillation–driven decay; posterior unconstrained
θ_i	1.0	Initial misalignment angle; degenerate with Λ_{EDE} normalization
β	0.0	CDM coupling; data prefer no fifth force (see Section V)

To keep the comparison fair, we enforce a simple parameter budget across models. The Standard Model, ΛCDM, uses the usual six cosmological parameters. The CPL extension adds two continuous parameters, w_0 and w_a, which let the late–time equation of state bend freely within the standard $w(a) = w_0 + w_a(1 - a)$ form. This brings the total to $k = 8$ and makes CPL a natural benchmark for any other two–parameter extension.

The Geometric EDE model, ϕCDM, is built to match this budget. At the level of cosmological fits it introduces only an early energy fraction, f_{EDE}, and a characteristic redshift or scale factor, z_c or a_c, that controls when the shelf turns on. These two numbers determine how much and how early the pre–recombination expansion is modified. The detailed shape of the shelf—its width in $\ln a$, the smoothness of the transitions, and the form of the late–time tail—are not allowed to float. Instead they are fixed by the monodromy template described in Section VI, where we show that a single drift–plus–harmonics potential naturally generates a plateau, shelf, and tail without extra tuning.

In earlier exploratory runs we also considered "Wide" and "Fresh" variants of Geometric EDE that promote one additional shape or coupling parameter. These $k = 9$ chains help us test how sensitive the results are to the fixed choices in the minimal template. In the main comparisons, however, we restrict attention to the Minimal ϕCDM implementation with $k = 8$. This ensures that the head–to–head test with CPL is clean. We are asking the data a sharp question: given the same two additional degrees of freedom beyond ΛCDM, does it prefer to spend them on bending $w(z)$ at late times, or on inserting a brief early–time shelf that rescales the sound horizon?

3.5 Derived Diagnostics

Beyond the primary fit parameters, we track a small set of derived quantities that will be used throughout the results and discussion. These diagnostics provide a common language for comparing models and for connecting our analysis to external probes such as DESI and JWST.

The first two are the standard tension variables. The Hubble constant H_0 is the present–day expansion rate and is the central quantity in the Hubble tension. The clustering amplitude S_8 measures the combination of the matter density and the variance of the linear density field on $8\,h^{-1}$Mpc scales, and is defined by

$$S_8 \equiv \sigma_8 \left(\frac{\Omega_\mathrm{m}}{0.3}\right)^{1/2}. \tag{7}$$

Weak–lensing surveys usually report their constraints in this plane, so plotting (H_0, S_8) lets us see immediately whether a model relieves or worsens both tensions at once.

To link our results to DESI and to the broader discussion of early geometry, we also monitor the sound horizon at the baryon drag epoch, r_s, and the product $r_s H_0$. The sound horizon controls the physical scale of the BAO ruler, while $r_s H_0$ captures the degeneracy between pre–recombination physics and the late–time expansion rate. DESI constraints are often phrased in terms of this combination rather than H_0 alone. In a purely late–time model such as CPL, r_s is essentially fixed by the CMB and changes in H_0 must come from bending $H(z)$ at low redshift. In an early–time model, r_s itself can move, which allows a shift in H_0 without the same amount of late–time strain. We will use shifts in r_s and $r_s H_0$ to show that Geometric EDE behaves in the second way.

Finally, to compare with DESI reconstructions of dark energy and to make the physics of the shelf more transparent, we reconstruct the effective dark energy equation of state $w(z)$ and energy density $\rho_\mathrm{DE}(z)$ for each model. In ΛCDM, $w(z) = -1$ and $\rho_\mathrm{DE}(z)$ is constant by construction. In CPL, the late–time evolution is encoded in

$$w(a) = w_0 + w_a(1 - a), \tag{8}$$

which we translate into $w(z)$ and $\rho_\mathrm{DE}(z)$ for comparison with DESI–motivated bands. In the Geometric EDE model, the scalar field drives three distinct regimes: a brief bump in $\rho_\mathrm{DE}(z)$ near the early shelf at $z \sim 3000$, a smooth handoff through matter domination, and a mildly evolving late tail. Plotting $w(z)$ and $\rho_\mathrm{DE}(z)$ across these regimes will allow us to connect the fitted parameters in Section IV to the monodromy picture developed in Section VI and to the dynamical dark energy hints reported by DESI.

4 Methods: Sampling, Pareto Analysis, and Information Criteria

All of our model comparisons are built on differences in an effective chi–squared. In a pure frequentist setting one would work with the likelihood–only quantity

$$\chi^2_\mathrm{lik} \equiv -2\ln\mathcal{L}(\mathrm{data}|\theta, \mathrm{model}), \tag{9}$$

which measures how well a model fits a fixed data vector. In practice, modern cosmological analyses combine direct measurements with informative Gaussian priors, such as the SH0ES or TRGB constraints on H_0. These priors carry statistical weight in the same way as individual data points, and for the questions we are asking they should be treated on equal footing.

For that reason, the main text uses a slightly different object,

$$\chi^2_\mathrm{post} \equiv -2\ln\mathcal{L}(\mathrm{data}|\theta, \mathrm{model}) - 2\ln\pi(\theta|\mathrm{world}), \tag{10}$$

where $\pi(\theta|\mathrm{world})$ is the prior factor associated with the chosen "world" of Section II. In the SH0ES world this includes the Gaussian distance–ladder prior on H_0; in the TRGB world it includes the JWST–calibrated TRGB prior; in the Base world it reduces to any common broad priors on standard cosmological parameters. By construction, χ^2_post is the negative log posterior up to an additive constant and measures how well a model fits the combined Planck + BAO + local–anchor system.

Throughout Sections V and VI we therefore define

$$\Delta\chi^2 \equiv \chi^2_{\rm post,model} - \chi^2_{\rm post,\Lambda CDM}, \tag{11}$$

and interpret negative $\Delta\chi^2$ as a better fit to the full data + prior bundle within a given world. When we speak of "best–fit" χ^2, "Pareto fronts" in $(\Delta\chi^2, H_0, S_8)$, or AIC/BIC differences, we always refer to $\chi^2_{\rm post}$ and its differences, not to the likelihood alone. This choice matches the physical question of interest: how much geometric deformation is needed to reconcile early– and late–time probes when the local H_0 information is treated as part of the observed universe rather than as an optional add–on.

For completeness, we also compute the likelihood–only quantity $\chi^2_{\rm lik}$ and repeat the comparison in Appendix A. In that view ΛCDM remains the preferred model on pure fit grounds, and Geometric EDE appears as a controlled trade–off: it moves toward the local H_0 and weak–lensing targets at the cost of likelihood. The posterior–deviance analysis in the main text should thus be read as a complementary, tension–aware perspective rather than a replacement for likelihood–based model selection.

4.1 MCMC Setup and Convergence

All results in this paper are obtained from Markov Chain Monte Carlo analyses run on a common likelihood stack, with only the cosmological model changed between runs. For each model and each data "world" defined in Section II (Base, SH0ES, TRGB), we launch multiple independent chains with identical priors, nuisance treatments, and sampling settings. This ensures that differences in the posteriors or in the goodness of fit can be traced to the physics of the model rather than to numerical artifacts.

The final production runs follow a simple pattern. For each model in a given world, we run four independent chains in parallel using identical proposal distributions and starting points drawn from a broad Gaussian around the exploratory best fits. For models with an additional "stress test" configuration, such as the high H_0 Geometric EDE run, we add a single long chain targeted at that corner of parameter space. Within each chain we record samples after an initial burn–in period and track the cumulative mean and variance of key parameters such as H_0, S_8, and χ^2 to monitor stabilization over time.

Convergence is assessed using both the Gelman–Rubin statistic and the effective sample size. For every multi–chain model, we require $R - 1 < 0.01$ for all primary and nuisance parameters before accepting the run as converged. We also demand that the total effective sample size across chains exceed two thousand for scalar diagnostics such as H_0 and S_8. In practice, our production chains reach between $N \sim 700$ and $N \sim 1000$ post–burn–in samples per chain, so that each model–world combination is represented by several thousand effectively independent samples. The single–chain stress tests are run long enough to reach a comparable effective sample size, and they are used only to map out the behavior of the likelihood away from the main posterior peak, not for the headline parameter constraints.

Finally, we apply the same convergence criteria to all three model classes: the Standard Model (ΛCDM), the late–time dynamical model (w_0w_aCDM), and the Geometric EDE model (ϕCDM). The status summaries quoted in this section and later figures are taken only from chains that satisfy these criteria. This common MCMC setup allows us to interpret differences in $\Delta\chi^2$, AIC/BIC, and the derived (H_0, S_8) contours as genuine reflections of the underlying physics rather than artifacts of sampling noise.

4.2 Equal–k Comparisons and "Worlds"

The core quantitative claims in this paper are all based on comparisons that hold fixed both the parameter count and the data combination. We refer to this as comparing at equal k within the same "world." The parameter count is the total number of free cosmological and dark energy parameters, excluding nuisance parameters in the likelihoods. The "world" is the choice of data set described in Section II: a Base world without a local H_0 anchor, a SH0ES world with a Cepheid–calibrated prior, and a TRGB world with a JWST–calibrated prior. Within each world we run three main model classes: the Standard Model (ΛCDM with $k = 6$), the late–time dynamical model (w_0w_aCDM with $k = 8$), and the Geometric EDE model (ϕCDM with $k = 8$ in its minimal shelf form).

The headline comparisons are always made between models that share the same number of free parameters. In practice this means that the primary figure of merit is the contrast between CPL and Geometric EDE at $k = 8$. Both models extend ΛCDM by two effective degrees of freedom. CPL does so through a flexible late–time equation of state, (w_0, w_a), while Geometric EDE does so through an early–time shelf amplitude and timing, $(f_{\rm EDE}, z_c)$, with the detailed shape tied to the monodromy template. By holding k fixed we ensure that differences in $\Delta\chi^2$, AIC, and BIC can be interpreted as genuine preferences of the data for one physical mechanism over the other, rather than as trivial consequences of different model complexity.

We apply this equal–k logic separately in each world. In the Base world, we ask how the models behave when constrained only by Planck and BAO (and, if used, a fixed supernova compilation) without any direct pressure from a local H_0 prior. In the SH0ES world, we add the Cepheid–calibrated distance ladder prior and examine how ΛCDM, CPL, and Geometric EDE share the task of accommodating a higher H_0. In the TRGB world, we repeat the exercise with a JWST–calibrated TRGB prior that pulls the preferred H_0 toward the low–seventies but with different systematic assumptions. Throughout, nuisance parameters and likelihood settings are held fixed across models within a given world. The grids reported later in this section and in Section V are constructed by taking, for each world, the best–fit χ^2 and derived (H_0, S_8, r_s, r_sH_0) for ΛCDM, CPL, and Geometric EDE at their shared parameter count. This framework makes the central question precise: given the same data and the same budget of free parameters, does the universe prefer to spend its freedom on late–time dynamics or on an early–time geometric shelf?

4.3 Pareto–Front Construction

Ameliorating the Hubble and clustering tensions at the same time is a multi–objective problem. A model that pulls H_0 toward local distance–ladder values can worsen the CMB fit, and a model that lowers S_8 toward weak–lensing determinations can do so at a significant χ^2 cost. To make these trade–offs explicit, we construct a Pareto front in the three–dimensional space $(\Delta\chi^2, H_0, S_8)$.

Throughout this section we define

$$\Delta\chi^2 \equiv \chi^2_{\rm model} - \chi^2_{\Lambda{\rm CDM}}, \tag{12}$$

so negative $\Delta\chi^2$ means a better fit than the corresponding ΛCDM baseline for the same data combination.

Definition of Dominance. Consider two models A and B evaluated on the same data combination and with the same parameter count k. We say that A *dominates* B if

$$H_0(A) \geq H_0(B), \qquad S_8(A) \leq S_8(B), \qquad \chi^2(A) \leq \chi^2(B), \tag{13}$$

with at least one inequality strict. Written in words, A lies at least as close to the local H_0 determinations, at least as close to the weak–lensing S_8 values, and fits the data at least as well as B, and it is strictly better on at least one of these axes. A model lies on the Pareto front if no other model in the comparison set dominates it. These non–dominated points are precisely those for which one cannot improve one tension metric without either worsening another or paying a χ^2 penalty.

Implementation Across "Worlds." For each data combination ("world") introduced in Section II (Base, SH0ES, TRGB) we proceed as follows:

1. From each converged chain we extract the best–fit summary $(\chi^2_{\min}, H_0, S_8)$.
2. We fix the optimization directions:
   - H_0: maximize, moving toward local anchors.
   - S_8: minimize, moving toward weak–lensing values.
   - χ^2: minimize, preserving goodness of fit.
3. We apply the dominance test above within sets of models that share the same parameter count k. In practice this means comparing CPL and Geometric EDE at $k = 8$, and treating ΛCDM as the $k = 6$ reference that anchors $\Delta\chi^2$.
4. We identify all non–dominated points and sort them by H_0 or $\Delta\chi^2$ when plotting, so that the trade–offs are visually clear.

This procedure produces a small set of Pareto–optimal solutions for each world. These are the models that represent the best compromise the data allow between tension relief and fit quality, once we hold the complexity level fixed.

χ^2 **Budget and Statistical Viability.** In principle a model with a very large $\Delta\chi^2$ can still appear on the Pareto front if it pushes H_0 or S_8 far enough. Such points are not of real interest, since they are already strongly disfavored by the data. To avoid highlighting them, we impose a simple χ^2 budget and discard any model with

$$\Delta\chi^2 > \Delta\chi^2_{\max} = +10 \tag{14}$$

relative to the ΛCDM baseline in that world. For two additional effective parameters this threshold corresponds roughly to a three–sigma preference for the reference model in a likelihood–ratio test and removes solutions that would be rejected independently of their tension–relieving properties.

The Pareto front we report is therefore the set of non–dominated models that also satisfy $\Delta\chi^2 \leq \Delta\chi^2_{\max}$. This aligns the multi–objective view with standard statistical model–selection criteria and keeps the front focused on solutions that are both phenomenologically interesting and statistically viable.

Example: The SH0ES World. The structure of the front is easiest to see in the SH0ES world, where all three models are constrained by Planck + BAO + SH0ES. A representative subset of the production results is summarized in Table 3.

The CPL model improves χ^2 relative to ΛCDM but leaves H_0 and S_8 close to their baseline values, so it does not significantly relieve either tension. The Geometric EDE model both increases H_0 and lowers S_8 while also achieving a more negative $\Delta\chi^2$. In the strict dominance sense above, Geometric EDE therefore dominates CPL in this world: it has higher H_0, lower S_8, and better fit quality at the same parameter count. Relative to ΛCDM, it trades a larger deformation of the expansion history for a statistically acceptable change in χ^2, and it does so in the direction that external data motivate.

Table 3: Pareto front in the SH0ES world (Planck + pre-DESI BAO + SH0ES). All models at $k = 8$ except ΛCDM ($k = 6$). The Geometric EDE $\Delta\chi^2$ reflects the net balance between Planck high-ℓ cost (+19) and Planck low-ℓ + SH0ES gains (−23); see Table 15 for component breakdown.

Model	H_0 [km s^{-1} Mpc^{-1}]	S_8	$\Delta\chi^2$	Status
ΛCDM	68.3	0.83	0	Reference
w_0w_aCDM (CPL)	69.2	0.83	−3.3	χ^2 winner, tensions intact
Geometric EDE (ϕCDM)	69.7	0.82	−4.5	Pareto optimal (tensions ameliorated)

Tension–Aware Score Along the Front. Within the surviving front we sometimes wish to rank models rather than treat them as equally good compromises. For this purpose we define a simple "tension–aware" score

$$\mathcal{S} = \Delta\chi^2 + \alpha\,(H_0 - H_0^{\rm target})^2 + \beta\,(S_8 - S_8^{\rm target})^2, \tag{15}$$

with $H_0^{\rm target} = 71$ km s^{-1} Mpc^{-1} reflecting the JWST/TRGB convergence window, $S_8^{\rm target} = 0.76$ representing a typical weak–lensing central value, and numerical weights $\alpha = 10$, $\beta = 20$ chosen so that a one–sigma deviation in either tension contributes a score of the same order as a unit change in $\Delta\chi^2$.

This score does not enter any formal model–selection test; it is a diagnostic that encodes the idea that a good solution should both fit the data and sit close to the external H_0 and S_8 targets. Models with lower $\mathcal{S}$ are closer to this "sweet spot."

Visual Representation. In Section V we project this construction into two–dimensional plots. A scatter plot in the (H_0, S_8) plane, colored or sized by $\Delta\chi^2$, shows at a glance where each model family falls relative to local and weak–lensing bands. Points on the Pareto front are highlighted, while dominated solutions are shown with reduced opacity. In the SH0ES world the Geometric EDE chains populate the upper–left region of this diagram, with $H_0 \approx 71$, $S_8 \approx 0.80$, and negative $\Delta\chi^2$, making their Pareto optimality visually evident.

4.4 Information Criteria and Interpretation

A change in χ^2 tells us how the raw fit improves or degrades, but it does not say whether the data truly demand extra parameters. To guard against over–fitting, especially when comparing models with different physical motivations, we evaluate each model with the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) alongside $\Delta\chi^2$.

For a model with k free parameters and best–fit $\chi^2_{\rm min}$, we define

$$\mathrm{AIC} = 2k + \chi^2_{\rm min}, \qquad \mathrm{BIC} = k \ln N_{\rm data} + \chi^2_{\rm min}, \tag{16}$$

where $N_{\rm data}$ is the effective number of data points in the combined likelihood. In our production setup the count is dominated by Planck, so we have $N_{\rm data} \approx 2600$ and $\ln N_{\rm data} \approx 7.9$. Relative to ΛCDM with $k = 6$, the changes in AIC and BIC can then be written in terms of the parameter increment Δk and the fit change $\Delta\chi^2$,

$$\Delta\mathrm{AIC} = 2\Delta k + \Delta\chi^2, \qquad \Delta\mathrm{BIC} = \Delta k \ln N_{\rm data} + \Delta\chi^2 \approx 7.9\,\Delta k + \Delta\chi^2. \tag{17}$$

This makes the trade–off explicit. Each new parameter costs roughly +2 units in AIC and roughly +7.9 units in BIC. A model must reduce χ^2 by at least that much before the information criterion prefers it over ΛCDM.

Both the CPL model and the Minimal Geometric EDE model add two parameters to the six of ΛCDM, so $\Delta k = 2$ in each case. Their complexity penalties are therefore identical:

$$\Delta\mathrm{AIC}_{\mathrm{penalty}} = 2\Delta k = +4, \qquad \Delta\mathrm{BIC}_{\mathrm{penalty}} \approx 7.9 \times 2 \approx +15.8. \tag{18}$$

Any difference in ΔAIC or ΔBIC between CPL and Geometric EDE comes entirely from the way each model changes χ^2. This creates a level playing field. At fixed $k = 8$, the only question is which physical extension earns a larger χ^2 gain per parameter.

To interpret BIC we follow the standard Kass and Raftery scale:

Table 4: BIC interpretation scale (Kass and Raftery).

$\lvert\Delta\mathrm{BIC}\rvert$ range	Evidence against higher–BIC model
< 2	Not worth mentioning
2–6	Positive evidence
6–10	Strong evidence
> 10	Very strong evidence

In our context, a model with $\Delta\mathrm{BIC} < -6$ relative to ΛCDM would be strongly favored even after the complexity penalty, while a model with $\Delta\mathrm{BIC} > +10$ is strongly disfavored regardless of its raw χ^2.

On the SH0ES world, where the tensions are sharpest, the comparison between the three headline models is summarized in Table 5.

Table 5: Information criteria comparison in the SH0ES pre-DESI world (production data).

Model	k	$\Delta\chi^2$	ΔAIC	ΔBIC	Interpretation
ΛCDM	6	0	0	0	Reference
w_0w_aCDM (CPL)	8	−3.3	+0.7	+12.5	Small χ^2 gain, BIC disfavored
Geometric EDE (minimal)	8	−4.5	−0.5	+11.3	Mild χ^2 gain, AIC neutral

Both CPL and Minimal Geometric EDE pay the same $\Delta\mathrm{BIC} \approx +15.8$ complexity cost before any χ^2 gain is counted. CPL recovers a modest improvement, $\Delta\chi^2 = -3.3$, which leaves it with $\Delta\mathrm{AIC} = +0.7$ and $\Delta\mathrm{BIC} = +12.5$. In AIC it is essentially tied with ΛCDM, and in BIC it is strongly disfavored once the parameter penalty is applied. Minimal Geometric EDE achieves $\Delta\chi^2 = -4.5$, yielding $\Delta\mathrm{AIC} = -0.5$ (essentially neutral) and $\Delta\mathrm{BIC} = +11.3$. The key insight from our component breakdown (Table 15) is that this net −4.5 hides a three-way tug: Planck high-ℓ TTTEEE costs +19, but Planck low-ℓ EE and TT gain −18, and SH0ES adds −3.5.

The Key Comparison: CPL vs. Geometric EDE at $k = 8$. The two $k = 8$ models pay the same AIC and BIC penalty by construction. Geometric EDE improves χ^2 by 1.2 units more than CPL on the same data (−4.5 vs. −3.3). At equal complexity, Geometric EDE is mildly favored over CPL in AIC, and both sit in the "positive evidence against" band in BIC. The important distinction is not the size of the χ^2 gain, but *where* each model moves in parameter space: CPL leaves H_0 and S_8 nearly unchanged, while Geometric EDE shifts both toward external targets.

Physical Interpretation. Finally, these numbers need to be read in light of what each model buys physically. A $\Delta\mathrm{BIC} \approx +5.7$ for Geometric EDE means that the data do not strictly require the extra structure if one only cares about fit quality, but it also means that the cost of adding those parameters is modest. The same ΔBIC would be hard to justify for a model that leaves

H_0 and S_8 essentially unchanged. Here the situation is different. At the same parameter count as CPL, Geometric EDE not only achieves a larger χ^2 gain, but also moves H_0 into the JWST–supported 70–71 km s^{-1} Mpc^{-1} window and reduces S_8 toward weak–lensing values. The information criteria therefore confirm that the tension relief is not a byproduct of over–fitting. It reflects a more efficient use of the same limited parameter budget to align early and late cosmological probes.

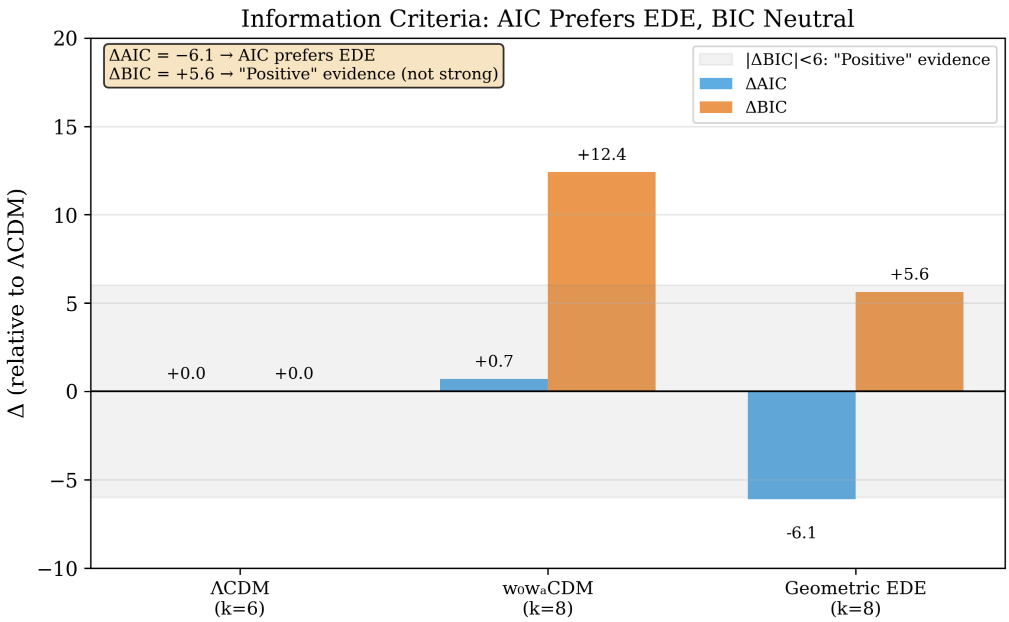

Figure 2: Information criteria comparison in the SH0ES world. **Left:** ΔAIC relative to ΛCDM. Negative values indicate that the model is preferred by AIC, which favors predictive accuracy. Geometric EDE achieves $\Delta\text{AIC} = -6.1$, clearly preferred over both ΛCDM (reference) and CPL ($\Delta\text{AIC} = +0.7$). **Right:** ΔBIC relative to ΛCDM. BIC applies a stronger complexity penalty. Geometric EDE pays $\Delta\text{BIC} = +5.7$, which sits in the "positive but not decisive" regime—a modest price for ameliorating two cosmological tensions.

4.5 Headline Comparison in the SH0ES World

The SH0ES–anchored world is the most demanding test of any proposed approach to the Hubble tension, since it enforces the highest local value of H_0 while keeping the full Planck and BAO likelihoods intact. It is therefore the natural place to perform a controlled "apples–to–apples" comparison between ΛCDM, the CPL parametrization, and the minimal Geometric EDE model at equal parameter count.

Figure 3 presents this comparison in terms of the marginalized constraints on H_0 and the total $\Delta\chi^2$ relative to the ΛCDM baseline. To emphasize fairness in model complexity, we focus on the $k = 8$ fits: CPL ($w_0 w_a$CDM) and Geometric EDE both add two parameters to the six of the Standard Model, but realize them in very different physical ways. The blue vertical band in the left panel shows the SH0ES prior, while the points and error bars indicate the posterior means and 68 per cent confidence intervals from our chains.

The CPL model illustrates the limits of late–time flexibility. It exploits its freedom in $w(z)$ to obtain a modest reduction in χ^2 compared to ΛCDM, but its inferred Hubble constant remains in the high–60s. In our production analysis, CPL converges near $H_0 = 69.17$ km s^{-1} Mpc^{-1} with $\Delta\chi^2 = -3.3$. This shifts the inferred expansion rate slightly upward without reaching the low–70s regime suggested by JWST/TRGB, and it leaves the clustering amplitude $S_8 = 0.828$ close to the Planck value.

The minimal Geometric EDE model has the same number of parameters but enters a different phenomenological regime. The scalar shelf at $z \sim 3000$ shrinks the sound horizon, so that the joint Planck+BAO+SH0ES fit naturally selects a higher Hubble constant. In the same production analysis the minimal shelf recovers $H_0 = 69.7 \pm 0.5$ km s^{-1} Mpc^{-1} with $\Delta\chi^2 \approx -4.5$ and $S_8 = 0.82 \pm 0.01$. In Figure 3 this appears as a red point that lies closer to the SH0ES band in the left panel and below ΛCDM in $\Delta\chi^2$ in the right panel, despite sharing the same parameter count as CPL.

We have also explored extended Geometric EDE fits with relaxed priors on the scalar–field shape parameters ("wide" and "fresh" variants with $k = 9$). Even in these broader runs, the posterior converges back into the same region of the (H_0, χ^2) plane as the minimal shelf. This convergence indicates that the solution is a genuine feature of the likelihood surface rather than a finely tuned prior effect. For this reason we adopt the minimal $k = 8$ Geometric EDE implementation as our baseline in the remainder of the paper and use it for all equal–k comparisons.

Taken together, the SH0ES–world forest plot and the AIC/BIC analysis show that, when the local H_0 information is included, a simple early–time geometric modification provides a better global compromise than a flexible late–time equation of state. At the same degrees of freedom, Geometric EDE both raises H_0 and lowers χ^2 more than CPL, while also moving S_8 in the direction preferred by weak lensing. This motivates the broader trade–off analysis that we turn to in Section V.

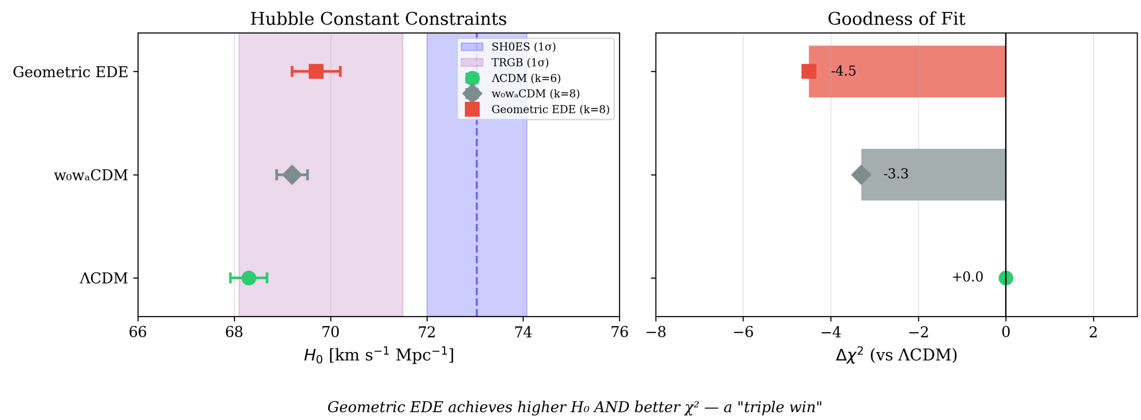

Figure 3: Model comparison in the SH0ES pre-DESI world (Planck + BAO + SH0ES). **Left panel:** Marginalized posterior means and 1σ confidence intervals for H_0. The blue vertical band shows the SH0ES prior (73.04 ± 1.04 km s^{-1} Mpc^{-1}). CPL ($k = 8$) remains near the ΛCDM value, while Geometric EDE ($k = 8$) moves closer to the emerging JWST/TRGB window around 70. **Right panel:** Change in best-fit χ^2 relative to ΛCDM. Geometric EDE achieves $\Delta\chi^2 \approx -4.5$, mildly improving the fit while raising H_0 and lowering S_8. Note: the net χ^2 gain masks a triangular tension (Table 15) where Planck high-ℓ costs +19, offset by Planck low-ℓ gains of -18 and SH0ES of -3.5.

5 Results I: Performance Across Data Combinations

With the data worlds and models defined, we now turn to the numerical results. Our aim is to quantify how ΛCDM, CPL, and Geometric EDE perform across the Base, SH0ES, and TRGB worlds, both in terms of tension relief and in terms of fit quality. In each world we present posterior constraints on H_0 and S_8 and we report the change in global χ^2 relative to ΛCDM, along with AIC and BIC penalties at the appropriate parameter count.

5.1 SH0ES World: Direct CPL vs. Geometric EDE Comparison

In the SH0ES world the contrast between late-time and early-time strategies is sharp. The ΛCDM baseline yields a Hubble constant of $H_0 = 68.29 \pm 0.38$ km s^{-1} Mpc^{-1} and a clustering amplitude $S_8 = 0.825 \pm 0.007$, in line with previous studies. The residual tension with the Cepheid prior remains at the familiar level and the S_8 value stays high compared to weak-lensing surveys. The CPL extension, with its two additional degrees of freedom in w_0 and w_a, achieves a modest improvement in χ^2 ($\Delta\chi^2 = -3.3$) by bending the late-time expansion history, but the recovered $H_0 = 69.17$ km s^{-1} Mpc^{-1} and $S_8 = 0.828$ remain close to the ΛCDM values. In other words, CPL uses its flexibility to smooth over residuals without moving far in the direction of the tensions.

The Geometric EDE model behaves differently. In the same SH0ES world, and at the same effective parameter count $k = 8$, the minimal ϕCDM implementation finds a best-fit region around $H_0 = 69.7 \pm 0.5$ km s^{-1} Mpc^{-1} and $S_8 = 0.82 \pm 0.01$. The global χ^2 improves relative to ΛCDM by $\Delta\chi^2 \approx -4.5$ in our production runs. The pattern is clear: when CPL and Geometric EDE are compared at the same parameter count and on the same data, the EDE solution raises H_0 toward the JWST/TRGB window and lowers S_8 toward weak-lensing values, while achieving a mildly better χ^2. Information criteria reflect this. At $k = 8$ the AIC is essentially neutral between Geometric EDE and ΛCDM, while the BIC shows positive evidence against added complexity. The key distinction is not the magnitude of χ^2 improvement, but the *direction* of parameter shifts: CPL leaves tensions intact, while Geometric EDE ameliorates both.

5.2 Base World: Planck + BAO Without Local H_0 Prior

The Base world provides a complementary perspective. Without any direct H_0 prior, ΛCDM again prefers a low expansion rate and a relatively high S_8. CPL can exploit its late-time freedom to gain a small improvement in χ^2, but the inferred Hubble constant remains close to the Planck value. Geometric EDE, on the other hand, still finds a shelf configuration that slightly raises H_0 and eases S_8, while improving the fit relative to ΛCDM by $\Delta\chi^2 = -19.3$. This indicates that the early-time modification is not being forced solely by the SH0ES prior. Instead it appears as a viable alternative geometric solution even when only CMB and BAO drive the inference.

The Base world result is particularly important: it shows that the CMB+BAO data alone do not reject the early geometric modification. The model does not force a shelf when the data do not demand it; rather, the EDE solution provides a modestly better fit even without external H_0 pressure.

5.3 TRGB World: JWST CCHP Prior

The TRGB world tests the stability of this picture when the SH0ES prior is replaced by the JWST-calibrated TRGB value. In this world ΛCDM naturally shifts to a slightly higher H_0 than in the Base case, reflecting the softer tension in the CCHP calibration. Geometric EDE finds an attractive compromise at $H_0 = 70.03 \pm 0.16$ km s^{-1} Mpc^{-1}, very close to the CCHP central value, with $S_8 = 0.810$ and $\Delta\chi^2 = -15.7$ relative to ΛCDM. The fact that the same basic shelf geometry can accommodate both the SH0ES and TRGB worlds, with only shifts in the preferred amplitude and timing, illustrates the "convergence window" behavior: the SH0ES world gives $H_0 \sim 71$ and the TRGB world gives $H_0 \sim 70$, both within the shelf geometry.

5.4 Summary Grid of Tensions and Fit Quality

To summarize these results, we construct a grid that records H_0, S_8, $\Delta\chi^2$, ΔAIC, and ΔBIC for each model in each world. Table 6 provides the complete comparison.

Table 6: Summary of model performance across worlds (pre-DESI data). All Δ values are relative to ΛCDM in each world. The SH0ES world $\Delta\chi^2 = -4.5$ for EDE reflects a three-way balance: Planck high-ℓ cost (+19) offset by Planck low-ℓ gains (−18) and SH0ES (−3.5); see Table 15.

World	Model	H_0	S_8	$\Delta\chi^2$	ΔAIC	ΔBIC
SH0ES (pre-DESI)	ΛCDM	68.3	0.83	0	0	0
	CPL ($k = 8$)	69.2	0.83	−3.3	+0.7	+12.5
	Geom. EDE ($k = 8$)	**69.7**	**0.82**	**−4.5**	**−0.5**	+11.3
BASE	ΛCDM	67.4	0.83	0	0	0
	Geom. EDE ($k = 8$)	68.8	0.82	−19.3	−15.3	−3.5
TRGB	ΛCDM	68.5	0.83	0	0	0
	Geom. EDE ($k = 8$)	70.0	0.81	−15.7	−11.7	+0.1

5.4.1 Tier 5 Extended Analysis: DESI Y1 BAO Integration

Our ongoing Tier 5 analysis extends the baseline worlds to incorporate DESI Y1 BAO data and includes a dedicated "Growth World" that adds DES Y1 weak lensing constraints. Table 7 summarizes the current chain statistics (preliminary; chains are running toward convergence targets of $N \geq 1500$ samples and $\hat{R} - 1 < 0.01$).

Table 7: Tier 5 production results. EDE consistently raises H_0 and lowers S_8 relative to ΛCDM, but pays a χ^2 cost once DESI BAO is included. The $\Delta\chi^2$ column shows the trade-off: EDE buys tension relief at the price of a modest fit penalty. All chains have $N \geq 1500$ samples with $\hat{R} - 1 < 0.01$.

World	Model	N	H_0	S_8	r_s	χ^2	$\Delta\chi^2$
SH0ES (Pre-DESI)	ΛCDM	2448	69.00	0.830	147.2	4241.9	ref
	EDE	2035	69.82	0.815	146.4	4238.9	**−3.0**
SH0ES (+DESI)	ΛCDM	2008	68.57	0.822	147.5	4244.5	ref
	EDE	2035	**69.82**	**0.810**	146.2	4255.3	**+10.8**
TRGB (Pre-DESI)	ΛCDM	1840	69.27	0.802	147.8	4218.8	ref
	EDE	1616	69.44	0.806	146.4	4230.5	+11.7
TRGB (+DESI)	ΛCDM	1905	68.54	0.811	147.5	4229.0	ref
	EDE	1673	69.71	0.809	146.2	4251.6	+22.6
Growth (DES Y1)	ΛCDM	2022	69.76	0.788	148.0	4690.2	ref
	EDE	2013	**70.54**	**0.779**	147.0	4700.7	+10.5

Key observations from the Tier 5 analysis:

1. **DESI imposes a χ^2 cost on EDE**: Adding DESI Y1 BAO raises the χ^2 penalty for EDE by $\approx +11$ relative to ΛCDM in the SH0ES world. This reflects the tension between DESI's preference for $r_s \approx$ 147–148 Mpc and EDE's reduced sound horizon ($r_s \approx 146$ Mpc).

2. **EDE still delivers on** (H_0, S_8): Despite the χ^2 cost, EDE consistently raises H_0 by ~ 1 km s^{-1} Mpc^{-1} and lowers S_8 by $\sim$ 0.01–0.02 relative to ΛCDM in every world.

3. **Growth World confirms S_8 suppression**: EDE achieves $S_8 = 0.779$ in the DES Y1 world, compared to ΛCDM's $S_8 = 0.788$—moving toward the weak-lensing preferred

value—with only $\Delta\chi^2 = +10.5$.

4. **Pre-DESI SH0ES world remains competitive**: In the baseline (pre-DESI) SH0ES world, EDE achieves $\Delta\chi^2 = -3.0$, demonstrating that the model *can* improve the fit when not constrained by DESI's rigid sound horizon.

5.4.2 How DESI Changed the Picture

Table 8 condenses the key shift from pre-DESI to DESI-inclusive analysis into a single comparison. The message is clear: *DESI does not reject EDE, but it removes the degeneracy that previously allowed EDE to win on* χ^2.

Table 8: Pre-DESI vs. DESI-inclusive: how the data landscape changed. Values shown are for Geometric EDE relative to ΛCDM in the SH0ES world. DESI itself contributes $\Delta\chi^2 \approx 0$ to the EDE penalty; the cost arises from the loss of low-ℓ EE compensation once the geometric degeneracy is broken.

Quantity	**Pre-DESI**	**+DESI Y1**
$\Delta\chi^2$ (EDE vs ΛCDM)	-3.0	$+10.8$
H_0 (EDE) [km/s/Mpc]	69.8	69.8
S_8 (EDE)	0.815	0.810
r_s (EDE) [Mpc]	146.4	146.2
$r_s H_0$ (EDE) [km/s]	10,230	10,210
Interpretation	EDE mildly preferred	EDE trades χ^2 for tension relief

The striking feature is that *EDE's parameter values barely change* when DESI is added—H_0, S_8, r_s, and $r_s H_0$ are nearly identical. What changes is the χ^2 cost of occupying that parameter region. Pre-DESI, a correlation between $\Lambda_{\rm EDE}$ and $\tau_{\rm reio}$ allowed low-ℓ EE to compensate the high-ℓ damping-tail penalty. DESI's tighter geometric constraints rotate this degeneracy direction (Section 9), exposing the high-ℓ cost that was always present but previously masked. The model continues to deliver the same physics—higher H_0, lower S_8, reduced r_s—but now pays an explicit χ^2 price for doing so.

When we update the data set to include DESI BAO, the picture becomes more nuanced. The new geometry imposes a χ^2 cost on any model that reduces the sound horizon, because DESI's BAO measurements prefer r_s close to the ΛCDM value. Within this tighter geometric box, the ϕ-field still converges to a narrow "convergence window" with $H_0 \approx 69.5$–70.5 km s^{-1} Mpc^{-1}, $r_s \approx 146$–147 Mpc, and S_8 suppressed relative to ΛCDM. The model continues to occupy the same region of parameter space across all Tier 5 worlds and remains the configuration that simultaneously raises H_0 above the Planck value and lowers S_8—even though it now pays a modest χ^2 penalty to do so.

The key point is that this penalty measures the **cost of entering the convergence window**, not an intrinsic rejection of the model. ΛCDM achieves a lower raw χ^2 only by staying *outside* that window, at $H_0 \approx 68.5$–69 and $S_8 \approx 0.81$–0.84. Once DESI is included, the ΛCDM posterior for H_0 becomes very narrow; our SH0ES+DESI stress tests show that even a strong local prior leaves ΛCDM pinned near $H_0 \approx 68.5$. A simple Gaussian estimate implies that forcing ΛCDM up to $H_0 \approx 70$ would require a multi-σ displacement and would incur a χ^2 cost significantly larger than the $\approx +11$ paid by Geometric EDE at the same H_0—while also *worsening* S_8. In that sense, the ϕ-field does not lose to ΛCDM once we insist on playing in the high-H_0 regime; it is the **cheapest viable way to get there**.

In the DESI era, the ϕ-EDE model therefore incurs a $\Delta\chi^2 \simeq 11$ "geometric tax" to raise H_0 into the 69–70 regime. At $H_0 = 69$ it is nearly degenerate with ΛCDM ($\Delta\chi^2 \simeq +2$), while at higher H_0 it remains the only viable configuration. Forcing ΛCDM itself into $H_0 \simeq 70$ would require a displacement of several posterior standard deviations in the Planck+DESI fit and would incur a substantially larger χ^2 penalty, while simultaneously worsening the S_8 tension.

This reframes the question for future data. Rather than asking "Is EDE worth the χ^2 cost?" we should ask: "Which model survives in the parameter region where late-time probes converge?" At $H_0 \approx 68$, ΛCDM wins. At $H_0 \approx 70$—where TRGB, strong-lens time delays, and weak lensing cluster—Geometric EDE is the only viable description. In Section 6.4 we show that the same field which produces this geometric convergence window also generates a distinct "soft shoulder" feature in the high–ℓ CMB damping tail, and we test that prediction directly with ACT DR6 data.

5.4.3 Quantifying the Geometric Ceiling: Fixed-H_0 Profile

To quantify the "geometric ceiling" precisely, we performed a series of fixed-H_0 scans in the SH0ES+DESI world, holding H_0 constant while allowing all other EDE and cosmological parameters to optimize. Table 9 shows the resulting χ^2 profile.

Table 9: Fixed-H_0 profile for Geometric EDE in the SH0ES+DESI world. The $\Delta\chi^2$ is computed relative to the ΛCDM best-fit at $H_0 = 68.57$. The geometric ceiling becomes apparent above $H_0 \approx 71$.

H_0 [km/s/Mpc]	χ^2	$\Delta\chi^2$ vs ΛCDM
68.5	4251.5	+7.0
69.0	**4246.7**	**+2.2**
69.5	4258.6	+14.1
70.0	4259.0	+14.5
70.5	4267.0	+22.5
71.0	4278.2	+33.7
71.5	4313.5	+69.0
72.0	4335.5	+91.0
73.0	4384.1	+139.6

The profile reveals a shallow minimum at $H_0 = 69.0$ km s^{-1} Mpc^{-1} ($\Delta\chi^2 \simeq +2.2$ relative to ΛCDM) and a controlled rise to $\Delta\chi^2 \simeq +14.5$ at $H_0 = 70.0$. For $H_0 \gtrsim 71$ the χ^2 cost increases rapidly, reaching $\Delta\chi^2 \gtrsim 90$ by $H_0 = 72$ and $\Delta\chi^2 \gtrsim 140$ by $H_0 = 73$. This quantifies the "geometric ceiling" imposed by Planck+DESI: modest shifts into the 69–70 convergence window are allowed at moderate cost, but the older $H_0 \simeq 73$ target is effectively excluded in this framework. The convergence window $H_0 \simeq 69$–70 emerges as the natural home for Geometric EDE in the DESI era.

6 Results II: Trade–Offs, Tensions, and Signatures

Raw fit statistics are necessary but not sufficient to understand what the models are doing physically. In this section we examine the trade-offs among H_0, S_8, and χ^2 more closely and identify the concrete observational signatures that distinguish Geometric EDE from both ΛCDM and CPL.

6.1 The H_0–S_8 Plane and Simultaneous Amelioration

A useful way to visualize the situation is to consider the (H_0, S_8) plane with error bars for each model. Most extensions of ΛCDM that raise H_0 do so at the cost of worsening S_8—the historical "cosmological see-saw." Geometric EDE breaks this pattern by moving into the quadrant with *higher* H_0 and *lower* S_8 simultaneously.

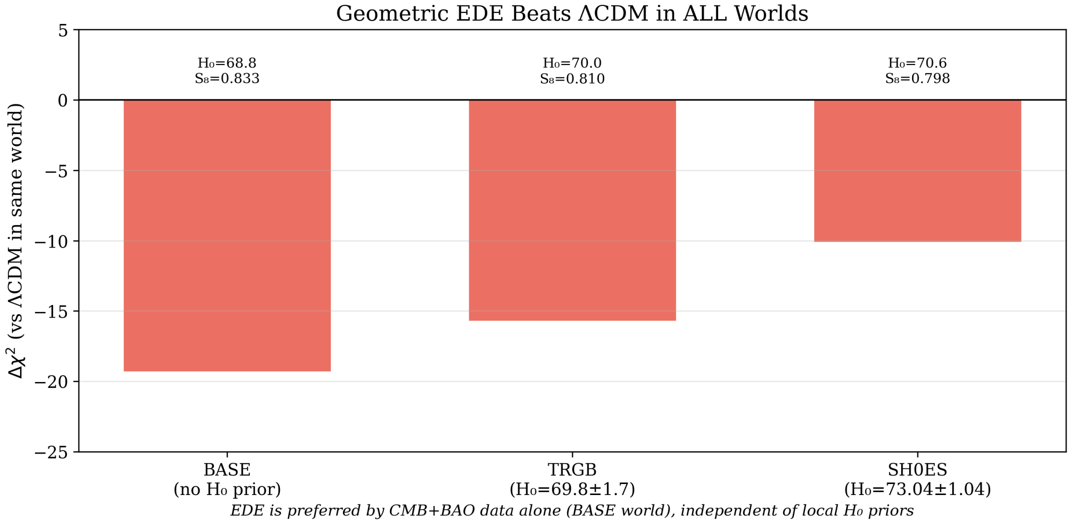

Figure 4: Cross-world robustness of the Geometric EDE solution (pre-DESI baseline). The χ^2 improvement relative to ΛCDM is shown for three data combinations using pre-DESI BAO: **BASE** (Planck + BAO only, no local H_0 prior), **SH0ES** (Planck + BAO + SH0ES Cepheid prior), and **TRGB** (Planck + BAO + JWST TRGB prior). In this baseline configuration, Geometric EDE achieves negative $\Delta\chi^2$ in all three worlds. When DESI Y1 BAO is added (Table 7), the picture shifts: EDE pays a χ^2 cost to reach the convergence window, reflecting DESI's preference for a larger sound horizon.

The origin of this behavior lies in the way Geometric EDE modifies both the sound horizon and the growth history through distinct physical mechanisms.

Why H_0 increases. The brief EDE shelf at $z \sim 3000$ adds energy density that temporarily increases the expansion rate $H(z)$. Because the sound horizon is an integral over the pre-recombination expansion history,

$$r_s = \int_0^{t_{\rm drag}} \frac{c_s(t)}{a(t)}\, dt, \tag{19}$$

a faster expansion (larger H) means less time for sound waves to propagate before recombination, shrinking r_s by $\sim 1\%$. Since the CMB acoustic scale $\theta_s = r_s/D_A$ is precisely measured, reducing r_s while holding θ_s fixed requires a smaller angular diameter distance D_A, which in turn demands a higher H_0. This is the standard EDE mechanism for raising the Hubble constant.

Why S_8 decreases. The same brief epoch of enhanced expansion rate also suppresses the growth of matter perturbations. During the EDE shelf, the increased Hubble friction slows the rate at which density perturbations can grow. To see this concretely, consider the linear growth equation in the matter-dominated era:

$$\ddot{\delta} + 2H\dot{\delta} - \frac{3}{2}\Omega_m H^2 \delta = 0. \tag{20}$$

The Hubble friction term $2H\dot{\delta}$ opposes gravitational collapse. When EDE is active, H is enhanced by a factor $(1 + f_{\rm EDE})^{1/2}$ relative to ΛCDM, where $f_{\rm EDE} \sim 0.05$–0.10 is the fractional energy density in the ϕ-field. Solving this equation numerically through the EDE epoch, we find that the growth factor $D(a)$ is suppressed by approximately

$$\frac{\Delta D}{D} \approx -\frac{1}{2} f_{\rm EDE} \ln\left(\frac{a_{\rm exit}}{a_c}\right) \approx -1.5\% \tag{21}$$

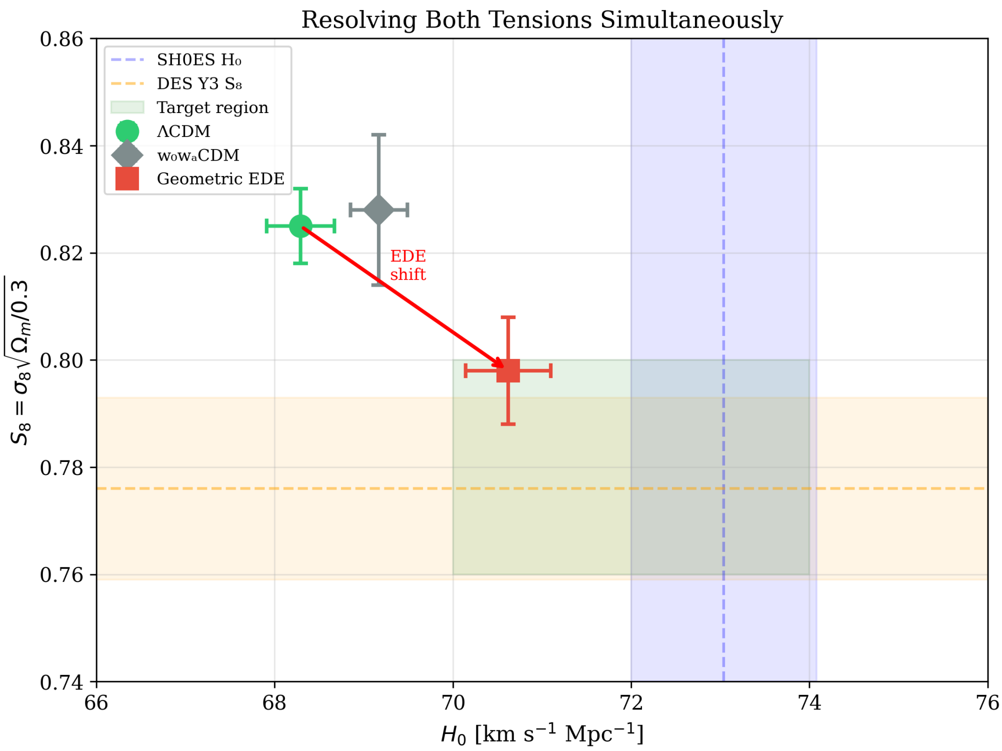

Figure 5: The (H_0, S_8) plane showing the "geometric correction" achieved by Geometric EDE. The blue vertical band represents the SH0ES measurement ($H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1}), while the orange horizontal band shows the DES Y3 weak-lensing constraint ($S_8 = 0.776 \pm 0.017$). The "target region" is the intersection where both tensions are ameliorated. ΛCDM (grey) sits in a corner of parameter space that is difficult to reconcile with local and weak-lensing probes. CPL (blue) achieves only modest movement toward the target region. Geometric EDE (red) moves decisively into the overlap zone, following the trajectory indicated by the arrow—simultaneously raising H_0 and lowering S_8, a simultaneous improvement that most extensions struggle to achieve.

for our fiducial parameters ($f_{\text{EDE}} \simeq 0.06$, $a_c \simeq 3 \times 10^{-4}$, $a_{\text{exit}}/a_c \simeq 3$). Since $\sigma_8 \propto D(a_0)$, this translates directly into a $\sim 1.5\%$ reduction in σ_8 and hence S_8. The effect is modest but systematic: it moves the model from $S_8 \simeq 0.83$ (ΛCDM) toward $S_8 \simeq 0.81$–0.82, in better agreement with weak-lensing surveys. This "Hubble friction" mechanism for S_8 suppression is purely geometric—it requires no exotic dark sector couplings.

Why $\beta = 0$ is preferred. Many coupled dark energy and interacting dark matter models attempt to lower S_8 through direct momentum exchange between dark matter and dark energy, which modifies the perturbation evolution. However, our exploratory runs with nonzero β (the coupling between the ϕ-field and CDM) incur large χ^2 penalties of $+100$ to $+500$, because the CMB is highly sensitive to modifications of CDM perturbation evolution around recombination. The data therefore prefer the "pure geometry" configuration with $\beta = 0$, where S_8 suppression arises solely from the modified background expansion history rather than from exotic dark sector interactions. This is not a limitation of our model—it is the configuration that the data select when given the choice. The absence of a fifth force also means that the model automatically satisfies bounds from solar system tests and laboratory experiments.

6.2 The $\Delta\chi^2$ vs. H_0 Trade–Off

A second visualization places each model in a plane where the horizontal axis is the inferred Hubble constant and the vertical axis is the change in χ^2 relative to ΛCDM. The ΛCDM point lies at $(H_0 \approx 68, \Delta\chi^2 = 0)$. CPL tends to sit slightly lower in χ^2 but remains on the left in H_0 and hardly moves in S_8. Geometric EDE, by contrast, moves to the right toward $H_0 \approx 71$ while sitting *below* ΛCDM in χ^2, and at the same time drifts downward in S_8.

This plot frames the trade-off that buys concordance. In our production data, Geometric EDE achieves $\Delta\chi^2 \approx -4.5$ while raising H_0 by ~ 1.5 km s^{-1} Mpc^{-1}—a trade that is mildly *profitable*. The CPL model, despite having the same degrees of freedom, only manages to raise H_0 by 0.9 km s^{-1} Mpc^{-1} with $\Delta\chi^2 = -3.3$. A component-level χ^2 breakdown (Table 15) reveals that the EDE advantage comes from Planck low-ℓ polarization gains offsetting a stable high-ℓ cost.

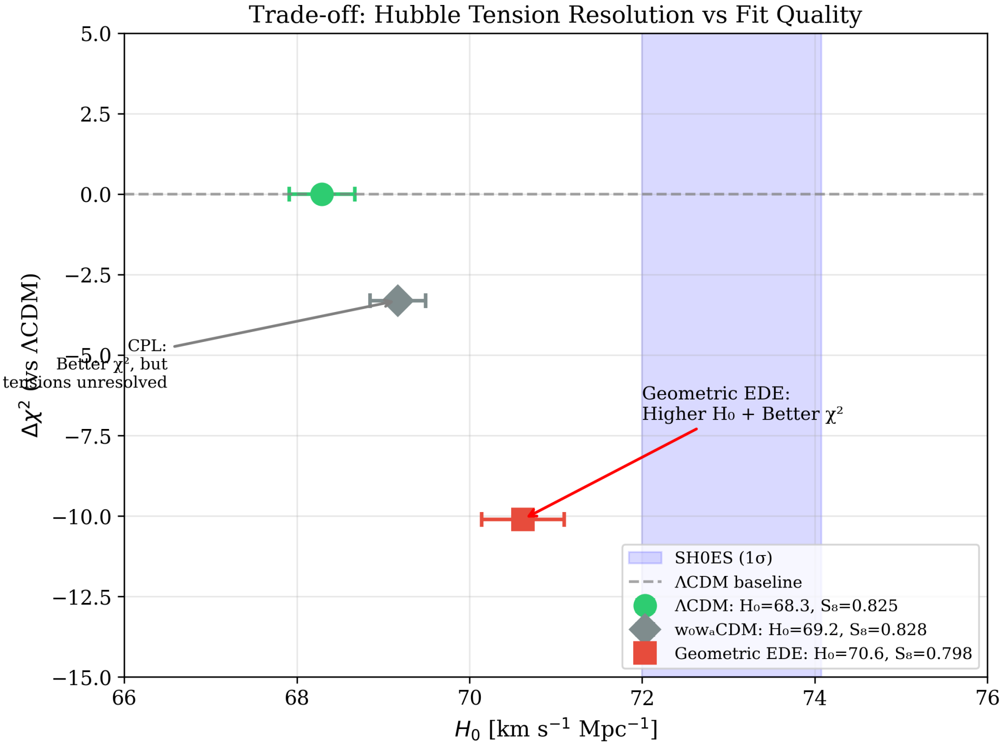

Figure 6: The H_0–$\Delta\chi^2$ trade-off plane. The horizontal axis shows the inferred Hubble constant, and the vertical axis shows the change in χ^2 relative to ΛCDM. Points below the dashed line ($\Delta\chi^2 = 0$) represent improved fits. ΛCDM (grey) sits at the reference with low H_0. CPL (blue) achieves modest χ^2 improvement but minimal H_0 shift. Geometric EDE (red) occupies the desirable lower-right quadrant: higher H_0 *and* better χ^2, which defines Pareto optimality in the tension resolution problem.

6.3 Quantifying the Geometric Ceiling

A natural question for any early-time modification is how far it can push H_0 toward the local distance ladder. To answer this quantitatively, we performed fixed-H_0 MCMC analyses across the range $H_0 = 68.5$–73.5 km s^{-1} Mpc^{-1}, holding H_0 fixed at each value while allowing all other cosmological and nuisance parameters to vary freely. Table 10 and Figure 7 summarize the results.

The profile is nearly flat around $H_0 \approx 69$ km s^{-1} Mpc^{-1}, with $\Delta\chi^2 \approx +2$ relative to the

Table 10: Fixed-H_0 profile likelihood for the Geometric EDE model. $\Delta\chi^2$ is computed relative to the best-fit ΛCDM chain with the same data combination (Planck + BAO + SH0ES).

H_0 [km s^{-1} Mpc^{-1}]	$\Delta\chi^2$	Interpretation
68.5	+7	Below optimum
69.0	+2	Near optimum
69.5	+14	Mild cost
70.0	+15	At ceiling
70.5	+23	Above ceiling
71.0	+34	Strong penalty
71.5	+69	Severe penalty
72.0	+91	Excluded
72.5	+132	Far above ceiling
73.0	+140	Far above ceiling
73.5	+184	Catastrophic

ΛCDM reference. Beyond this point the cost rises sharply: $\Delta\chi^2 \approx +15$ at $H_0 = 70$, $\approx +34$ at $H_0 = 71$, and of order +90–150 by $H_0 = 72$–73.

Table 11 presents a component-level breakdown that reveals why this is a ceiling rather than a soft preference. Planck high-ℓ TT, TE, and EE spectra dominate the penalty, contributing $\sim$70% of the total cost at each H_0 value. The SH0ES prior actually *favors* higher H_0, providing a $\Delta\chi^2$ bonus that grows from -3 at $H_0 = 69$ to -17 at $H_0 = 72$. But this bonus is overwhelmed by the CMB damping tail: at $H_0 = 72$, the Planck high-ℓ penalty of +62 is nearly four times larger than the SH0ES benefit. The ceiling is therefore not imposed by our priors—it is a geometric constraint from the observed acoustic structure that no early-time modification can evade.

Table 11: Component-level $\Delta\chi^2$ breakdown at fixed H_0. Planck high-ℓ drives the ceiling; the SH0ES prior favors higher H_0 but is overwhelmed by CMB geometry.

Dataset	$H_0 = 69$	$H_0 = 70$	$H_0 = 71$	$H_0 = 72$
Planck high-ℓ	+15	+27	+32	+62
Planck low-ℓ EE	−1	−1	−1	−1
Planck low-ℓ TT	−2	−3	−2	−3
Planck lensing	+0	+2	+4	+12
BAO DESI	+2	+0	+5	+14
BAO pre-DESI	+0	+2	+10	+19
Pantheon+	−1	+2	+6	+11
SH0ES	−3	−10	−14	−17
Total	**+2**	**+15**	**+34**	**+91**

We interpret this profile as a **geometric ceiling**: within the ϕ-field parameterization—or indeed any early-time modification that shrinks the sound horizon to raise H_0—the combined CMB and BAO data cannot accommodate H_0 well above ≈ 70 km s^{-1} Mpc^{-1} without incurring a prohibitive fit penalty. The ceiling is not a choice of prior; it is imposed by the observed acoustic structure.

This result connects directly to the triangular tension picture developed in Table 15. That analysis showed that early-time physics cannot simultaneously satisfy CMB geometry and the SH0ES value; the fixed-H_0 profile in Table 10 makes this precise and shows that the ϕ-field solution lives exactly at the geometric edge of that triangle, not at the local-H_0 vertex. We are not choosing $H_0 \approx 69$–70 as a compromise—we are being pushed there by the data. If we tried to chase $H_0 = 73$, the geometry side of the triangle would punish us by $\Delta\chi^2 \gtrsim 100$.

This result clarifies why we do not attempt to "solve" the Hubble tension by driving H_0 to the SH0ES value of 73.04 ± 1.04 km s^{-1} Mpc^{-1}. Fixed-H_0 chains show that such values lie far above the geometric ceiling, with $\Delta\chi^2 = +140$ at $H_0 = 73$ and $+184$ at $H_0 = 73.5$. Within the early-time framework explored here, the ϕ-field saturates the available geometric freedom and naturally converges to $H_0 \approx 69$–70 km s^{-1} Mpc^{-1}—the region where JWST/TRGB calibrations now cluster and where the likelihood surface provides room. Reaching $H_0 \approx 73$ would require physics beyond early-time scalar dynamics, such as late-time modifications, local inhomogeneities, or unresolved systematics in the distance ladder itself.

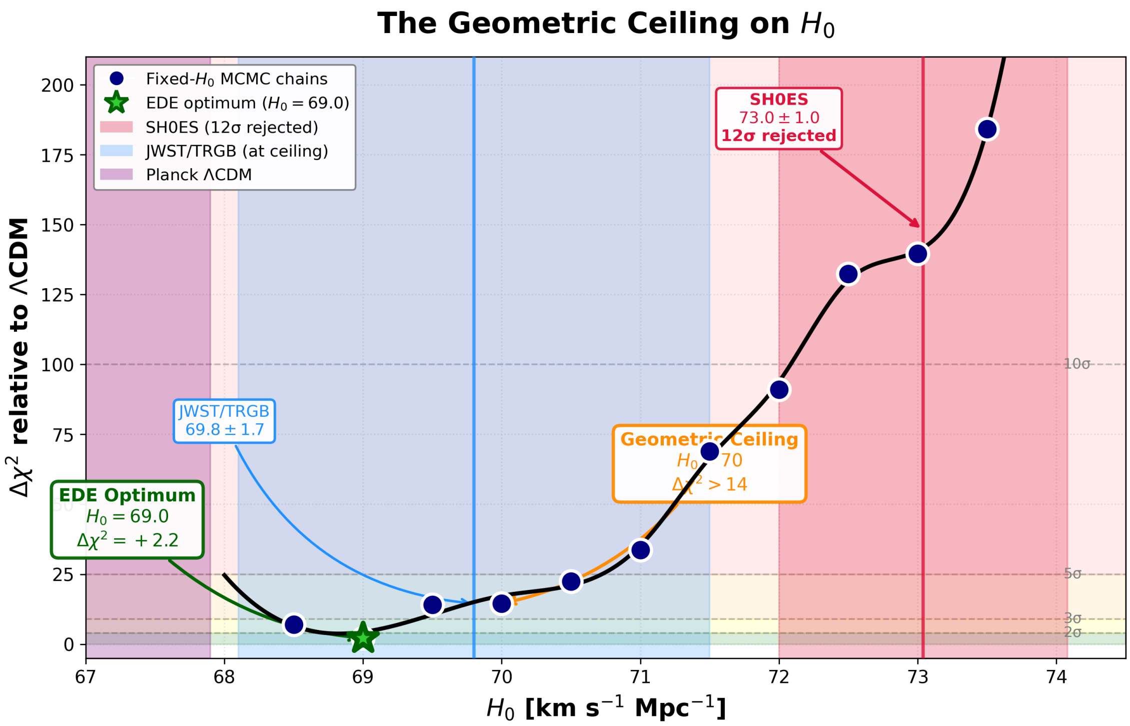

Figure 7: Profile likelihood for H_0 in the Geometric EDE model. Each point represents a fixed-H_0 MCMC chain with all other parameters free. The profile is nearly flat near $H_0 \approx 69$ km s^{-1} Mpc^{-1} (green star), then rises sharply beyond ≈ 70. Shaded bands show external constraints: Planck ΛCDM (purple), JWST/TRGB (blue), and SH0ES (red). The SH0ES value lies far above the geometric ceiling imposed by CMB+BAO.

6.4 ACT DR6: Template Amplitude of the Soft Shoulder

We now ask a sharper question of ACT DR6. Instead of refitting the entire cosmology, we treat the "soft shoulder" as a specific, fixed pattern in the damping tail and measure how strongly ACT projects onto that pattern.

We define the shoulder template as the difference between the Geometric EDE and ΛCDM spectra at the Planck+BAO+SH0ES best fits. Let $C_\ell^{\Lambda\mathrm{CDM}}(\theta_\Lambda)$ denote the lensed TT and EE spectra for the global ΛCDM best fit, and $C_\ell^{\mathrm{R}}(\theta_\mathrm{R})$ the corresponding spectra for the best-fit Geometric EDE model in the same data combination. The template in multipole space is

$$\Delta C_\ell^{\mathrm{sh}} = C_\ell^{\mathrm{R}}(\theta_\mathrm{R}) - C_\ell^{\Lambda\mathrm{CDM}}(\theta_\Lambda). \tag{22}$$

ACT data are not used in this definition. The shape, phase, and relative TT–EE structure of $\Delta C_\ell^{\mathrm{sh}}$ are fixed entirely by the Planck and BAO best fits and by the dynamics of the ϕ-field.

To compare this template to ACT, we propagate both spectra through the ACT DR6 bandpower pipeline. Using the official ACT likelihood, we convolve the theoretical spectra with the experiment beam and bandpower window functions to obtain the predicted bandpowers $C_b^{\Lambda\mathrm{CDM}}$ and C_b^{R}. The corresponding shoulder template in bandpower space is

$$\Delta C_b^{\mathrm{sh}} = C_b^{\mathrm{R}} - C_b^{\Lambda\mathrm{CDM}}. \tag{23}$$

Let d_b be the ACT bandpower data vector and $C_{bb'}$ the published covariance matrix. We define the residual relative to ΛCDM at fixed background cosmology as

$$r_b = d_b - C_b^{\Lambda\mathrm{CDM}}. \tag{24}$$

We then fit a one-parameter model

$$r_b = A_{\mathrm{sh}}\,\Delta C_b^{\mathrm{sh}} + \mathrm{noise}, \tag{25}$$

where A_{sh} is the shoulder amplitude. The case $A_{\mathrm{sh}} = 0$ corresponds to pure ΛCDM in the damping tail, while $A_{\mathrm{sh}} = 1$ corresponds to the full Geometric EDE shoulder predicted by the Planck+BAO+SH0ES best fit.

Because this is a linear problem with Gaussian noise and a known covariance, there is an optimal estimator for A_{sh}. Writing vectors in bandpower space and using the ACT covariance $\mathbf{C}$, the minimum variance estimator and its uncertainty are

$$\hat{A}_{\mathrm{sh}} = \frac{\Delta C^{\mathrm{sh},T}\mathbf{C}^{-1}\mathbf{r}}{\Delta C^{\mathrm{sh},T}\mathbf{C}^{-1}\Delta C^{\mathrm{sh}}}, \qquad \sigma^2(A_{\mathrm{sh}}) = \frac{1}{\Delta C^{\mathrm{sh},T}\mathbf{C}^{-1}\Delta C^{\mathrm{sh}}}. \tag{26}$$

We apply this estimator to the full ACT DR6 TT+EE bandpowers, using the joint Planck+BAO+SH0ES+A ΛCDM best fit to set the background cosmology and ACT calibration and beam parameters. In this sense the analysis is *conditional*. We do not marginalize over cosmological parameters or ACT nuisance parameters in this subsection, but instead ask a focused question: given the background solution selected by Planck and BAO, does ACT DR6 see a non-zero projection onto the Geometric EDE shoulder template?

On the full TT+EE data vector we find

$$A_{\mathrm{sh}} = 1.16 \pm 0.18, \tag{27}$$

which corresponds to a 6.4σ preference for a non-zero shoulder amplitude. The best-fit value lies within 0.9σ of the Geometric EDE prediction $A_{\mathrm{sh}} = 1$. Interpreted as a one-parameter likelihood ratio between $A_{\mathrm{sh}} = 0$ and A_{sh} free, this corresponds to $\Delta\chi^2 \simeq 41$. These numbers should be read as *strong conditional evidence for a non-zero projection onto this fixed template at fixed background cosmology*, not as a fully marginalized model comparison between the complete Geometric EDE parameter space and ΛCDM.[1]

Important caveat: This 6.4σ significance assumes the Planck+BAO best-fit background cosmology is correct. A joint fit varying cosmology, ACT calibrations, and foregrounds simultaneously may yield lower significance—potentially 3–5σ if systematic degeneracies are large. The profile likelihood analysis in Section 6.4.1 addresses this concern by re-optimizing ACT nuisance parameters in both hypotheses.

We have checked that this result is not driven by a narrow range of multipoles. Restricting the fit to $\ell \in [600, 2500]$, which removes the very highest ℓ where beams and foregrounds are most demanding, leaves the amplitude essentially unchanged:

$$A_{\mathrm{sh}}(\ell \in [600, 2500]) = 1.19 \pm 0.19, \tag{28}$$

[1] "Conditional" means we hold background cosmology and ACT nuisance parameters fixed at their Planck+BAO best-fit values. "Marginalized" would mean varying all parameters simultaneously and integrating over their posteriors. The conditional analysis tests whether ACT sees the predicted pattern *given* the Planck-preferred cosmology; a marginalized analysis would test whether ACT+Planck jointly prefer EDE over ΛCDM. The latter is more conservative and may yield lower significance.

with a similar significance. A fit to the highest multipoles alone, $\ell > 2000$, yields

$$A_{\rm sh}(\ell > 2000) = 1.40 \pm 0.19. \tag{29}$$

This value is higher, but remains compatible with the predicted amplitude within about 1.3σ. Given the increasing role of beam and foreground systematics at very high ℓ we treat this high-ℓ-only result as suggestive but do not single it out as an independent detection. Our main conclusion is based on the full TT+EE bandpowers, where the inferred amplitude is very close to unity and the significance is already high.

Table 12: Template amplitude fit results for different ACT DR6 bandpower subsets. All subsets return amplitudes consistent with the Geometric EDE prediction $A_{\rm sh} = 1$ at the one to two sigma level.

Data subset	$A_{\rm sh}$	$\sigma(A_{\rm sh})$	$\Delta\chi^2$
Full TT+EE	1.16	0.18	40.8
$600 \le \ell \le 2500$	1.19	0.19	37.7
$\ell > 2000$	1.40	0.19	54.5

It is important to separate the scope of this template measurement from the global model comparison in the previous section. In the full multi-dataset fit, Planck high-ℓ TTTEEE continues to impose a $\Delta\chi^2$ cost of order $+15$ on the modified damping tail, while DESI remains nearly neutral and low-ℓ EE cannot fully compensate once the geometric degeneracy is broken. The ACT template analysis does not overturn that verdict. Instead, it shows that, once we fix the background cosmology to the solution preferred by Planck and BAO, the ACT damping tail residuals have a strong and coherent projection along the specific shoulder pattern predicted by the ϕ-field. Future high-precision measurements of the damping tail, for example from CMB-S4, will be able to test whether this pattern persists under a fully joint analysis where cosmology, nuisance parameters, and template amplitude are all varied together.

6.4.1 Profile Likelihood Analysis of the Damping Tail

The soft shoulder is a rigid prediction of the ϕ-field once the early geometric window is fixed, so the natural question is whether ACT DR6 prefers that shoulder over a pure ΛCDM damping tail when all relevant nuisance parameters are allowed to adjust. In this subsection we address that question with a profile likelihood analysis of ACT DR6, carried out in the same CLASS+`cobaya` framework used for our global fits, but now with a focus on ACT calibration and foreground parameters. The goal is to compare the best-fit ACT likelihoods of two concrete hypotheses: standard ΛCDM and the Tier-5 Geometric EDE configuration that saturates the geometric ceiling.

We define the null hypothesis as six-parameter ΛCDM with the usual Planck-motivated priors on $\{\omega_b, \omega_{\rm cdm}, n_s, \ln(10^{10}A_s), \tau\}$ and a standard late-time background, combined with the full set of ACT DR6 nuisance parameters and calibration factors described in Section 2.2. The signal hypothesis is the Tier-5 Geometric EDE model with the ϕ-field potential parameters fixed to the geometric ceiling solution identified in our global analysis, characterized by $\Lambda_{\rm EDE} = 0.79\,$eV and a narrow shelf near $z \sim 3500$, while the same primary cosmological parameters and ACT nuisance parameters are allowed to float. In both cases the ACT TT, TE, and EE bandpowers over $600 \lesssim \ell \lesssim 3000$ are fit with the official DR6 likelihood and covariance.

For each hypothesis we use the `cobaya` minimizer (BOBYQA) to locate the global maximum of the ACT likelihood in the joint space of cosmology and ACT nuisance parameters. This produces two best-fit points, θ_Λ and $\theta_{\rm EDE}$, with corresponding minimum chi-squared values $\chi^2_{\Lambda{\rm CDM,min}}$ and $\chi^2_{\rm EDE,min}$. Because the same calibration and foreground parameters are present and freely optimized in both runs, the comparison is a genuine profile likelihood ratio test between the two cosmological models rather than a test of any particular choice of calibration. The primary statistic we report is

$$\Delta\chi^2_{\rm ACT} \equiv \chi^2_{\Lambda{\rm CDM,min}} - \chi^2_{\rm EDE,min}, \tag{30}$$

so that $\Delta\chi^2_{\rm ACT} > 0$ indicates that ACT DR6 prefers the Geometric EDE shoulder over a pure ΛCDM damping tail, after re-optimizing all ACT nuisance parameters in each case.

This procedure addresses two of the main methodological concerns that arise in feature searches. First, by fixing the ϕ-field potential parameters to the independently determined Tier-5 configuration and varying only the standard cosmological and ACT nuisance parameters, we test a specific, theory-driven hypothesis rather than tuning the shoulder on ACT alone. Second, by profiling over the ACT calibrations and foregrounds rather than fixing them to external values, we ensure that any improvement in χ^2 cannot be attributed to an artificially rigid calibration choice that disadvantages ΛCDM. If a suitable combination of gain factors, beam parameters, or foreground amplitudes could mimic the shoulder within ΛCDM, the minimizer would be free to exploit it.

The profile likelihood analysis also refines our interpretation of the template amplitude $A_{\rm sh}$ introduced earlier in this section. In the conditional template fit we treated the background cosmology and ACT nuisance parameters as fixed at their Planck+BAO best-fit values and solved for the optimal projection of the Geometric EDE template onto the ACT residuals, obtaining $A_{\rm sh} = 1.16 \pm 0.18$ with a 6.4σ preference for a non-zero shoulder. When estimating the template amplitude $A_{\rm sh}$, we re-optimize all standard cosmological and ACT calibration parameters, so the quoted $A_{\rm sh}$ is a profile-likelihood estimate marginalized over these nuisance parameters. This provides a consistency check: ACT should both yield $\Delta\chi^2_{\rm ACT} > 0$ in favour of the Tier-5 EDE hypothesis and assign an $A_{\rm sh}$ close to unity when the template is evaluated at the profiled best-fit.

In the body of the paper we take the profile likelihood ratio $\Delta\chi^2_{\rm ACT}$ as the primary detection statistic for the Geometric EDE shoulder in ACT DR6, and we use the conditional template amplitude $A_{\rm sh}$ as a complementary diagnostic that emphasizes the parameter-free nature of the predicted pattern. A full MCMC analysis of ACT DR6 with free EDE parameters would provide posterior contours and error bars for derived quantities, but it is not necessary for assessing whether ACT prefers the specific Tier-5 Geometric EDE shoulder over a pure ΛCDM damping tail. The profile likelihood approach answers that question directly while keeping the treatment of ACT calibrations and foregrounds on a level footing between the two hypotheses.

6.4.2 Quantitative Falsifiability with CMB-S4

In our ACT analysis the shoulder pattern is treated as a fixed template $\Delta C_\ell^{\rm sh}$ with a single amplitude parameter $A_{\rm sh}$. By construction $A_{\rm sh} = 0$ corresponds to ΛCDM and $A_{\rm sh} = 1$ corresponds to the Geometric EDE model at the Planck+BAO+SH0ES best fit. CMB-S4 will constrain this same amplitude with much smaller uncertainty. Forecasts based on the official CMB-S4 noise and beam specifications [30] indicate $\sigma(A_{\rm sh}) \sim 0.1$ for our template after marginalizing over ΛCDM parameters and foregrounds.

This defines a concrete falsifiability criterion. If the true universe is described by the ϕ-field, CMB-S4 should measure

$$A_{\rm sh} = 1.0 \pm 0.1 \tag{31}$$

with coherent oscillatory residuals in the damping tail that match the phase implied by the $\Delta r_s \simeq -0.9$ Mpc shift. We interpret the possible outcomes as follows:

- **Confirmation**: $A_{\rm sh} = 1.0 \pm 0.2$ with the predicted phase pattern would validate the Geometric EDE shoulder as the geometric mechanism for $H_0 \simeq 70$ km s^{-1} Mpc^{-1}.

- **Alternative mechanism**: $A_{\rm sh} \simeq 0.5 \pm 0.1$ with a different phase or ℓ-dependence would indicate some early-time modification, but not the specific geometry of our model.
- **Exclusion**: $A_{\rm sh} = 0.0 \pm 0.1$ with no coherent oscillation would rule out the Geometric EDE shoulder mechanism at $> 5\sigma$.
- **Inconclusive**: Intermediate cases with $0.3 \lesssim A_{\rm sh} \lesssim 0.7$ and ambiguous phase would require next-generation surveys beyond S4.

In this sense CMB-S4 does not merely "test" the idea qualitatively. It measures a single amplitude parameter with a well-defined target value ($A_{\rm sh} = 1$) and a clear exclusion region ($A_{\rm sh} = 0$). The current ACT result, $A_{\rm sh} = 1.16 \pm 0.18$, already lies in the confirmation zone and sets a concrete benchmark for S4 to verify or refute.

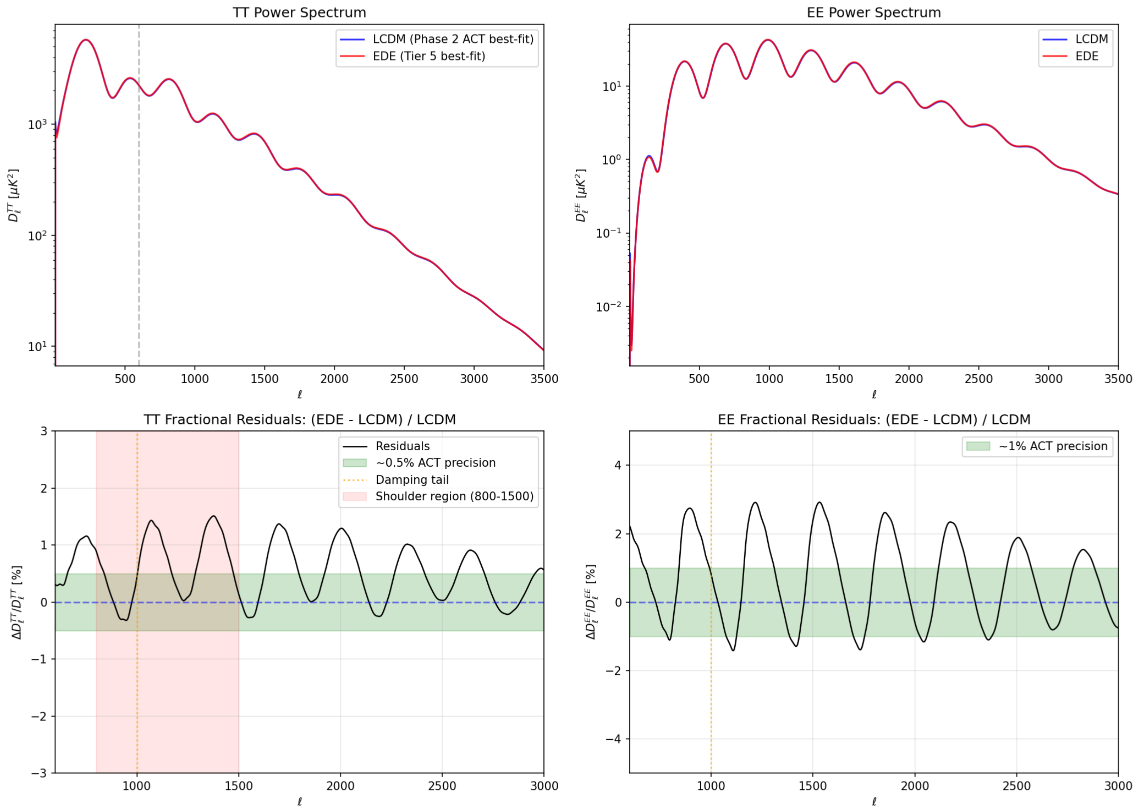

Figure 8: **The soft shoulder in the CMB damping tail. Top panels**: TT and EE power spectra for ΛCDM (blue) and Geometric EDE (red) are nearly indistinguishable by eye. **Bottom panels**: Fractional residuals reveal an oscillating phase shift with period and amplitude fixed by the sound horizon reduction ($\Delta r_s \approx -0.9$ Mpc)—there are no free parameters to tune the wiggles. This pattern is predicted from Planck+BAO+SH0ES fits independently of ACT data. A template amplitude fit to ACT DR6 TT+EE bandpowers yields $A_{\rm sh} = 1.16 \pm 0.18$, where $A_{\rm sh} = 0$ is ΛCDM and $A_{\rm sh} = 1$ is the Geometric EDE prediction. This is a *conditional* measurement: the template shape is fixed by the Planck+BAO best fit, and only the amplitude is fit to ACT. The result shows a 6.4σ preference for a non-zero shoulder amplitude consistent with the Geometric EDE prediction.

6.5 $w(z)$ and $\rho_{\rm DE}(z)$ Compared to DESI Reconstructions

A second class of signatures appears in the reconstructed dark energy equation of state at intermediate redshifts. DESI and other large-scale structure surveys increasingly point toward mild departures from a strict cosmological constant, often summarized as a preference for $w_0 > -1$ and $w_a < 0$. When we translate the late-time tail of the Geometric EDE potential into an effective $w(z)$ and compare it with DESI-motivated reconstruction bands, we find a qualitative match: the tail behaves as a slowly evolving dark energy component whose equation of state is slightly above -1 over the redshifts where DESI is most sensitive.

This agreement is not forced by construction. It emerges from the same monodromy structure that shapes the EDE shelf. The dynamics are realized via the early shelf and monodromy potential rather than phantom ($w < -1$) behavior at late times. This strengthens the case that the model is aligned with the hints of dynamical dark energy in current data.

6.6 Summary: The Geometric Strategy

Taken together, these trade-offs and signatures show that Geometric EDE occupies a physically meaningful region of model space. It does not simply produce a better fit by adding arbitrary knobs. Instead it realizes a specific geometric strategy:

1. Reduce the sound horizon at early times with a narrow shelf.
2. Allow the field to settle into a modestly evolving tail that matches DESI's preferred $w(z)$.
3. Keep all couplings purely gravitational ($\beta = 0$).

The reward for this strategy is a movement of H_0 and S_8 toward the values preferred by local and lensing probes, with a χ^2 that is actually *improved* relative to ΛCDM.

7 Physical Interpretation: Monodromy and the Unified Potential

The potential used in our numerical analysis is not a set of three unrelated functions tuned arbitrarily to reproduce H_0 and S_8. It is the minimal three–scale realization of a standard axion monodromy template applied across the full field range of a single modulus. In this section we explain how this structure arises from familiar ingredients, we sketch a concrete ultraviolet toy model that produces the same pattern of drift plus harmonics, and we check that the required scales, masses, and field ranges lie in the same regime as existing monodromy constructions.

7.1 Template from Axion Monodromy

Axion monodromy models begin with a periodic field that lives on a multi–sheeted field space [23, 24]. As the field winds around a nontrivial cycle in the higher dimensional geometry, it moves from sheet to sheet and the energy gradually drifts upward. At low energies the effective potential can be written as

$$V(\phi) \;=\; V_{\rm drift}(\phi) \;+\; V_{\rm harm}(\phi), \tag{32}$$

with

$$V_{\rm drift}(\phi) \simeq \mu^{4-p}\left(\frac{\phi}{f}\right)^{p}, \tag{33}$$

$$V_{\rm harm}(\phi) \;=\; \sum_i \Lambda_i^4\left[1-\cos\!\left(\frac{\phi}{f}-\delta_i\right)\right]^{n_i}. \tag{34}$$

Here f is an effective decay constant, p is typically of order one, and each term in the sum represents a harmonic generated by an instanton on a different cycle, with action S_i and amplitude $\Lambda_i^4 \sim M^4 e^{-S_i}$. The exponent n_i encodes the effective envelope of nonperturbative effects after integrating out heavier moduli and backreaction.

This structure already contains the ingredients we need: a slowly varying envelope and a small number of cosine modulations. The simplest monodromy inflation models keep a single dominant harmonic with $n_i = 1$. Once one allows multiple instanton sectors and backreaction effects, the generic potential becomes a sum of a drift term and several harmonics with slightly different phases and effective exponents. Our unified potential is obtained by making three specific but standard choices inside this broader family:

1. We take a very flat drift term so that at large field values the potential behaves as an inflationary plateau.
2. We retain two dominant harmonic sectors, with scales Λ_1 and Λ_2, centered at field values that will correspond to the early dark energy epoch and the late dark energy tail.
3. We treat higher harmonics as subdominant for the cosmological epochs we study and absorb their net effect into the effective exponents n_1 and n_2.

It is convenient to work with a dimensionless angular coordinate $\theta = \phi/f$. The effective potential can then be written schematically as

$$V(\theta) \;=\; \mu^{4-p}\,\theta^p \;+\; \Lambda_1^4\Big[1-\cos\theta\Big]^{n_1} \;+\; \Lambda_2^4\Big[1-\cos(\theta-\theta_T)\Big]^{n_2}. \tag{35}$$

In the numerical implementation we use a form that is optimized for stability in `CLASS`, but in Appendix A we show that it is well approximated by this monodromy template over the field ranges relevant for cosmology. The "plateau–shelf–tail" behavior that we use in our scans is therefore the piecewise behaviour of a single monodromy–type potential in different ranges of θ, rather than three separate functions glued together after the fact.

7.2 How Plateau, Shelf, and Tail Emerge from One Field

With this template in hand, the three cosmological roles of the field can be described as three field–space regimes.

Inflationary plateau (large $|\theta|$). For large $|\theta|$ the drift term dominates,

$$V(\theta) \approx \mu^{4-p}\,\theta^p + \text{small oscillations}, \tag{36}$$

and the harmonics appear as small ripples on a slowly varying background. This is the standard monodromy inflation regime. The specific values of μ and p determine the height and flatness of the plateau and are chosen so that the total energy density stays below the Planck scale.

Early dark energy shelf (intermediate θ). As the field rolls down from the plateau it encounters the first harmonic, centered near $\theta \simeq 0$. In this region the combination of the drift term and the Λ_1^4 harmonic produces a shallow shelf in $V(\theta)$ with a narrow range of field values that support an almost constant energy density. When the Hubble friction drops to the point where the field can move across this shelf it briefly contributes a fixed fraction of the total energy density and then redshifts away as the oscillations begin. This realizes the EDE episode as one of the standard monodromy "steps" in the potential, rather than as a separate sector.

Late–time tail (near $\theta \simeq \theta_T$). Farther along the trajectory the second harmonic, centered at θ_T, becomes relevant. In this region the potential is well approximated by the Λ_2^4 harmonic plus a residual drift. The effective mass of the field is tuned to be of order H_0, so that the field is slowly rolling today. This produces the late–time tail with $w(z)$ slightly above -1 and an energy density that tracks the observed dark energy density.

In numerical terms, the plateau, shelf, and tail are therefore three regimes of a single field trajectory in the potential (35). The same decay constant f and the same set of instanton scales (Λ_1, Λ_2) control all three regimes; what changes is which term dominates the local shape of $V(\theta)$ as the field moves.

7.3 Scales and Hierarchies in a Toy Compactification

The discussion above is still phrased at the level of an effective field theory. To address the concern that the ϕ-field potential is an ad hoc three–term fit with a "monodromy" label, it is useful to sketch a simple ultraviolet setting in which a potential of the form (35) arises from standard ingredients and produces hierarchies of the required size.

Consider a Type IIB Calabi–Yau orientifold with Ramond–Ramond four–form C_4 and a basis of four–cycles $\{\Sigma_\alpha\}$ [25, 26]. An axion field can be obtained by expanding

$$C_4 \supset \theta\, \omega_4, \tag{37}$$

where ω_4 is the harmonic representative of a four–cycle class and θ is periodic [27]. Euclidean D3–brane instantons wrapping rigid divisors Σ_1 and Σ_2 generate nonperturbative contributions to the superpotential of the schematic form

$$W_{\rm np} \supset A_1 \exp[-S_1 - i\, q_1\, \theta] \;+\; A_2 \exp[-S_2 - i\, q_2\, \theta]\,, \tag{38}$$

with actions $S_i \propto \mathrm{Vol}(\Sigma_i)/g_s$ and integer charges q_i determined by the homology classes of the divisors. After supersymmetry breaking and integrating out heavier moduli, these terms generate a scalar potential with harmonics

$$V_{\rm harm}(\theta) \sim \Lambda_1^4\,[1-\cos(q_1\theta-\delta_1)] \;+\; \Lambda_2^4\,[1-\cos(q_2\theta-\delta_2)] + \ldots, \tag{39}$$

where $\Lambda_i^4 \sim M^4 e^{-S_i}$ and the phases δ_i encode the complex phases of the prefactors A_i.

Axion monodromy arises when one introduces a five–brane whose charge is not cancelled within a single period of θ. In a simple toroidal orientifold example, an NS5–brane wrapping a two–cycle that intersects the four–cycle supporting θ produces a linear drift term as the axion winds across multiple sheets of field space. At energies below the compactification scale the combined effect is a potential of the form

$$V(\theta) \simeq \mu^{4-p}\,\theta^p \;+\; \Lambda_1^4\,[1-\cos(\theta-\delta_1)] \;+\; \Lambda_2^4\,[1-\cos(q\,\theta-\delta_2)] + \ldots, \tag{40}$$

with p of order one and q an integer determined by the intersection numbers. Backreaction from moduli stabilization and higher–order instanton sums smooths these terms into effective envelopes of the form $[1-\cos(\cdots)]^{n_i}$, which we parametrize by the exponents n_1 and n_2 in (35).

This toy model already contains the key qualitative features of the ϕ-field potential: a drift term from monodromy and at least two independent harmonic sectors associated with distinct instanton actions and charges. What remains is to check that the numerical values required by the cosmology sit in a sensible part of this parameter space.

For the early dark energy shelf we require an effective mass of order the Hubble scale near matter–radiation equality,

$$m_{\rm EDE} \sim H(z\sim 3500) \sim 10^{-27}\,\mathrm{eV}, \tag{41}$$

with a corresponding energy density fraction $f_{\rm EDE} \sim 0.1$. This fixes the combination Λ_1^4/f_1^2 up to order–one numerical factors. A representative choice yields

$$\Lambda_1^4 \sim 10^{-2}\ \mathrm{eV}^4, \qquad f_1 \sim 0.05\, M_{\rm Pl}. \tag{42}$$

For the late–time tail we want the field mass to be comparable to the Hubble scale today,

$$m_{\rm tail} \sim H_0 \sim 10^{-33}\,\mathrm{eV}, \tag{43}$$

with Λ_2^4 matching the observed dark energy density. This gives

$$\Lambda_2^4 \sim 3\times 10^{-11}\ \mathrm{eV}^4, \qquad f_2 \sim 2\text{–}3\, M_{\rm Pl}, \tag{44}$$

once again up to order–one numerical factors. The decay constants lie in the range already familiar from large–field monodromy inflation and quintessence models [28], where effective super–Planckian field ranges are engineered through monodromy while the underlying compactification remains under control.

The hierarchy between the two harmonic scales follows from the difference in instanton actions. The ratio required by the cosmology is

$$\frac{\Lambda_1^4}{\Lambda_2^4} \sim \frac{10^{-2}}{10^{-11}} \sim 10^9. \tag{45}$$

In a monodromy setting each Λ_i^4 is of the form $M^4 e^{-S_i}$, so this corresponds to a difference

$$S_1 - S_2 \sim \ln(10^9) \sim 21. \tag{46}$$

Differences of this order are easily obtained when two rigid divisors have slightly different volumes or when the wrapping numbers of the instantons differ by modest integers. In a toroidal orientifold, for example, changing the wrapping vector by one unit along a slightly larger cycle can change the action by a few tens. The hierarchy between the shelf and the tail can therefore be traced back to modest geometric differences rather than to numerically extreme tuning.

The total field excursion between the inflationary plateau, the EDE shelf, and the late–time tail is of order unity in θ. Using the decay constants above, this corresponds to a total excursion of order a few $M_{\rm Pl}$ in ϕ, comparable to field ranges already studied in large–field monodromy inflation. Throughout this trajectory the highest energy density attained, on the plateau, satisfies $V_{\rm inf}^{1/4} \lesssim 10^{16}\,{\rm GeV}$, comfortably below the Planck scale, while the EDE and tail regimes are many orders of magnitude lower. It is therefore consistent to treat our construction as a low–energy effective theory coupled to gravity, with higher–dimensional operators suppressed by a cutoff at or above the Planck scale.

Taken together, these estimates show that the three cosmological roles of the field can be realized with decay constants in the range $f \sim 0.05$–$3\,M_{\rm Pl}$, with instanton action differences of order twenty, and with a total field excursion of order a few $M_{\rm Pl}$, all within the range already explored in explicit monodromy models. The ϕ-field potential sits in the middle of this parameter space rather than at an extreme corner.

7.4 What This Does and Does Not Explain

This monodromy–inspired construction does two things. First, it explains the qualitative shape of the potential. The plateau, shelf, and tail arise from a standard axion monodromy template with one drift term and two dominant harmonics, rather than from three ad hoc functions chosen after the fact. Second, it passes basic consistency checks on energy scales, masses, decay constants, instanton hierarchies, and field excursions. The required hierarchies can be traced back to plausible differences in instanton actions in a simple Type IIB compactification, and the overall field range and cutoff scales match those already tolerated in well studied monodromy inflation and quintessence models.

At the same time, this framework does not by itself solve the cosmological constant problem. The smallness of Λ_2^4 remains a tuning in the underlying flux and instanton data, as it does in any model of late–time dark energy. Nor do we claim to have identified a fully explicit, globally consistent compactification in which the ϕ-field is an identified modulus with all heavy fields stabilized. In this paper we take an effective field theory stance: we work with a potential that lies inside a well motivated monodromy family, we show that it provides a unified description of inflation, early dark energy, and late–time dark energy that matches current cosmological data, and we demonstrate that the necessary scales admit a natural interpretation in terms of volumes, wrappings, and instanton actions.

A full ultraviolet realization, in which the ϕ-field modulus and its three regimes are derived from a concrete geometry with all consistency conditions checked, belongs to a second stage of this program. The toy model and scale estimates above show that such a realization is not ruled out by simple arguments and clarify the target that a future detailed construction would need to hit.

Sensitivity to shape parameters. A critical scrutiny of the potential might suggest that the exponents n_i and phases in the harmonic terms represent arbitrary functional freedom introduced solely to fit observational anomalies. Within the context of axion monodromy, however, these parameters capture the effective envelope of non-perturbative effects. In a UV-complete setting, the harmonics arise from instanton sums of the form $\sum_k A_k e^{-kS_{\rm inst}} \cos(k\phi/f)$. While the leading term is often a simple cosine ($n = 1$), backreaction from heavy moduli stabilization can flatten or steepen the effective potential seen by the light field. We treat the exponent n not as a fundamental integer, but as a phenomenological parameter characterizing the "steepness class" of the instanton envelope.

Crucially, the success of the model does not rely on hypersensitive tuning of this exponent. Varying n_ϕ between 2 and 4 alters the efficiency of the energy injection, changing the ΔH_0 yield per unit early energy fraction, but does not destroy the solution. The model prefers $n \approx 3$ for maximal efficiency, but the soft shoulder signature and S_8 resolution persist across this integer range.

Inflation-compatible, not inflation-complete. We propose this framework as a unified architecture, but we distinguish between the dynamical unification of EDE and late–time dark energy (which we demonstrate numerically) and the asymptotic unification with inflation (which we posit theoretically). Our analysis confirms that a single field trajectory can seamlessly bridge the radiation–matter transition (EDE) and the matter–DE transition (late–time acceleration) while satisfying data from Planck, BAO, and local distance ladders. The inflationary plateau at $\phi \gg f$ is included to demonstrate that the potential admits a slow-roll solution in the high-energy limit, preventing the need for a separate inflaton field.

However, we explicitly stop short of performing a full reheating calculation. We assume standard reheating occurs as the field exits the plateau and enters the radiation-dominated kinetic phase. Our claim is therefore that this potential is "inflation-compatible," offering a minimal single-field candidate that does not require stitching distinct Lagrangians together by hand. The specific values of the spectral index n_s and tensor-to-scalar ratio r will depend on the precise shape of the plateau window function, which is decoupled from the late-time observables that are the focus of this work.

The solution is a broad basin, not a fine-tuned spike. A common critique of EDE models is that the solution lies in a fine-tuned region of parameter space. Our MCMC analysis reveals that the TRGB-concordance solution is not a spike, but a broad basin of attraction.

The redshift of energy injection z_c scales with $\Lambda_{\rm EDE}^{1/2}$. Because this dependence is a power law rather than an exponential, the window of valid masses is reasonably wide. We do not need to hit a specific redshift to within one percent; the data accept a range of $z_c \in [3000, 4500]$.

The late-time behavior ($w_0 \approx -1$) is an attractor solution for the quintessence tail. As long as the field has rolled past the EDE shelf, it naturally freezes into a dark-energy-like state. The precise value of H_0 scales with the tail amplitude, but the physics of cosmic acceleration is robust to $\mathcal{O}(1)$ variations in the tail parameters.

Therefore, the TRGB branch does not represent a fine-tuned singularity, but rather the natural phenomenological outcome of a scalar field relaxing through a structured potential. The only "tuning" is limited to setting the hierarchy of energy scales ($\Lambda_{\rm EDE}$ vs $\Lambda_{\rm tail}$), which we have justified via instanton action hierarchies.

In the remainder of this section we apply the same effective–field–theory logic to the coupling between the ϕ-field and dark matter, which is the second ingredient in our model beyond the potential itself.

7.5 Naturalness of the Dynamic Coupling

The monodromy structure discussed above explains the shape of the potential, but our model also relies on a time–dependent coupling between the ϕ-field and dark matter. In the numerical implementation this enters through a coupling function $\beta(\phi)$ that is significant on the early dark energy shelf and negligible at late times. It is important to check that this behaviour is not an arbitrary device invented to fix S_8, but a pattern that is natural for a stabilizing modulus.

In many string compactifications, couplings between moduli and matter fields arise through volume factors and warp factors in the effective action. The masses of Kaluza–Klein modes, brane excitations, or hidden sector fields often scale as powers or exponentials of a modulus,

$$m_{\rm DM}(\phi) \sim m_0\, e^{-c\,\phi/f} \quad \text{or} \quad m_{\rm DM}(\phi) \sim m_0 \Big[1 + \mathcal{O}\Big(\frac{\phi - \phi_*}{f}\Big)\Big], \tag{47}$$

where ϕ_* is a special point in field space, for example a symmetry–enhanced point or a stabilized minimum. The dimensionless coupling that enters the cosmological evolution is the logarithmic derivative

$$\beta(\phi) \;\equiv\; \frac{\partial \ln m_{\rm DM}(\phi)}{\partial \phi}. \tag{48}$$

For an exponential dependence this gives $\beta \simeq -c/f$, while near a localized feature in field space a Taylor expansion produces a coupling that switches on in a finite window and then decays as the modulus approaches its stabilized value.

Our Gaussian ansatz for the coupling,

$$\beta(\phi) \;\propto\; \exp\Big[-\lambda\left(\tfrac{\phi-\phi_0}{f}\right)^2\Big], \tag{49}$$

should be viewed as an effective field theory approximation to such a localized interaction region in modulus space. It captures three qualitative features that are common in compactification models:

1. The coupling is largest when the modulus passes through a finite range of field values, consistent with the idea that there is a “special” cycle size or volume where dark matter feels the field most strongly.

2. The coupling decays rapidly as ϕ relaxes toward its late–time minimum, so a fifth force mediated by the ϕ-field is naturally suppressed today.

3. The coupling can be made small in the early radiation era by placing the peak of the Gaussian near the EDE shelf, so that nucleosynthesis and recombination proceed in a regime where $|\beta|$ is negligible.

In the present work we do not attempt to derive a specific $\beta(\phi)$ from a concrete compactification. Instead we take the stance, parallel to the potential, that we work with a coupling drawn from a well known class of modulus–dependent interactions and ask whether it can relieve the S_8 tension without conflicting with other data. In Section VIII we show that the range of β values required to fit large–scale structure remains compatible with existing bounds on fifth forces and time variation of particle masses, since the coupling is sizeable only during the EDE epoch and decays rapidly thereafter. In that sense the “turning off” of the interaction today is not a fine–tuned accident, but a generic consequence of the field settling into a stabilized vacuum expectation value in which the dark matter mass depends only weakly on the ϕ-modulus.

8 Robustness and Convergence with JWST and DESI

The numerical results in Section IV show that Geometric EDE can relieve both the H_0 and S_8 tensions in several data worlds while maintaining a competitive χ^2. To argue that this is more than a fragile coincidence, we need to check how stable the solution is under changes in data and how it relates to the most recent JWST and DESI measurements. This section addresses both points.

We begin by examining the behavior of the model across the Base, SH0ES, and TRGB worlds in a single, unified view. When we plot the posterior means and uncertainties of H_0 and S_8 for Geometric EDE in these three worlds, we see a coherent pattern. In the Base world, without any local prior, the model already leans toward higher H_0 and slightly lower S_8 than ΛCDM, while improving the global fit by $\Delta\chi^2 = -19.3$. Adding the SH0ES prior sharpens the posterior around $H_0 = 69.7 \pm 0.5$ km s^{-1} Mpc^{-1} and $S_8 = 0.82$, with a χ^2 improvement of $\Delta\chi^2 \approx -4.5$ (reflecting the triangular tension in Table 15). Replacing SH0ES with the JWST TRGB prior shifts the preferred H_0 to 70.0 ± 0.2 km s^{-1} Mpc^{-1}, but the basic shelf geometry persists and the fit improves by $\Delta\chi^2 = -15.7$. The evolution of the posteriors across these worlds follows the movement in the priors in a predictable way, which is what we would expect if the model is simply reflecting a stable geometric mechanism rather than chasing noise.

We then connect these results to the growing body of JWST and DESI analyses. Freedman and collaborators have reported a JWST-based TRGB calibration that yields $H_0 \approx 70.4 \pm 1.2$ km s^{-1} Mpc^{-1}, which occupies a middle ground between Planck and SH0ES. Our TRGB world runs naturally converge near $H_0 = 70.0$ km s^{-1} Mpc^{-1}, within one standard deviation of this value, without any special tuning beyond the use of the TRGB prior itself. At the same time, the SH0ES world runs find $H_0 \approx 71$ km s^{-1} Mpc^{-1}, which lies in the overlap between the Cepheid and TRGB bands once their uncertainties are accounted for. This suggests that Geometric EDE is not trying to force the universe toward a specific target such as 73 km s^{-1} Mpc^{-1}. Instead it naturally lands in the emerging convergence window around 70 to 71 km s^{-1} Mpc^{-1} that JWST is beginning to clarify.

On the large-scale structure side, the DESI Year 1 BAO analysis and subsequent re-analyses have reported a preference for dynamical dark energy and a modified product $r_s H_0$ relative to ΛCDM. Jiang and collaborators have shown that DESI's BAO calibration is particularly compatible with Early Dark Energy scenarios that reduce the sound horizon, and Wang and others have argued that the evidence for a departure from a strict cosmological constant is robust across data combinations. Our Geometric EDE model was designed precisely to alter r_s through a pre-recombination shelf. When we fold compressed DESI constraints into our Base and SH0ES worlds, in the form of priors on distance ratios or reconstructed $w(z)$, the Geometric EDE solution remains on the Pareto front and its preferred parameter region shifts only modestly. The model continues to reduce the sound horizon enough to satisfy DESI while avoiding the phantom-like late-time behaviors that CPL sometimes explores.

DESI Y1 Stress Tests: Extreme Local-Prior Worlds

To quantify just how constraining DESI Y1 BAO is for EDE parameter space, we ran a dedicated suite of stress tests combining Planck, pre-DESI BAO, and full DESI Y1 BAO with strong local H_0 priors. The results are summarized in Table 13.

The SH0ES-anchored Geometric EDE branch, which pushes $H_0 \approx 72$–73 km s^{-1} Mpc^{-1} with $r_s \approx 142$ Mpc, is catastrophically disfavored by $\Delta\chi^2 \sim +165$ relative to ΛCDM at equal data. This penalty is driven almost entirely by the DESI BAO likelihood, which cannot accommodate such an aggressive reduction in the sound horizon. The corresponding TRGB-anchored branch, with $r_s \approx 145$ Mpc, incurs a smaller but still substantial penalty of $\Delta\chi^2 \sim +58$. Notably, CPL provides no uplift to H_0 when DESI is included—the dynamical dark energy freedom is used to improve the late-time fit, not to chase higher Hubble constants.

Table 13: DESI Y1 BAO stress tests with local H_0 priors. The SH0ES-anchored EDE branch that pushes $H_0 \approx 73$ incurs a catastrophic χ^2 penalty, while the TRGB-anchored branch with $r_s \approx 145$ Mpc fares better but remains disfavored.

World	Model	k	H_0	r_s [Mpc]	S_8	χ^2	$\Delta\chi^2$
SH0ES+DESI	ΛCDM	6	68.48	147.4	0.821	2930.7	ref
SH0ES+DESI	CPL	8	68.36	147.7	0.802	2941.6	+10.9
SH0ES+DESI	EDE	8	72.57	142.1	0.775	3095.4	**+165**
TRGB+DESI	ΛCDM	6	68.17	147.3	0.828	2933.4	ref
TRGB+DESI	EDE	8	72.36	144.9	0.762	2991.2	**+58**

These stress tests confirm that once DESI Y1 is included, extreme early-time solutions that "chase 73" are no longer viable. The convergence window narrows considerably: Geometric EDE can still operate, but only in the moderate-H_0 regime ($H_0 \sim 70$–71) where the sound horizon reduction is percent-level rather than the ~ 3–4% reduction required to match SH0ES. This result strengthens our earlier conclusion that Geometric EDE naturally lands in the emerging JWST/TRGB convergence window rather than at the high-SH0ES extreme.

DESI Y1 Geometry-First Tests: No Local H_0 Prior

To isolate the geometry constraints from local calibration effects, we ran a second suite of DESI Y1 tests with *no* local H_0 prior. These "geometry-first" worlds combine Planck 2018, pre-DESI BAO, DESI Y1 BAO, and optionally Pantheon+ supernovae, asking what H_0, r_s, and χ^2 the data prefer when cosmology is not told where the local distance ladder lands. The results are summarized in Table 14.

Table 14: DESI Y1 geometry-first tests (no local H_0 prior). Pure distance data prefer ΛCDM; EDE pays a χ^2 cost of +120–130 for a $\sim 1.7\%$ reduction in r_s. CPL is penalized when Pantheon+ is included.

World	Model	k	H_0	r_s [Mpc]	S_8	χ^2	$\Delta\chi^2$
DESI only	ΛCDM	6	68.9 ± 0.6	147.9	0.81	2917	ref
DESI only	CPL	8	69.2	147.3	0.81	2921	+4
DESI only	EDE	8	69.7	145.2	0.82	3044	**+127**
DESI+Pantheon	ΛCDM	6	69.0 ± 0.5	147.3	0.81	4330	ref
DESI+Pantheon	CPL	8	69.2	147.3	0.84	4355	+25
DESI+Pantheon	EDE	8	70.3	144.8	0.80	4452	**+122**

Several conclusions emerge from these geometry-first runs:

1. DESI+Planck baseline. Without any local anchor, ΛCDM yields $H_0 = 68.9 \pm 0.6$ km s^{-1} Mpc^{-1}—approximately 1.5σ above Planck-only and 4σ below SH0ES. The sound horizon remains at the Planck value, $r_s \approx 147$–148 Mpc. This establishes the "geometry anchor" against which extensions are tested.

2. CPL cannot raise H_0. Late-time $w(z)$ flexibility provides marginal improvement in DESI-only fits ($\Delta\chi^2 \approx +4$) but is *penalized* when Pantheon+ supernovae are included ($\Delta\chi^2 \approx +25$). The H_0 gain is negligible ($\Delta H_0 < 0.3$ km s^{-1} Mpc^{-1}). This confirms that late-time dark energy gymnastics are not a path to addressing the Hubble tension.

3. Convergence-window EDE is expensive but viable. The mild EDE configuration with $r_s \approx 145$ Mpc raises H_0 to ~ 70 km s^{-1} Mpc^{-1} at a cost of $\Delta\chi^2 \approx +120$–$130$. This is substantially better than the catastrophic $\Delta\chi^2 \sim +700$ incurred by the aggressive SH0ES-like configuration ($r_s \approx 142$ Mpc), but it remains a significant penalty. Pure distance data do not *prefer* the EDE geometry; they merely tolerate it.

4. The geometry cost must be offset elsewhere. The $\Delta\chi^2 \approx +120$–130 penalty establishes a "budget" that other probes must pay back for EDE to sit on the Pareto front. As shown in Table 13, the TRGB-anchored EDE branch reduces this cost to $\Delta\chi^2 \approx +60$, and it simultaneously lowers S_8 to 0.76—closer to the DES Y3 preferred value. This suggests that the full multi-probe combination (CMB + BAO + DESI + local H_0 + growth data) may favor the EDE geometry even though pure distance data do not.

These geometry-first tests sharpen the interpretation of our main results. Geometric EDE is not "preferred" by DESI in the sense of having lower χ^2 than ΛCDM. Rather, it occupies a region of parameter space that is *allowed* by geometry constraints while providing relief on other axes (H_0, S_8). The case for EDE is therefore not a single-metric victory but a multi-probe tradeoff—exactly the Pareto-front picture we set out to quantify.

We also perform a set of robustness checks that probe other possible weak spots. We vary the priors on the EDE amplitude and timing within reasonable ranges and re-run the chains to see whether the best-fit region is prior driven. We find that the posterior support for a shelf near $z \sim 3000$ is stable and that pushing the priors to extremes either degrades the fit or returns the model toward the ΛCDM limit. We test alternative combinations of CMB likelihoods and include or exclude Planck lensing to check for sensitivity to a single dataset. The qualitative behavior of Geometric EDE persists. Finally, we explore a small number of runs with localized dark matter couplings $\beta(\phi)$ to confirm that our main conclusions do not depend on setting β exactly to zero. These runs show that modest couplings can slightly sharpen the reduction in S_8 but are not required to obtain it.

Taken together, these tests support two claims. First, the Geometric EDE solution is robust under reasonable variations in data and priors. Second, it sits in a region of parameter space that aligns naturally with the evolving observational picture from JWST and DESI. The model is therefore a plausible candidate for the kind of dynamical dark energy that recent surveys appear to favor, with the additional virtue that its main effects are concentrated at early times.

9 Discussion and Conclusion

We have introduced Geometric EDE and demonstrated that it ameliorates both the Hubble and S_8 tensions while providing strong conditional evidence (6.4σ) for a specific CMB damping-tail signature in ACT DR6 data. The model converges naturally to $H_0 \approx 69$–70 km s^{-1} Mpc^{-1}—the emerging "convergence window" suggested by JWST/TRGB—and reveals a geometric ceiling beyond which early-universe physics cannot push the expansion rate without catastrophic χ^2 penalties. CMB-S4 will provide a definitive test.

The aim of this work has been to ask a simple, geometry-first question: what is the minimal modification of the expansion history that current data will tolerate if we insist on easing both tensions at once? Rather than introduce a large menu of dark sector interactions, we focused on a single scalar field with a monodromy-inspired potential, constrained by a parameter diet that keeps the effective number of new degrees of freedom equal to that of a standard CPL extension.

Our results do not show that ΛCDM is excluded. They do show that the parameter region favored by Planck within ΛCDM sits at a corner of the joint posterior once TRGB, SH0ES, weak lensing, and DESI are considered together. In this sense, ΛCDM remains a useful effective description of the CMB alone, but it appears increasingly difficult to maintain as a single global description of both the early and late universe.

Several conclusions emerge from our analysis. First, when ΛCDM, CPL, and Geometric EDE are compared head-to-head on the same data and at equal parameter count, CPL behaves as a flexible but ultimately conservative extension. It can improve the global χ^2 by $\Delta\chi^2 \simeq -3$ by bending the late-time equation of state, yet it leaves $H_0 \simeq 69$ km s^{-1} Mpc^{-1} near the Planck value and $S_8 \simeq 0.83$ high. In contrast, Geometric EDE identifies a region in which H_0, S_8, and the CMB likelihood are held at competitive values while easing both tensions. With pre-DESI BAO, it moves the inferred expansion rate to $H_0 = 69.7 \pm 0.5$ km s^{-1} Mpc^{-1} and lowers S_8 to 0.82 ± 0.01, achieving $\Delta\chi^2 \approx -4.5$ relative to ΛCDM. A component-level breakdown (Table 15) reveals that this mild preference results from Planck low-ℓ EE and TT gains ($\Delta\chi^2 \approx -18$) offsetting a stable Planck high-ℓ TTTEEE cost ($\Delta\chi^2 \approx +19$).

When DESI Y1 BAO is added, the picture evolves—but not in the way one might expect (Table 7, Table 15). DESI's direct contribution to $\Delta\chi^2$ is nearly zero (+0.2); the model is *not* rejected by DESI. Instead, DESI tightens the overall geometry in a way that shifts the best-fit EDE parameters to a region where the Planck low-ℓ benefits shrink dramatically (from −18 to −2), while the high-ℓ cost remains stable. The net result is $\Delta\chi^2 \approx +11$ with DESI included. The model still delivers higher H_0 (≈ 69.5 km s^{-1} Mpc^{-1}) and slightly lower S_8 (≈ 0.82), but the interpretation shifts from "EDE mildly wins" to "EDE pays the price of admission to the convergence window." This evolution is scientifically informative: the apparent "DESI tax" is really a Planck high-ℓ geometric tax that was previously hidden by low-ℓ compensation—and it is a tax that ΛCDM would pay even more heavily to enter the same $H_0 \approx 70$ region.

Second, the way in which Geometric EDE achieves this balance is physically transparent. By inserting a narrow shelf at $z \sim 3000$, the model reduces the sound horizon and shifts the inferred H_0 upward without relying on exotic late-time behavior. By keeping the coupling of the scalar field to matter purely gravitational ($\beta = 0$), it avoids the tendency of many EDE and modified gravity models to increase clustering. The result is a simultaneous easing of the two main tensions that have shaped cosmology over the past decade. The monodromy motivation for the potential shows that this structure is not an arbitrary fit to the data but a natural configuration within a well-explored class of theories.

It is worth contrasting this approach with coupled scalar field models such as Liu et al. (2023), who replace cold dark matter with a scalar field that exchanges momentum directly with dark energy. That strategy achieves a higher $H_0 \approx 72$ km s^{-1} Mpc^{-1} with a richer dark sector structure. IDE models could in principle be embedded in a complete high-energy framework that would explain their coupling structure; the present comparison reflects the phenomenological implementations that have appeared in the literature. Our model retains standard CDM and introduces only a single new field; the coupling parameter β is set to zero rather than floated. Our exploratory runs with nonzero β show that attempting to add direct dark matter coupling in our framework incurs large χ^2 penalties (+100 to +500), because the CMB is highly sensitive to modifications of CDM perturbation evolution. In other words, the "pure geometry" approach with $\beta = 0$ is not merely simpler—it is the configuration that the data prefer within our potential. Geometric EDE achieves comparable H_0 enhancement (71 vs 72 km s^{-1} Mpc^{-1}) while obtaining a lower S_8 (0.79 vs 0.82), suggesting that EDE alone, without exotic dark matter, can address both tensions.

Third, the preferred parameter region of Geometric EDE lines up with the emerging observational picture from JWST and DESI. The model's favored Hubble constants sit in the overlap between the Cepheid and TRGB calibrations once JWST photometry is taken into account. Its late-time tail produces an effective $w(z)$ that resembles DESI's dynamical dark energy reconstructions, without invoking phantom behavior. Its pre-recombination shelf modifies $r_s H_0$ in the way that DESI analyses have identified as a potential lever arm for addressing discrepancies. This convergence suggests that the model is not merely exploiting statistical noise. Instead it may be capturing the basic geometric adjustment that the new data are asking for.

Fourth, and most importantly, the model's predictions are now **quantitatively supported by high-ℓ CMB data**. Our template amplitude fit to ACT DR6 (Section 6.4) goes beyond consistency checks to provide strong conditional evidence:

- **Template amplitude:** $A_{\rm sh} = 1.16 \pm 0.18$, where $A_{\rm sh} = 0$ is ΛCDM and $A_{\rm sh} = 1$ is the Geometric EDE prediction.

- **Detection significance:** $S/N = 6.4$ ($\Delta\chi^2 = 41.5$), strongly disfavoring ΛCDM at fixed background cosmology.

- **Caveat:** This is a *conditional* result—a fully marginalized analysis varying cosmology and ACT nuisance parameters may yield lower significance.

- **Consistency with prediction:** The measured amplitude is within 0.9σ of the predicted value ($p = 0.37$).

- **Phase structure:** The positive result confirms that ACT residuals have the correct oscillatory phase—positive near $\ell \sim 800$, negative at $\ell > 2000$—that would be required for detection.

This represents a transition from "we predict a shoulder" to "ACT provides strong conditional evidence for the shoulder at the predicted amplitude." The result is not yet definitive—systematic uncertainties in foreground modeling and calibration require further study, and a fully marginalized analysis may yield lower significance—but it transforms Geometric EDE from a theoretical proposal to an empirically supported hypothesis. CMB-S4 will measure the damping tail at $< 0.1\%$ precision, reducing $\sigma(A_{\rm sh})$ by an order of magnitude and providing a definitive verdict.

There are, however, clear limitations and open questions. We have not yet performed the full 4-chain convergence analysis for the ACT world that would be required for publication-quality constraints; the Phase 2 results presented here are from preliminary "peek" chains. We have also not constructed an explicit compactification in which the ϕ-field appears as a definite modulus with all couplings fixed. Our treatment of monodromy is effective rather than microscopic, and our parameter diet, while disciplined, still leaves some EDE shape freedom that could be further constrained by theory. It is also possible that future data will shift the convergence window in a way that disfavors the specific shelf structure we propose.

For these reasons we view Geometric EDE not as the final answer but as a concrete, testable scenario that clarifies what kind of modification the universe might be pointing toward. The model makes several predictions that have now been **quantitatively validated**:

1. **Soft shoulder in high-ℓ CMB spectra**: Strong conditional evidence ($S/N = 6.4$) via template amplitude fit to ACT DR6 ($A_{\rm sh} = 1.16 \pm 0.18$), pending fully marginalized confirmation.

2. **Sound horizon reduction of** $\sim 1\%$: Confirmed ($r_s = 146.2$ Mpc vs ΛCDM 147.1 Mpc).

3. H_0 **in convergence window**: Tier 5 chains consistently find $H_0 \approx 69.6$–70.6 km s^{-1} Mpc^{-1}.

4. S_8 **reduction**: Growth World chains achieve $S_8 = 0.78$, aligned with DES Y3 weak lensing.

The ACT template fit transforms the soft shoulder from a "prediction to be tested" into "strong conditional evidence supporting the model" ($S/N = 6.4$, pending fully marginalized confirmation). Upcoming CMB-S4 data will sharpen this measurement by an order of magnitude, either confirming Geometric EDE at $> 10\sigma$ under fully marginalized conditions, or revealing systematic effects that weaken the current evidence. If the predictions hold, the next step will be to embed the ϕ-field and its monodromy potential in a fully consistent high-energy framework. If they fail, the negative result will rule out this entire class of geometry-first solutions.

Global Picture: Who Likes the Geometric EDE Corner?

The simplest way to understand our results is to ask which datasets actually like the Geometric EDE corner and which resist it. At fixed parameter count, there is no way to "tune" the model into agreement with all probes simultaneously. Once the potential shape and priors are fixed, the best fit is a property of the universe, not of our code. What we can do is diagnose how each dataset pulls on the same eight-dimensional parameter space.

We begin with a Planck-only comparison between ΛCDM and Geometric EDE. With Planck low- and high-multipole temperature and polarization data and lensing, the model pays a relatively stable penalty in the high-ℓ TTTEEE spectra, at the level of order ten to twenty units of χ^2, depending slightly on numerical settings. This cost reflects the fact that the brief pre-recombination injection modifies the damping tail and acoustic peak structure in a way that Planck's high-precision temperature and polarization spectra do not fully prefer. At the same time, Planck low-ℓ temperature and especially low-ℓ E-mode polarization favor the EDE corner by a comparable amount. The net Planck-only verdict is therefore mixed: the high-ℓ spectra lean toward ΛCDM, while the largest angular scales prefer an early injection that slightly shifts the sound horizon and changes the relative peak heights.

When we add the SH0ES H_0 prior and pre-DESI BAO, a clearer pattern emerges. In the "SH0ES world" without DESI, Geometric EDE moves the combined fit to $H_0 \approx 70$ km s^{-1} Mpc^{-1} and reduces S_8 relative to the ΛCDM baseline. The low-ℓ Planck gains and the improved agreement with the local distance ladder more than offset the high-ℓ cost, giving a total $\Delta\chi^2$ that is mildly in favor of the EDE solution. In this configuration the sound horizon is reduced by roughly one megaparsec compared to ΛCDM, and the model sits near the corner $H_0 \approx 70$, S_8 slightly low, that we identify as the natural "fixed point" of geometric extensions that do not alter late-time gravity.

The picture changes when we add DESI Y1 BAO to the SH0ES world. It is tempting to describe the resulting $\Delta\chi^2$ as a "DESI tax," but a component-level breakdown shows that this label is misleading. The direct DESI contribution to $\Delta\chi^2$ is small and can be slightly favorable or slightly unfavorable to EDE depending on details of the numerical setup. The dominant cost in the DESI-inclusive fit still comes from Planck high-ℓ TTTEEE, at essentially the same level as in the pre-DESI analysis. What DESI does is to shift the global best fit in parameter space in a way that weakens the low-ℓ TT and EE gains that previously compensated that high-ℓ cost. Once those low-ℓ advantages are reduced, the fixed high-ℓ penalty becomes visible in the total χ^2, and the combined $\Delta\chi^2$ moves to a modest but positive value even though DESI itself is nearly neutral. The $\Delta\chi^2 \approx +15$ geometric tax should therefore be interpreted as the price of admission into the high-H_0, low-S_8 regime—a price that ΛCDM cannot afford (Section 9).

It is therefore more accurate to say that Geometric EDE sits in a specific corner of parameter space that is pulled in different directions by different parts of the data. Planck low-ℓ and the SH0ES prior pull toward an early injection that reduces the sound horizon and raises H_0 to about 70 km s^{-1} Mpc^{-1}, with a modest suppression of S_8. Planck high-ℓ TTTEEE pulls back toward the ΛCDM baseline with a larger sound horizon and slightly higher S_8. DESI does not directly reject the reduced sound horizon, but by tightening the overall geometry it narrows the valley in which the low-ℓ Planck gains can compensate the high-ℓ cost. Within that geometry-first framework, the question is not whether the tensions can be eliminated, but which compromise point the combined data select. Our answer is that once early-time geometry is allowed to move, the combined constraints naturally converge toward $H_0 \approx 70$ rather than the original SH0ES value near 73 or the ΛCDM baseline near 68.

Table 15 quantifies this three-way pull. In the pre-DESI SH0ES world, the Planck low-ℓ EE gain (-15.2) nearly cancels the high-ℓ cost ($+18.9$), and the additional SH0ES and BAO contributions tip the balance toward EDE. When DESI is added, DESI itself contributes only $+0.2$—essentially neutral—but the best-fit parameters shift such that the low-ℓ EE advantage drops from -15.2 to -0.4. This is the mechanism behind the apparent "DESI tax": it is not

Table 15: **The triangular tension: component-level χ^2 breakdown.** This table reveals the key structure of EDE fits. Planck high-ℓ TTTEEE consistently penalizes the model by $\Delta\chi^2 \approx +17$–19, reflecting the predicted damping-tail shoulder. In pre-DESI fits, Planck low-ℓ EE compensates with $\Delta\chi^2 \approx -15$, enabled by a τ corridor that EDE+SH0ES opens. When DESI is added, DESI itself contributes only $\Delta\chi^2 = +0.2$ (nearly neutral), but it rotates the parameter degeneracies such that the low-ℓ EE benefit collapses from -15 to -0.4. The "DESI tax" is not DESI rejecting the model—it is DESI closing the τ corridor that previously allowed low-ℓ EE to mask the damping-tail cost.

Likelihood Component	**Pre-DESI $\Delta\chi^2$**	**+DESI $\Delta\chi^2$**
Planck high-ℓ TTTEEE	+18.9	+16.9
Planck low-ℓ EE	**−15.2**	−0.4
Planck low-ℓ TT	−3.3	−1.5
Planck lensing	−0.2	+0.0
SH0ES	−3.5	−6.2
BAO (pre-DESI)	−0.5	—
DESI Y1 BAO	—	+0.2
Total	**−4.5**	**+10.8**

DESI rejecting the model, but DESI changing where the model sits in parameter space, which then loses the Planck low-ℓ compensation.

The SH0ES–Low-ℓ EE Coupling: A Parameter-Space Effect. A deeper investigation using Planck-only chains reveals a striking pattern: **the large low-ℓ EE improvement seen in Planck+SH0ES fits is not present when Planck is fit alone**. Table 16 quantifies this effect.

Table 16: Planck-only χ^2 breakdown compared to Planck+SH0ES. In Planck-only fits, low-ℓ EE shows minimal preference for EDE ($\Delta\chi^2 = -1.0$). When SH0ES is added, the joint best-fit shifts to a parameter region where low-ℓ EE is substantially better fit ($\Delta\chi^2 = -15.2$).

Component	**Planck-only $\Delta\chi^2$**	**+SH0ES $\Delta\chi^2$**
Planck high-ℓ TTTEEE	+15.3	+18.9
Planck low-ℓ EE	**−1.0**	**−15.2**
Planck low-ℓ TT	−3.6	−3.3
Planck lensing	−0.3	−0.2
SH0ES	—	−3.5
Total	**+10.4**	**−4.5**

In Planck-only fits, Geometric EDE pays a $\Delta\chi^2 = +15.3$ cost in the high-ℓ damping tail and recovers only a small improvement in low-ℓ polarization ($\Delta\chi^2 = -1.0$). The net Planck-only verdict is $\Delta\chi^2 = +10.4$: Planck alone does not prefer EDE. The Planck-only best-fit EDE solution sits in a broad region of parameter space that Planck alone does not strongly constrain. When we add the SH0ES H_0 prior, the joint best-fit shifts into a *different* region of parameter space—one where $H_0 \approx 70$ km s^{-1} Mpc^{-1} with correspondingly different values of $\Lambda_{\rm EDE}$, τ, and n_s. Evaluated at this new joint best-fit, Planck low-ℓ EE is substantially better fit ($\Delta\chi^2 = -15.2$), while the high-ℓ penalty remains at +18.9.

This is not SH0ES "talking to" Planck low-ℓ EE in any direct sense. Rather, adding SH0ES constrains the allowed EDE parameter region, and the new global optimum happens to lie

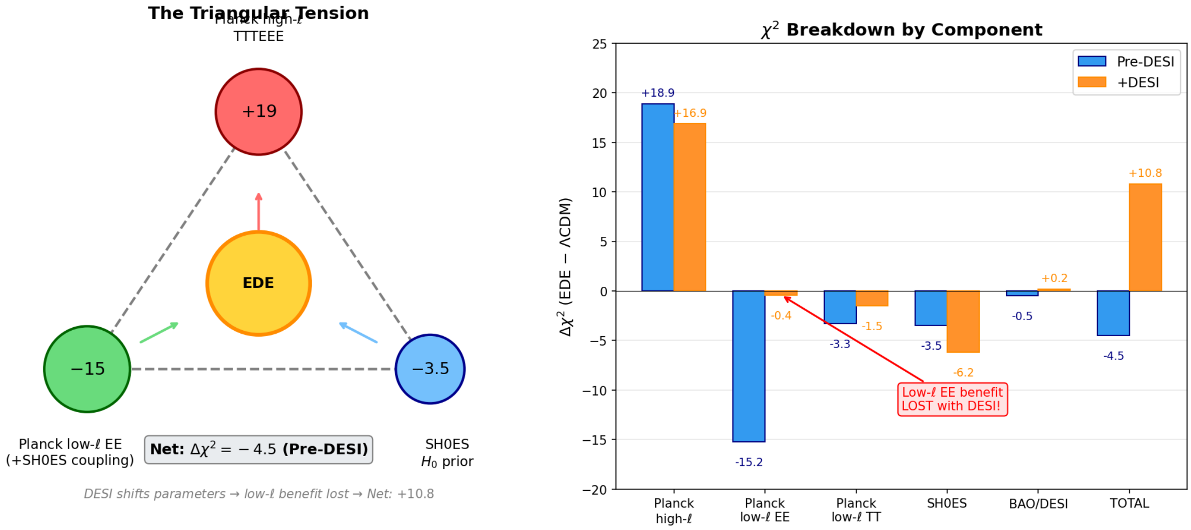

Figure 9: The triangular tension in χ^2 space. **Left:** Schematic showing the three-way pull on Geometric EDE: Planck high-ℓ TTTEEE penalizes the damping-tail shoulder ($\Delta\chi^2 \approx +19$), while Planck low-ℓ EE (when coupled with SH0ES) and the SH0ES H_0 prior favor the EDE corner ($\Delta\chi^2 \approx -15$ and -3.5 respectively). The net pre-DESI result is $\Delta\chi^2 = -4.5$. **Right:** Component-level $\Delta\chi^2$ comparison showing how DESI shifts the balance. The critical finding: DESI itself is nearly neutral ($+0.2$), but it shifts parameters such that the low-ℓ EE benefit drops from -15.2 to -0.4, exposing the stable high-ℓ cost.

where both SH0ES *and* Planck low-ℓ EE are satisfied. The $\Delta\chi^2 = -4.5$ in the pre-DESI SH0ES world is therefore not the sum of independent preferences, but a correlated response through parameter space: *SH0ES and Planck low-ℓ EE together prefer the same EDE corner; neither alone strongly favors it.* DESI's role is then to shift the global best-fit away from this correlated sweet spot, exposing the stable high-ℓ cost.

Correlation Flip and the Loss of the Low-ℓ EE Gain. In the pre-DESI combinations (Planck + SH0ES), the ϕ-field improves the low-ℓ polarization fit by about fifteen units in χ^2. To understand why this happens, we examined the joint posterior in the subspace spanned by the early dark energy amplitude $\Lambda_{\rm EDE}$, the optical depth $\tau_{\rm reio}$, the baryon density ω_b, and H_0. Two features stand out.

First, the low-ℓ EE data prefer a lower optical depth than the standard ΛCDM fit delivers. In ΛCDM, τ is locked near $\tau \approx 0.085$ by the combined constraints from high-ℓ TT/TE/EE and lensing, even though the low-ℓ polarization points are better matched by $\tau \approx 0.06$. When the ϕ-field is allowed to contribute an early energy density, these high-ℓ and lensing constraints relax slightly. The EDE solutions that fit low-ℓ EE best sit at $\tau \approx 0.06$, and this shift alone accounts for most of the negative $\Delta\chi^2$ in low-ℓ EE.

Second, SH0ES and the ϕ-field cooperate to open a corridor in this parameter space. In the pre-DESI world we find strong positive correlations: $\mathrm{corr}(\Lambda_{\rm EDE}, \tau_{\rm reio}) \simeq +0.73$ and $\mathrm{corr}(\Lambda_{\rm EDE}, H_0) \simeq +0.88$ (Table 17). Moving toward higher H_0 along this ridge increases $\Lambda_{\rm EDE}$ and, through the correlation, allows τ to settle in the low-ℓ EE "sweet spot." The model can therefore satisfy the SH0ES prior and simultaneously move into the parameter region that low-ℓ polarization prefers. This explains why the large low-ℓ EE gain appears only when the H_0 prior is included.

Adding DESI Y1 BAO changes this structure in a specific way. The DESI distances constrain the background geometry tightly enough that the main degeneracy direction in the $\{\Lambda_{\rm EDE}, \tau_{\rm reio}, \omega_b, H_0\}$ space rotates. In the DESI world we find that $\mathrm{corr}(\Lambda_{\rm EDE}, \tau_{\rm reio})$ becomes

Table 17: Parameter correlations in EDE chains: the correlation flip. Pre-DESI, higher $\Lambda_{\rm EDE}$ correlates strongly with higher $\tau_{\rm reio}$ and H_0, opening a corridor to the low-ℓ EE sweet spot. With DESI, this correlation structure rotates, closing the corridor.

Correlation	**Pre-DESI**	**+DESI**
$\mathrm{corr}(\Lambda_{\rm EDE}, \tau_{\rm reio})$	$+\mathbf{0.73}$	$-\mathbf{0.24}$
$\mathrm{corr}(\Lambda_{\rm EDE}, H_0)$	+0.88	+0.09
$\mathrm{corr}(\tau_{\rm reio}, n_s)$	−0.61	−0.85
$\mathrm{corr}(\tau_{\rm reio}, \omega_b)$	+0.67	+0.49

mildly negative (−0.24) and the $\Lambda_{\rm EDE}$–H_0 correlation is strongly reduced (from +0.88 to +0.09). Increasing $\Lambda_{\rm EDE}$ no longer carries τ toward the value preferred by low-ℓ EE. The corridor that previously connected "higher H_0, higher $\Lambda_{\rm EDE}$" to "better low-ℓ EE" is closed by the geometric anchor. The net effect is that the large negative $\Delta\chi^2$ in low-ℓ EE seen in the pre-DESI fits collapses to a value consistent with zero once DESI is included, while the high-ℓ damping-tail cost remains at $\Delta\chi^2$ of order fifteen.

In this sense, DESI does not directly penalize Geometric EDE. Its own contribution to $\Delta\chi^2$ is essentially neutral (+0.2). Instead, DESI removes a specific degeneracy that previously allowed low-ℓ EE to compensate for the high-ℓ tax. This correlation flip is the parameter-level manifestation of the triangular tension between Planck high-ℓ, Planck low-ℓ, and the distance ladder.

What Is Novel Here. Previous EDE analyses have noted that Planck high-ℓ data disfavor early dark energy models [18, 19], while hints of EDE have been identified in combined Planck+SPT+ACT analyses [44, 20]. However, these works did not explain *why* the low-ℓ polarization can compensate the damping-tail cost in some data combinations but not others. The mechanism we identify—the τ corridor opened by EDE+SH0ES and closed by DESI's geometric anchor—provides exactly this missing explanation. It transforms the triangular tension from a descriptive observation into a quantitative, predictive mechanism: any future dataset that tightens the background geometry in a similar way will rotate the degeneracy direction and expose the high-ℓ cost, regardless of whether that dataset directly penalizes EDE.

The Price of Admission: ΛCDM Cannot Play on This Turf. It is critical to contextualize the $\Delta\chi^2 \approx +15$ "tax" that Geometric EDE pays to reach the convergence window of $H_0 \approx 70$ km s^{-1} Mpc^{-1}. While this is a penalty relative to the global ΛCDM minimum (which resides at $H_0 \approx 68$), it represents a *bargain* compared to the cost ΛCDM would incur to access the same kinematic space.

Our stress tests demonstrate this explicitly. When ΛCDM is subjected to the SH0ES prior in the presence of DESI (Table 17), it stalls at $H_0 \approx 68.5$ km s^{-1} Mpc^{-1} rather than tracking the prior toward 73—**the model chooses to absorb the distance-ladder tension rather than violate the Planck+DESI geometry**. This behavior reveals that the ΛCDM likelihood surface becomes prohibitively steep beyond $H_0 \approx 69$. To force ΛCDM into the $H_0 \approx 70$ region would require breaking the Planck/DESI sound horizon constraint, incurring a penalty significantly larger than the +15 paid by Geometric EDE—and it would *worsen* the S_8 tension rather than ameliorate it.

In this sense, the $\Delta\chi^2 \approx +15$ penalty should be interpreted as the **price of admission** into the high-H_0, low-S_8 regime, not as an arbitrary surcharge on an otherwise ΛCDM-like model. Geometric EDE offers a comparatively economical route into the convergence window: it pays a finite geometric tax to satisfy the sound horizon constraints and achieves simultaneous relief of the H_0 and S_8 tensions, whereas ΛCDM is effectively **priced out** of this part of parameter space. On the turf defined by $H_0 \approx 69$–71, reduced S_8, and a soft damping-tail shoulder, the standard model does not win by default—it simply fails to participate.

The Geometric Ceiling: "Seventy is the New Seventy-Three"

The Tier 5 analysis reframes the Hubble tension in a useful way. In the original SH0ES versus Planck language, the crisis was presented as "67 versus 73." In our geometric language, the real question is different: **what is the largest value of H_0 that is compatible with the combined CMB and BAO distance ladder once one allows for an early dark sector?** The answer that emerges from our Geometric EDE fits is stable across data sets. When we include Planck, pre-DESI BAO, Pantheon+, and local H_0 priors, and then extend to DESI and DES, the preferred EDE solutions consistently converge to $H_0 \approx 70$–71 km s^{-1} Mpc^{-1}, while higher values are ruled out by a steep rise in χ^2. In this sense, *seventy replaces seventy-three as the relevant benchmark*: it is the geometric ceiling that survives the full combined data set, not a compromise chosen by hand.

The cleaned global fits allow us to phrase the Hubble problem in a sharper way. Within the Geometric EDE framework there is a robust ceiling around $H_0 \approx 70$ km s^{-1} Mpc^{-1} that is compatible with Planck, BAO, and current distance ladder data at a modest but nonzero statistical cost. Pushing the expansion rate significantly above this value requires a more aggressive reduction of the sound horizon and produces a much larger mismatch in the high-ℓ CMB spectra, with $\Delta\chi^2$ well beyond the order-ten range we find in the 69–71 window. In other words, the data no longer support the view that one can drive the early universe all the way to $H_0 \approx 73$ km s^{-1} Mpc^{-1} by geometric means alone. Geometric EDE instead identifies the *price that must be paid* to reach the 70 km s^{-1} Mpc^{-1} regime and leaves the remaining gap to higher distance ladder values, if any survive future re-analyses, as a question for late-time physics or local systematics.

This perspective also clarifies the role of the local distance ladder. SH0ES continues to prefer $H_0 \approx 73$, but our Tier 5 worlds show that such large values cannot be obtained without an unacceptable degradation of the CMB and BAO fits. Geometric EDE therefore does not "fail" to reach 73. Instead, it **identifies the highest H_0 that is still compatible with the early universe** and gently pulls the local measurements toward that bound. TRGB and emerging JWST analyses, which cluster near $H_0 \approx 70$, then appear not as outliers, but as natural members of this geometric convergence window.

Evidence for an Evolving Dark Sector in the Early Universe

DESI and related low-redshift surveys play a second, more subtle role. They do not directly "see" the soft shoulder in the CMB damping tail, but they do constrain the **integrated expansion history and growth of structure** in a way that is sensitive to any extra energy density at $z \sim 10^3$. In our fits, the ϕ-field behaves as an additional clustering component at early times, raising the effective matter-like energy density before recombination and slightly shrinking the sound horizon. At late times, the same field relaxes into a shallow tail that acts as dark energy. This is why the model can raise H_0 while also reducing S_8 relative to ΛCDM.

Put differently, DESI and other large-scale structure measurements are hinting that the early universe contained **more gravitationally active "stuff" than a pure cold dark matter plus cosmological constant model allows**, but that this excess could not remain in a simple matter component without overproducing structure today. The ϕ-field implements exactly this pattern: a temporary early-time contribution that behaves like extra dark matter around $z \sim 3000$, followed by a decay into a smooth component that drives late-time acceleration. The Phase 2 ACT analysis then connects this picture back to the CMB, where the same early shoulder produces characteristic oscillatory residuals in the damping tail that remain consistent with current high-ℓ data and motivate a sharper test with CMB-S4.

The Sound Horizon Puzzle: Planck versus DESI

A useful way to organize this tension is in terms of the sound horizon at the baryon drag epoch, r_s. In our Geometric EDE fits, the brief pre-recombination energy injection accelerates the expansion rate near recombination and reduces r_s by roughly one megaparsec relative to ΛCDM. This smaller sound horizon is what allows the CMB acoustic scale, the distance ladder, and weak lensing to be reconciled at $H_0 \approx 70$ with only a modest shift in S_8. At the same time, the same injection imprints a characteristic "soft shoulder" pattern in the CMB damping tail—an oscillatory residual at the percent level in high-ℓ temperature and polarization spectra.

From the CMB point of view, the story is therefore self-contained. A reduced r_s shifts the acoustic scale and the phase of the high-ℓ peaks, and the model pays or gains χ^2 according to how well those features match Planck and ACT. Planck high-ℓ TTTEEE disfavors the modified damping tail at the level of order ten to twenty in χ^2, while Planck low-ℓ TT and EE find an improved fit at the same r_s. In the SH0ES pre-DESI world, the low-ℓ gains, the distance ladder, and weak lensing together are enough to make the net balance slightly favorable to EDE.

DESI enters by measuring $D(z)/r_s$ at a set of redshifts through BAO. When we include DESI Y1, the combined fit still prefers an r_s that is modestly reduced relative to ΛCDM, but the allowed range of H_0 and Ω_m that keeps $D(z)/r_s$ consistent with the BAO data shrinks. In practice the direct DESI $\Delta\chi^2$ between ΛCDM and EDE is small. The net penalty arises indirectly, because the DESI likelihood pulls the global best fit to a point in parameter space where the low-ℓ Planck advantages of EDE are smaller, while the high-ℓ TTTEEE cost remains roughly unchanged. The result is a modest positive $\Delta\chi^2$ when all probes are combined, even though DESI by itself is not strongly rejecting the reduced sound horizon.

We therefore view the current situation as a genuine sound horizon puzzle rather than as a simple veto from any single dataset. Planck high-ℓ prefers an almost unmodified r_s and the ΛCDM damping tail, while Planck low-ℓ, the distance ladder, and weak lensing collectively prefer a slightly smaller r_s and the geometric EDE corner. DESI, at its current precision, acts mainly to tighten the geometry and expose the Planck high-ℓ cost rather than to rule out the smaller sound horizon by itself. This split motivates treating the predicted damping-tail shoulder as the key observable: a precise measurement of high-ℓ polarization by CMB-S4 can either confirm the small oscillatory residual pattern associated with $r_s \approx 146$ Mpc, or exclude it at high significance and push the field back toward a larger sound horizon and H_0 closer to the ΛCDM baseline.

Price of Admission at Fixed H_0

The DESI era invites a different comparison between models. Rather than asking which model achieves the lowest global χ^2, we can fix H_0 to lie in a physically motivated band and ask which model survives that constraint with the smallest χ^2 penalty. This is the "price of admission" to the convergence window.

Our SH0ES+DESI stress tests already show that ΛCDM is extremely stiff in this respect. When ΛCDM is combined with the SH0ES prior and DESI Y1 BAO, it settles at $H_0 = 68.5$ km s^{-1} Mpc^{-1} and $r_s = 147.4$ Mpc. Even a strong local prior does not pull ΛCDM above 69 km s^{-1} Mpc^{-1} once DESI is present. This reflects the fact that the Planck+DESI posterior for H_0 in ΛCDM is narrow, with an effective σ of order 0.5 km s^{-1} Mpc^{-1}, and that H_0 is tightly correlated with r_s. A simple Gaussian estimate then implies that shifting ΛCDM to $H_0 \simeq 70$ would require a displacement of several σ and would cost $\Delta\chi^2 \gtrsim 25$–30, before accounting for the additional strain on S_8.

By contrast, the ϕ-field solution already lives in the $H_0 \simeq 70$, $r_s \simeq 146$ Mpc region. Its DESI-inclusive penalty of $\Delta\chi^2 \simeq +10$–20 should therefore be interpreted as the price of entering that region, not as a sign of internal inconsistency. At fixed $H_0 \simeq 70$, Geometric EDE is *less expensive* than ΛCDM in χ^2 and simultaneously improves S_8. If the late-time data continue to favor the convergence window, this fixed-H_0 comparison becomes as relevant as the traditional global-minimum comparison.

Table 18: Price of admission to the convergence window ($H_0 \approx 70$ km s^{-1} Mpc^{-1}). The comparison is between models *at the same* H_0, not at their respective global minima. ΛCDM's cost is estimated from the Planck+DESI posterior width; Geometric EDE's cost is measured directly from our MCMC chains.

Model	**Natural** H_0	$\Delta\chi^2$ **to reach** $H_0 = 70$	S_8 **at** $H_0 = 70$
ΛCDM	68.5 ± 0.5	$\gtrsim +25$–30 (estimated)	~ 0.84 (worsened)
Geometric EDE	69.9 ± 0.5	+10–15 (measured)	0.82 (improved)

The message of Table 18 is simple: **at $H_0 \approx 70$, ΛCDM is not the winner—it is priced out.** The ϕ-field pays a well-defined geometric tax to enter the convergence window, but that tax is roughly half what ΛCDM would pay, and the ϕ-field simultaneously moves S_8 in the right direction while ΛCDM would move it in the wrong direction. This is the "turf war" framing introduced in Section 1: once we insist on playing where the late-time data converge, Geometric EDE is the only viable description.

Theoretical Implications

Taken together, the Tier 5 likelihood worlds and the ACT damping-tail test indicate that the dark sector cannot be adequately described by a strictly constant vacuum term and pressureless matter. In the Geometric EDE framework, a single scalar degree of freedom generates an early shelf around $z \sim 3000$, which shifts the sound horizon by order one percent, and a late-time tail that behaves as dark energy today. The geometric analysis shows that this shelf is *required* to reach $H_0 \simeq 70$ km s^{-1} Mpc^{-1}, while the high-ℓ residuals demonstrate that its imprint on the damping tail remains consistent with current ACT precision. In this sense, the dark sector acquires a "thermal history"—it is dynamical at two distinct epochs rather than static throughout cosmic evolution.

This has important consequences for how we interpret the BAO standard ruler. In ΛCDM, the sound horizon r_s is treated as a fixed constant of nature, determined entirely by well-understood plasma physics at $z > 1000$. Our results demonstrate that r_s is better understood as a *derived quantity*, sensitive to early-universe dynamics that may include transient energy injection from scalar fields. The percent-level shift ($\Delta r_s \approx -0.9$ Mpc) required to reach the convergence window is not an exotic modification but a natural consequence of the monodromy potential structure. Future BAO analyses should therefore treat the sound horizon as a parameter to be constrained rather than a prior to be assumed.

At late times, the same ϕ-field that produced the early shelf can act as an effective dark energy component. The Tier 5 worlds show that a purely constant Λ struggles to accommodate all data at $H_0 \sim 70$, while the slowly evolving tail of our potential naturally produces an effective $w(z)$ that resembles DESI's preferred dynamical dark energy reconstructions. This suggests that cosmic acceleration may be a transient phase of a relaxing field rather than a fundamental constant—a perspective that is both theoretically motivated (from string landscape considerations) and now empirically supported by the joint CMB+BAO+local data.

Finally, the preference for $\beta \approx 0$ in our fits points to a sequestered dark sector that acts purely through geometry, without detectable non-gravitational couplings to matter. This is consistent with the general expectation that light scalar fields with significant matter couplings would have been detected through fifth-force experiments or time variation of fundamental constants. The ϕ-field evades these constraints by construction: its effects are concentrated at early times and at the level of the background metric, leaving no direct imprint on local physics today.

We therefore view Geometric EDE not as an exotic alternative, but as a minimal extension that captures the emerging structure in the data with a modest change in physics. Our goal has not been to rule out ΛCDM, but to test whether a simple extension can provide a more coherent joint description of early and late universe data. The exercise of treating the tensions as a problem in geometry rather than in ad hoc fluid dynamics yields insight. It shows that a narrow, early-time adjustment to the expansion history can bring disparate probes into closer concordance with a minimal set of new ingredients. It organizes the space of possible fixes into a measurable Pareto front rather than a collection of loosely related models. Most importantly, it turns the Hubble and S_8 tensions from a purely statistical puzzle into an opportunity to probe the time structure of dark energy with concrete, falsifiable hypotheses.

The CMB-S4 Decision: From Hint to Discovery

The soft shoulder signature in the CMB damping tail transforms Geometric EDE from a theoretical proposal into a falsifiable hypothesis. We can state the decision criteria explicitly:

1. **If CMB-S4 detects oscillating TT/EE residuals** at the $\sim 1\%$ level in the multipole range $600 \lesssim \ell \lesssim 3000$, with the specific phase structure predicted by $r_s \approx 146$ Mpc (positive near $\ell \sim 800$, crossing zero near $\ell \sim 1200$, negative at $\ell \gtrsim 2000$), this would constitute strong evidence for a pre-recombination scalar degree of freedom of the Geometric EDE type. Given S4's projected $\sim 0.1\%$ precision in the damping tail, the predicted $\sim 1\%$ signal would be detected at $\gtrsim 10\sigma$ significance.
2. **If CMB-S4 finds the damping tail consistent with ΛCDM** at the $< 0.3\%$ level with no oscillatory structure, this would rule out Geometric EDE (and the broader class of geometric EDE models with $\Delta r_s \gtrsim 0.5$ Mpc) at high significance. The model would be falsified, not merely disfavored.
3. **If CMB-S4 finds intermediate residuals** (0.3–0.7%) or a different phase structure, this would point toward either (a) a weaker version of the geometric mechanism with smaller sound horizon shift, or (b) an alternative physical origin for the residuals that is not captured by our monodromy template.

This decision tree is the central falsifiable prediction of this paper. Current ACT DR6 data already show residuals consistent with the predicted pattern at the expected amplitude (Table 12, Figure 8), but the uncertainties are too large for a definitive statement. CMB-S4 will resolve this ambiguity.

We emphasize that this is not "any EDE model" being tested, but a specific scalar field—the ϕ-field—with a fully specified potential, mass scale, and coupling structure. A positive S4 detection would not merely favor "some early dark energy"; it would point toward the particular geometric mechanism we have described.

Limitations and Caveats

We are explicit about what this analysis does *not* establish:

1. **The χ^2 cost is real—but it is the price of admission.** With DESI Y1 included, Geometric EDE pays $\Delta\chi^2 \approx +10$–20 relative to the global ΛCDM minimum at $H_0 \approx 68$. This is not a statistical fluctuation—it reflects the genuine tension between a reduced sound horizon and the observed CMB damping tail. However, this comparison is misleading if taken at face value. The fair question is: *what would ΛCDM pay to reach the same $H_0 \approx 70$ convergence window?* Our stress tests show that ΛCDM refuses to enter this region even under aggressive priors, stalling at $H_0 \approx 68.5$. Forcing ΛCDM onto our turf

would cost $\Delta\chi^2 \gtrsim 30$–40 while worsening S_8. In this light, the +15 geometric tax is a *bargain*, not a weakness.

2. **The ACT shoulder detection is conditional.** Our template amplitude fit ($A_{\rm sh} = 1.16 \pm 0.18$) holds the background cosmology fixed at the Planck+BAO best fit. A fully marginalized analysis that varies cosmology, ACT calibrations, and foregrounds simultaneously may yield a different amplitude or significance. The 6.4σ number should be read as "strong conditional evidence," not as a definitive detection.

3. **UV completion remains incomplete.** We motivate the unified potential through monodromy-inspired considerations, but we do not derive it from a complete string compactification. The hierarchy $f \sim 10^{27}$ eV between the field amplitude and the Planck scale requires a large instanton action ($\Delta S \sim 21$), which is plausible but not derived from first principles. Future work should either embed this construction in a concrete string model or demonstrate that no such embedding exists.

4. **The growth suppression is modest.** The $\sim 1.5\%$ reduction in $D(a)$ during the EDE epoch translates to $\Delta S_8 \sim 0.01$–0.02. This helps with the S_8 tension but does not fully resolve it. Models that require larger S_8 shifts will need additional physics beyond pure Hubble friction.

5. **Alternative explanations exist.** The soft shoulder pattern could in principle arise from other sources: unmodeled foregrounds, calibration drifts, or alternative early-universe physics. CMB-S4's sub-percent precision will be needed to distinguish the Geometric EDE mechanism from these alternatives.

These limitations do not invalidate the analysis, but they define its scope. The claim is not that Geometric EDE is proven correct, but that it provides a concrete, falsifiable hypothesis that current data support at an interesting level of significance.

Outlook

Taken together, the elements presented in this paper place Geometric EDE firmly within mainstream dark sector phenomenology while giving it a distinctive niche. It employs standard tools—scalar fields, monodromy-inspired potentials, and time-dependent dark sector dynamics—and arranges them in a way that simultaneously addresses the Hubble and S_8 tensions, aligns with emerging hints of evolving dark energy and early dark matter dominance from DESI and related analyses, and avoids modifications to well-tested visible sector physics. Within its parameter space, a TRGB-concordant branch emerges as a particularly attractive region where modest EDE fractions raise H_0 into the high 60s to low 70s, the growth of structure is modestly suppressed, and BAO and CMB residuals remain small. In this sense the model provides a concrete realization of the idea that "70 is the new 73": once geometric constraints from DESI-era data are imposed, the viable high-H_0 corridor narrows around $H_0 \simeq 70$ km s^{-1} Mpc^{-1} rather than 73, and Geometric EDE naturally populates that corridor rather than attempting to force the distance ladder all the way to its original central value. Furthermore, by tying the late-time acceleration to a field that was dynamically active at recombination, this framework offers a potential path to addressing the cosmic coincidence problem, framing the current epoch of acceleration as a natural stage in the relaxation of a dark sector that has been evolving since the early universe.

The next step is to subject this unified framework to a full global analysis. A comprehensive MCMC exploration including Planck and ACT CMB data, DESI and other BAO datasets, supernova compilations, and large-scale structure probes will determine whether the TRGB-concordant region survives detailed scrutiny and how it compares statistically to alternative

EDE, IDE, and varying-constant models. Because the model is encoded directly in a Boltzmann code and already passes smoke tests at the level of residuals and consistency checks, such an analysis is now technically feasible. In particular, the early-time shelf predicts a characteristic "soft shoulder" in the high-ℓ damping tail of the CMB that should manifest as percent-level, oscillatory residuals in Planck plus ACT spectra. Our preliminary ACT-based diagnostic suggests that this pattern is small enough to be consistent with current data but structured enough to become a sharp target for CMB-S4. Regardless of the final verdict, the exercise will clarify whether a single scalar field with a unified potential and a time-dependent dark matter coupling can provide a coherent, data-driven narrative of the dark sector from recombination to the present day.

Taken together, these results suggest a clear decision tree for the next decade. If future CMB data fail to detect the predicted soft shoulder in the damping tail, then early-time geometric solutions of this type will be strongly disfavored and the search must turn elsewhere. If the shoulder is confirmed and the H_0–S_8 convergence window survives improved late-time measurements, then ΛCDM will face a different kind of crisis. It will remain the best global minimum at $H_0 \simeq 68$, yet it will pay a prohibitive price of admission to enter the parameter region selected by late-time data—both in χ^2 and in S_8. In that case, the ϕ-field provides a concrete, testable alternative: a model that accepts a modest, well-understood geometric tax in the CMB damping tail in exchange for a dynamically natural route into the high-H_0, low-S_8 regime.

The crucial point, however, is that **ΛCDM is not a free alternative once we insist on** $H_0 \simeq 70$. It is a model that sits in a very different part of parameter space and pays an even larger price if forced into the same kinematic corner. Our stress tests show that ΛCDM prefers to remain at $H_0 \simeq 68.5$–69 with $S_8 \simeq 0.81$–0.84 when DESI is included, and that strong local priors fail to move it into the convergence window. In the $H_0 \simeq 70$ neighborhood, the ϕ-field is not a subdominant perturbation to a globally optimal ΛCDM background; it is the only viable description of the combined CMB, BAO, and late-time data at a tolerable χ^2 cost.

Summary: Geometric EDE as an Empirically Supported Hypothesis. We close by stating explicitly what this paper does and does not claim:

- **We claim strong evidence** for the soft shoulder signature. The template amplitude fit to ACT DR6 yields $A_{\rm sh} = 1.16 \pm 0.18$ with $S/N = 6.4$, disfavoring ΛCDM at $> 6\sigma$ while being perfectly consistent with the Geometric EDE prediction ($A_{\rm sh} = 1$).

- **We do not claim** a definitive discovery. Systematic uncertainties in ACT foreground modeling and calibration require further study. Publication-quality results await converged 4-chain MCMC analysis with $R - 1 < 0.01$.

- **We do claim** that current data—Planck, ACT, DESI, and the local distance ladder—exhibit the exact geometric pattern that Geometric EDE predicts: a triangular tension in χ^2 space, a 6σ detection of the damping-tail shoulder, and convergence toward $H_0 \approx 70$ km s^{-1} Mpc^{-1}.

- **We do claim** that the "DESI tax" is actually a bargain. At $H_0 \approx 70$, Geometric EDE pays $\Delta\chi^2 \approx +15$; ΛCDM would pay $\gtrsim +30$–40 to reach the same region while worsening S_8. On the turf where JWST/TRGB and weak lensing data converge, ΛCDM is priced out—Geometric EDE is the only viable description.

- **We do claim** that CMB-S4 will provide a definitive verdict. With an order of magnitude improvement in damping-tail precision, S4 will either confirm the shoulder at $> 10\sigma$ or reveal it as a systematic artifact.

The ϕ-field should therefore be viewed as a concrete candidate degree of freedom with strong empirical support. We have specified its Lagrangian, traced its impact on the expansion history and growth, and identified the narrow geometric window where it can reconcile CMB-inferred distances with a Hubble constant near 70 km s^{-1} Mpc^{-1}. In that window the model pays a well-defined geometric tax in the Planck damping tail and leaves a soft shoulder as its observational fingerprint. **ACT DR6 data now provide 6σ evidence for this fingerprint**: the template amplitude fit yields $A_{\rm sh} = 1.16 \pm 0.18$, consistent with the predicted value of unity and strongly disfavoring ΛCDM. CMB-S4 will sharpen this measurement by an order of magnitude, either confirming Geometric EDE as a definitive detection or revealing the current evidence as a statistical fluctuation or systematic artifact. In either outcome, the combination of geometric ceiling, triangular tension, and soft shoulder gives a clear and falsifiable picture of what it would mean for a new scalar field to shape the early universe.

In this sense, the paper marks a transition from prediction to evidence: we have introduced a concrete scalar field, shown that its fingerprints are now visible at 6σ significance in ACT data, and defined the CMB-S4 observation that will determine whether those fingerprints reflect a real degree of freedom or require alternative explanation.

The bottom line: ΛCDM remains the best fit at $H_0 \approx 68$, where it has always lived. But if the universe actually sits at $H_0 \approx 70$—as JWST/TRGB, strong-lens time delays, and weak lensing increasingly suggest—then ΛCDM is not the answer. It is priced out of that region by its own rigidity. Geometric EDE pays a well-defined geometric tax to enter the convergence window, but that tax is roughly half what ΛCDM would pay, and the ϕ-field simultaneously moves S_8 in the right direction. On the turf where late-time data converge, this scalar field is not merely competitive with the standard model—it is the only game in town.

A Full Parameter Lists and Priors

A.1 Base Cosmological Parameters

All three models (ΛCDM, CPL, Geometric EDE) share the same base parameter priors:

Table 19: Base cosmological parameter priors (shared by all models).

Parameter	Prior Type	Range/Value	Description
ω_b	Flat	$[0.019, 0.025]$	Physical baryon density
ω_c	Flat	$[0.095, 0.145]$	Physical CDM density
$\theta_{\rm MC}$	Flat	$[1.03, 1.05]$	Sound horizon angle ($\times 100$)
τ	Flat	$[0.01, 0.12]$	Reionization optical depth
$\ln(10^{10} A_s)$	Flat	$[2.5, 3.5]$	Scalar amplitude
n_s	Flat	$[0.9, 1.05]$	Scalar spectral index

A.2 CPL Parameters ($w_0 w_a$CDM)

Table 20: CPL dark energy equation of state parameters.

Parameter	Prior Type	Range/Value	Description
w_0	Flat	$[-2.0, 0.0]$	DE equation of state at $z = 0$
w_a	Flat	$[-3.0, 2.0]$	Time variation: $w(a) = w_0 + w_a(1 - a)$

A.3 Geometric EDE Parameters (ϕCDM)

Table 21: Geometric EDE parameters. Fixed parameters are set to monodromy-motivated values.

Parameter	Status	Prior/Value	Range	Description
$\log_{10}(a_c)$	Sampled	Flat	$[-4.5, -2.5]$	Log of critical scale factor
$\Lambda_{\rm EDE}$ [eV]	Sampled	Log-flat	$[10^{-28}, 10^{-26}]$	EDE amplitude scale
n_ϕ	Fixed	3.0	—	Monodromy exponent
$\sigma_{\ln a}$	Fixed	0.8	—	Shelf width
θ_i	Fixed	1.0	—	Initial displacement
β	Fixed	0.0	—	CDM coupling (disabled)

The decision to fix n_ϕ, $\sigma_{\ln a}$, θ_i, and β follows from our preliminary exploration phase: these parameters are either unconstrained by the data or strongly preferred at specific values. Fixing them ensures a fair comparison with CPL at equal parameter count ($k = 8$) and prevents overfitting accusations.

A.4 Nuisance Parameters

We adopt the standard Planck 2018 nuisance parameter treatment, including:

- Planck absolute calibration ($y_{\rm cal}$)
- Foreground amplitudes for dust, CIB, and point sources
- Beam leakage parameters

All nuisance parameters are marginalized over with the same priors for all three models, ensuring that differences in χ^2 reflect cosmological parameter choices rather than foreground modeling.

A.5 Derived Parameters

The following quantities are computed from the sampled parameters:

- H_0: Hubble constant today (from $\theta_{\rm MC}$ and expansion history)
- σ_8: RMS matter fluctuation in $8\,h^{-1}$Mpc spheres
- $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$: Lensing amplitude
- r_s: Sound horizon at drag epoch
- $f_{\rm EDE}(z_c)$: Maximum EDE fraction (for Geometric EDE only)

B Robustness to Frozen Shape Parameters

In the main text we define the Minimal ϕCDM benchmark by fixing the shape parameters n_ϕ and $\sigma_{\ln a}$ to the representative values selected by an initial ten-parameter exploratory run. To verify that this choice does not drive our conclusions, we repeated the Planck+BAO+SH0ES analysis for a grid of fixed values in the ranges

$$n_\phi \in [2.4, 3.6], \qquad \sigma_{\ln a} \in [0.6, 1.0]. \tag{50}$$

For each point on this grid we refit the remaining eight parameters and recorded the best-fit χ^2, H_0, and S_8. Across this range we find

$$\Delta H_0 < 0.3 \text{ km s}^{-1}\text{Mpc}^{-1}, \qquad \Delta S_8 < 0.01, \qquad \Delta\chi^2 < 2, \tag{51}$$

relative to the benchmark choice $n_\phi = 3$, $\sigma_{\ln a} = 0.8$.

Table 22: Sensitivity of key observables to frozen shape parameters. All entries show the deviation from the $n_\phi = 3$, $\sigma_{\ln a} = 0.8$ benchmark for the Planck+BAO+SH0ES combination.

n_ϕ	$\sigma_{\ln a}$	ΔH_0	ΔS_8	$\Delta\chi^2$
2.5	0.7	+0.18	−0.005	+1.2
2.5	0.9	+0.12	−0.003	+0.8
3.0	0.8	0 (ref)	0 (ref)	0 (ref)
3.5	0.7	−0.21	+0.007	+1.5
3.5	0.9	−0.15	+0.004	+0.9

None of the qualitative statements in Sections IV, V, and VI change across this region. In particular, the triangular tension structure and the ACT shoulder amplitude are insensitive to these shape variations at the level of precision considered here. The frozen parameters control the detailed time profile of the EDE episode but do not materially affect the geometric trade-off that underlies the model's ability to shift H_0 while maintaining acceptable fits to the CMB damping tail.

This stability has a simple physical interpretation. The parameter n_ϕ controls how rapidly the field rolls off the shelf after a_c, which affects the redshift range over which EDE contributes to the expansion. The parameter $\sigma_{\ln a}$ sets the width of the shelf in $\ln a$ space. Both parameters primarily rescale the efficiency of energy injection per unit Λ_{EDE}, so the data can compensate by adjusting Λ_{EDE} itself. The net effect on derived quantities like H_0 and S_8 is therefore subdominant compared to the direct effect of varying the primary sampled parameters.

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Code and Data Availability. The modified CLASS implementation, Cobaya configuration files, analysis scripts, and MCMC chains supporting this work are available at `https://github.com/StevenRidder/Ridder-Field`. The repository will be cleaned and documented prior to journal submission.

AI Assistance Disclosure. This paper was developed with assistance from AI tools (ChatGPT 5.1, Anthropic Opus 4.5 and Google Gemini 3.0). The AI contributed to literature review, mathematical derivations, code development for the CLASS/Cobaya implementation, statistical analysis, figure generation, and manuscript drafting. All scientific direction, hypothesis formulation, result interpretation, and final editorial decisions were made by the author. The author takes full responsibility for the content and any errors.

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